

AE4 salivary transport

Complete mathematical analysis, parameter provenance and diagnosis of model inadequacy

Technical report based on the completed Codex Tasks 43 and 44

15 September 2026

This report consolidates the completed analysis of the frozen Task 40 model and the prescribed Task 41 inverse family. It includes the full mathematical exposition, expanded computational methods, complete numerical case tables, parameter provenance and the original electronic evidence. It is a technical basis for article development, not a new model or a claim that the biological mechanism has been validated.

Immutable scientific inputs

Task 43: 547f113d9123ab1976774639a69483faf40cef34

Task 44: e5fa9840bf96be3147c6118daee4942468c78f8b

Numerical replay: f783785a863440df459e9c5530beba4c7eb59110

The source branches, production code, original outputs and existing manuscript are not modified by this report. Exact results, local numerical findings, finite duration calculations and unresolved biological questions are explicitly distinguished.

Abstract

The reconstructed AE4 salivary transport model was examined using complete parameter provenance, invertible nondimensionalisation, conditional algebraic closure results, conserved amount identities, equilibrium continuation, local stability and sensitivity analysis, source vector projections and an inverse CaCC recruitment calculation. Carbon, alkalinity, both finite volumes, lumen composition, algebraic pH and electrical closure, and AE4 regulation remain present. Thirteen production states are represented by eleven independent dynamic coordinates after exact charge elimination. At the frozen parameter point the stationary AE4 null secretion deficit is approximately 0.94617%, while the original 600 s cumulative deficit is 3.8575%. Local NKCC chloride compensation is approximately 92.901%; the integrated replacement of missing AE4 chloride over 60 to 600 s is 88.507%. These are distinct observables. The system already contains slow modes of approximately 346 and 460 s in WT and null. A physiological NBC absent root exists at much lower AE4 loading, so the alkalinity obstruction is conditional on a specified loading demand. The prescribed Task 41 family requires approximately 89.51% AE4 dependent stimulated CaCC recruitment at its first numerical comparison crossing, but that target selected construction later violates the pH gate. The analysis diagnoses a mismatch between a particular mathematical reconstruction and the biological phenotype. It does not prove that NKCC1 is the unique defective component, that compensation must vanish in all protocols, or that a better model has already been identified. Full source attribution and electronic numerical evidence accompany the technical account.

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Part I

Purpose, interpretation and computational methods

1 Question, scope and standard of inference

The scientific question is why AE4 loss produces a substantial sustained salivary secretion deficit even though NKCC1 remains available as a chloride loading pathway, whereas AE2 loss has comparatively little effect in the experimental comparison. The work documented here did not construct a successful replacement model. It analysed a particular conservation explicit reconstruction and established exactly how that reconstruction responds to AE4 perturbation. The distinction matters: a mathematically coherent response can still be an inadequate account of the biological experiment.

The main mathematical input is the completed Task 44 analysis, including its final report, all included TeX fragments, numerical tables, verification records and executable analysis scripts. The parameter input is the completed Task 43 audit. Their final branch commits are pinned on the title page. Both refer to the frozen Task 40 production configuration; Task 44 additionally examines the single Task 41 CaCC recruitment family. Earlier Tasks 37 to 42 supply the frozen reference context. Task 46 model discovery is outside the scope of this report, and no prospective candidate is treated as a result.

This is a source grounded synthesis. Part II retains the complete Task 44 mathematical exposition, rather than replacing it by a summary. Part III retains the completed Task 43 provenance account. Part IV supplies the full equilibrium case tables, spectra, modal contributions, expression sensitivities, all extended parameter sensitivity metrics, inverse scan and both inverse sensitivity observables, followed by detailed parameter records. Part V explains verification, coverage and the implications for article development. The accompanying electronic archive contains the original reports, scripts and machine readable payloads unchanged, with an individual hash manifest. Large arrays, all nearby trial records and per diagnostic residuals are preserved there rather than rounded into a few headline numbers.

1.1 Four levels of conclusion

An *exact or proved* result follows from the stated equations under its explicit domain assumptions. It need not hold after a kinetic law, charge convention or transported source vector is changed. A *local numerical* result describes a solved equilibrium, derivative or spectrum at a specified parameter point. It is not a proof of global uniqueness or a statement over unknown physiological parameter ranges. A *finite duration numerical* result describes an initial condition, stimulus, integration method and observation window. It is not an equilibrium statement. An *unresolved* issue is one for which the existing sources do not establish the proposed conclusion. The completed analysis contains all four categories, and the report does not promote one into another.

In particular, repeated numerical reproduction establishes consistency of a computation with its saved inputs. It does not independently validate the biological mechanism. Similarly, a physiological gate is an inherited admissibility criterion used in this investigation, not a newly measured confidence interval. A case can pass these gates while having a separate thermodynamic qualification.

Source record: `evidence/task44/FINAL_STATUS.md`; `evidence/task44/report.tex`; `evidence/task43/FINAL_STATUS.md`.

2 What system was actually analysed?

The production model represents an acinar cell, a finite lumen and a fixed extracellular bath. Its thirteen evolving coordinates are the five intracellular amounts of sodium, potassium, chloride, total inorganic carbon and total alkalinity, the cell volume, the corresponding five luminal amounts, the lumen volume and an AE4 regulatory fraction. The fixed bath is an external reservoir; the lumen has an outflow sink. Consequently, conservation means that every internal and boundary source is accounted for. It does not mean that each amount in the open cell and lumen is individually constant.

Total inorganic carbon is the sum of carbon dioxide, bicarbonate and carbonate. Alkalinity records bicarbonate, twice carbonate, the deprotonated noncarbonate buffer, hydroxide and the negative contribution of hydrogen. Internal protonation changes speciation but does not create total inorganic carbon or alkalinity. Proton extrusion and bicarbonate or carbonate transport have different source signatures. Retaining those signatures is central to the analysis, rather than a cosmetic reformulation of a chloride model.

At each evaluation, carbonate equilibrium determines pH and species concentrations from amounts and volume. The electrical equations determine apical and basolateral voltage from the current balance. Neither pH nor voltage is an independently frozen constant. Water flow responds to the current osmotic state, and lumen outflow removes both water and solutes. AE4 regulation remains dynamic. The calcium stimulus is prescribed, not generated by a calcium signalling subsystem.

Two charge identities permit exact elimination of the two alkalinity coordinates from the numerical independent state. This changes thirteen stored production coordinates into eleven independent dynamic coordinates, without discarding the alkalinity balances or their effects. In particular, the eliminated intracellular alkalinity is reconstructed from sodium, potassium, chloride and fixed anion amount; luminal alkalinity is reconstructed from luminal ionic charge. This is not the reduction in *Marty.tex*. No carbon pool, volume or luminal composition has been held constant to obtain the eleven dimensional system.

The independent representation is important for equilibrium differentiation. Keeping redundant charge coordinates would introduce directions that do not represent freely variable dynamics on the analysed charge manifold. The computed nonsingularity and stability statements refer to that manifold and the declared regulatory linearisation, not to an unconstrained enlargement of the model.

2.1 Notation used in the inherited analysis

The mathematical chapters use N for NKCC1 cycle flux, H for NHE1 flux, E for AE2 flux, B for NBC cycle flux, J_4 for AE4 cycle flux, $P = P_a + P_b$ for pump cycle flux, and C_a for apical chloride efflux. The symbol H used as a flux must not be confused with hydrogen concentration, which is written h or $[H^+]$. Each NKCC1 cycle transports two chlorides; the parent AE4 source transports one chloride per scalar cycle. Q denotes lumen outflow, while its time integral is the accumulated secreted volume. Amount fluxes are in fmol/s, volumes in pL, concentrations in mM and voltage in V unless a table states otherwise.

Source record: `evidence/task44/exact_results.tex`; `evidence/task44/core.py`.

3 Why the numerical outputs diagnose an inadequate phenotype

The experimental account carried into Task 44 reports an approximately $35 \pm 4.7\%$ reduction in ten minute secretion following AE4 deletion, with the quoted uncertainty identified as a standard error rather than a confidence interval. The initial response is comparatively preserved and the difference

develops subsequently. That account comes from the source review in the completed analysis, with its stated review limits; this synthesis does not claim a new experimental reanalysis [Peña Münzenmayer et al., 2015a].

Against that context, the frozen Task 40 model yields only a 3.8575% cumulative null deficit over 600 s. In a separate constant stimulus equilibrium calculation the null deficit is approximately 0.94617%. Neither number should replace the other. The former includes the initial stored material, changing state and finite lumen response. The latter compares stationary outputs on solved local branches. Both describe small effects for this particular configuration.

The compensation measurements explain an important part of the behaviour. At the WT equilibrium, the local gain is

$$-\frac{2 \, dN^*/de_4}{dJ_4^*/de_4} \simeq 0.92901.$$

This is an incremental equilibrium derivative, not the fraction of the entire knockout recovered. Over 60 to 600 s in the finite experiment, NKCC1 supplies an additional approximately 36.33701 fmol of chloride, compared with approximately 41.05550 fmol of missing AE4 chloride. The replacement fraction is therefore approximately 88.507%. The same extra NKCC1 loading is approximately 23.163% of WT NKCC1 loading. The three percentages have different denominators and different experimental meanings.

These results support the statement that the analysed network largely compensates the lost AE4 chloride input. They do not establish that a single parameter, transporter law or molecular process is uniquely responsible for the mismatch with experiment. Pump, NHE1, NBC, carbonate state, ionic storage, voltages and water flow are coupled. The sensitivity analysis identifies several uncertain quantities with substantial effects. A mechanistic diagnosis should therefore distinguish the observed compensation from an unproved claim that NKCC1 alone is defective.

3.1 What the chloride identity does and does not explain

The exact equal routing identity is

$$C_a = 6P - H + 2\dot{n}_{Na,i} - \dot{n}_{A,i} - \dot{n}_{Cl,i}.$$

At stationary intracellular amounts the three derivatives vanish and the explicit AE4 term is absent. The proof uses a linear combination of the sodium, alkalinity and chloride balances. It neither fixes nor removes carbon, pH, volume or lumen dynamics.

The identity constrains the ways that secretion can respond; it does not evaluate the state dependent pump and NHE1 rates. The total derivative with respect to AE4 expression remains nonzero through the equilibrium state. The small computed null deficit is consequently a numerical result of the adopted model and parameter point, not a universal consequence of the cancellation. Nor is C_a identical to fluid secretion: luminal chloride concentration, paracellular return and changing chloride stores intervene between apical export and volume outflow.

A correct article claim is therefore that the cancellation reveals a structural constraint and the subsequent full calculation quantifies strong compensation. A stronger statement that the identity proves a small phenotype for every allowed parameter set is not supported.

Source record: `evidence/task44/integrated_context.tex`; `evidence/task44/output/compensation_comparison.json`; `evidence/task44/exact_results.tex`.

4 Three corrections that change the scientific interpretation

4.1 Alkalinity support is conditional, not universal NBC necessity

The parent stationary alkalinity requirement is $2J_4 = H + 2B - E$. The exact finite duration budget additionally includes the change in the intracellular alkalinity store. Given a specified positive AE4 demand and upper bounds on the admissible non NBC support, that balance provides a genuine obstruction. However, it does not require the demand itself to remain fixed when the whole system changes.

Task 44 finds an NBC absent physiological mathematical root. Its AE4 cycle flux is approximately 0.00632651 fmol/s rather than 0.07507245 fmol/s at the reference WT root, and its stationary secretion is approximately 3.035% lower. This is not a contradiction of the conditional budget. The system avoids the larger alkalinity demand by operating at substantially lower AE4 loading. The defensible conclusion is that extra alkalinity support is required for the *specified larger loading regime under the stated capacity assumptions*, not that all physiological secretion mathematically requires an NBC pathway. The pathway's molecular identity remains unknown.

4.2 Slow modes exist and affect the output

The slowest local decay times are approximately 345.82 s in WT and 459.92 s in the null. The modal calculation also evaluates how an expression perturbation excites each mode and how that mode projects onto secretion. Slow modes have nonzero contributions to the secretion response, even when their contributions partly cancel those of other modes in the stationary derivative.

It would therefore be incorrect to infer a missing slow regulator simply because a particular secretion curve has the wrong timing. Local eigenvalues, input projection, output projection, nonnormal state amplification, initial conditions and finite perturbation size must be considered together. The source provides a local linear response calculation, not a validated linear approximation to the complete WT to null experiment. The correction of the regulatory eigenvalue from a spurious 60 s mode to the proper 30 s mode concerns differentiation across a numerical clamp; it does not remove the slower chemical modes.

4.3 The CaCC construction is a conditional inverse requirement

In the prescribed Task 41 family, the stimulated CaCC conductance is multiplied by $1 - a\rho(1 - e_4)$. WT is unchanged for any ρ , while a fully stimulated null retains $1 - \rho$ of parent conductance. This is an explicitly target selected family, not independent evidence of AE4 control of CaCC recruitment.

The first numerical crossings of a 30.3% cumulative deficit require approximately 89.50727% AE4 dependent recruitment with adaptive integration and 89.50806% with the original sampled observable. A finite ordered scan and local bracket refinement do not prove global monotonicity or exclude every unsampled earlier crossing. Furthermore, separate continued stimulus null cases exceed the upper pH gate near 1485 s. The construction demonstrates what this particular added coupling can achieve during its declared 600 s interval, and where it ceases to satisfy the inherited physiological criterion. It does not establish a universal lower bound, sustained physiological validity or a biological mechanism.

Source record: `evidence/task44/report.tex`; `evidence/task44/equilibrium_results.tex`; `evidence/task44/inverse_results.tex`.

5 Thermodynamics, experimental comparability and the limits of blame

The source vector experiment changes transported coefficients while retaining the parent scalar AE4 carrier law. An inherited carrier law with locally balanced edges does not automatically become a thermodynamically valid implementation of every substituted cycle. Task 44 explicitly records an opposed signed flux and affinity in C0 WT and in several sodium supported alternatives at the tested states. This qualification belongs beside any assertion of physiological acceptance, not in an unrelated footnote.

Thus a numerical root can satisfy all inherited concentration, volume and current residual gates and still fail a separate cycle affinity check. Current closure and stoichiometric charge conservation are necessary bookkeeping properties, but they are not a proof of full energetic consistency. The report retains both sets of diagnostics for every source class. It does not silently repair the scalar laws or reject inconvenient rows from the mathematical analysis.

The NKCC1 assay requires equally careful interpretation. The source review reports no detectable genotype difference in the initial isolated NKCC1 dependent chloride uptake under bicarbonate free, inhibitor treated conditions. This is not a direct measurement of the total integrated NKCC1 chloride loading during the complete stimulated network experiment [Peña Münzenmayer et al., 2015a]. The present strong compensation is an experimental tension that needs matched protocol testing. A nondetection in one assay does not establish a universal identity of NKCC fluxes for every intracellular state and stimulus.

The original simulations also initialise all three expression cases at the same inherited WT state. That is not the same preparation as established knockout animals with potentially different resting stores. The 5% expression result in this repository is a prescribed model intervention; it must not be called an independently measured expression phenotype unless an appropriate experimental source is supplied. These distinctions prevent model predictions and historical protocol choices from being reclassified as biological observations.

The current NKCC1 law has a rational saturating form with recorded forward and reverse bounds. Strong network compensation is therefore not, by itself, evidence that the law has unlimited flux. The unresolved question is whether its effective capacity, dependence on the ionic state, regulation and coupling to the rest of this reconstruction are appropriate for the biological preparation. The Task 43 audit and Task 44 sensitivities make that question specific, but do not settle it.

Interpretation. The results justify seeking a better constrained mechanistic model. They do not already prove which replacement is correct. The purpose of the next reconstruction is to test candidate explanations of this diagnosed failure, rather than to impose a desired compensation value and label the resulting secretion curve a discovery.

Source record: `evidence/task44/literature_comparison.tex`; `evidence/task44/equilibrium_results.tex`; `evidence/task44/output/dimensionless_groups.json`; `evidence/task43/report.tex`.

6 Expanded account of the computational analysis

This chapter explains the executable operations behind the final mathematical report. It is derived from the pinned scripts and their saved records. It does not introduce new fitted values, new trajectories or an unreported reconstruction of an unavailable failed attempt.

6.1 Recovery of the actual production configuration

The entry point `core.py::reference` loads the historical Task 40 validation helper, obtains its nested production objects and reads the frozen WT state from the separate resting state file. It checks the state digest and the six stimulated parameter object digests against the frozen inputs. The source manifest is compared with Git blob hashes for the relevant production files. The audit reports 150 manifest entries, with 149 matches and one instructional metadata difference in `AGENTS.md`; that difference is not described as a numerical model change.

The nested model comprises the selected whole cell model, the NKCC stimulation wrapper and the minimal NBC wrapper, with a separate AE4 source class wrapper when required. A clone used for equilibrium calculations replaces the input by constant full stimulation while retaining the selected kinetic parameters. The parameter audit distinguishes this active configuration from dataclass defaults, legacy fields and constructor reference tuples. This is why a table of defaults would not reproduce the calculation.

For a buffer parameter perturbation, the code adjusts an unused constructor reference chloride value to preserve the constructor's fixed charge consistency. The actual frozen conserved initial condition used in the inverse trajectories is not replaced by that reference tuple. Consequently, a changed buffer amount may change the initial algebraic pH even though the conserved state is held fixed. These two operations must not be conflated with a new genotype resting solve.

Source record: `evidence/task44/core.py::reference,clone;evidence/task44/output/source_audit.json;evidence/task43/audit.py.`

6.2 Invertible scaling and independent reconstruction

The scaling object uses bath sodium for C_0 , the frozen cell and lumen volumes for V_0 and L_0 , and the initial fully stimulated WT NKCC cycle flux for J_0 . The flux scale is a cycle rate, not a chloride rate. Distinct amount scales are used for the cell and lumen. The time scale is $t_0 = C_0 V_0 / J_0$ and the water flux scale is J_0 / C_0 . The complete diagonal state scaling retains the regulatory fraction with scale one.

The independent index list is (0, 1, 2, 3, 5, 6, 7, 8, 9, 11, 12) in the zero based production ordering. Expansion reconstructs intracellular and luminal alkalinity from the exact charge identities before the production evaluator is called. Thus the independent vector field still depends on carbonate speciation and electrical closure at each evaluation. A second, explicit dimensionless right hand side and a separately assembled thirteen state right hand side provide checks that are stronger than merely printing the same production output twice.

The recorded dimensional versus dimensionless comparison has a maximum scaled state discrepancy of approximately 7.524×10^{-10} and a dimensional cumulative water recovery discrepancy of approximately 1.443×10^{-15} pL. These are agreement measures at saved comparison points. They do not establish a singular perturbation approximation or a global integration error theorem. The exact verification additionally reconstructs all laws and source terms at 84 random valid states, using seed 45044 and twelve states per source class. Those finite checks supplement the proofs; they do not replace them.

Source record: `evidence/task44/core.py::Scaling;evidence/task44/verify_exact.py;evidence/task44/output/exact_verification.json;evidence/task44/output/rescaling_verification.json.`

6.3 Four different experiments, not one interchangeable simulation

The first experiment reproduces the frozen finite production protocol. Its saved reference uses a resting observable at exactly $t = 0$ and begins production integration at 10^{-6} s with the shared

conserved state. The audited production solver is Radau with relative tolerance 10^{-7} , amount tolerance 10^{-10} fmol, volume tolerance 10^{-12} pL, regulatory tolerance 10^{-10} and maximum step 2 s. The dose labels do not generate the prescribed calcium and beta inputs through a dose response model.

The second experiment is the separate constant full stimulus equilibrium problem. The code evaluates that vector field at an arbitrary fully stimulated time, conventionally 1 s; the `time_s` field in an equilibrium table is not an equilibration time or an experimentally observed delay. Equilibria are solved by a nonlinear root method, not identified by declaring a finite trajectory endpoint stationary.

The third experiment is the local expression step about a solved equilibrium. It uses the linearised state and output maps and therefore describes a derivative response to a small parameter perturbation. The fourth is the Task 41 inverse family, integrated through 600 s with an additional accumulated outflow variable and separately examined under continued stimulation. A conclusion established for one of these experiments is not automatically a conclusion about another.

The parent integration adapter uses Radau, relative tolerance 10^{-8} , absolute tolerance 10^{-11} and maximum step 2 s for its standard analysis integration; the dimensionless tolerances are scaled consistently. The inverse verification uses the tighter relative tolerance 2×10^{-9} and maximum step 1 s, including an independent thirteen state integration with subsequent quadrature of dense outflow. These analysis settings are not presented as the frozen production tolerances.

Source record: `evidence/task43/output/numerical_controls.json;evidence/task44/core.py::integrate;evidence/task44/inverse_analysis.py;evidence/task44/inverse_verify.py.`

6.4 Equilibrium solution and physiological acceptance

For each source class and expression setting, ten positive independent chemical and volume coordinates are represented by their logarithms. The regulatory equilibrium is set to one. The hybrid nonlinear root method uses `xtol=1e-10` and a limit of 1500 function evaluations. The returned candidate is expanded to all thirteen production coordinates, and all eleven independent equations are then checked. In the root helper, a maximum independent residual below 10^{-7} is the numerical acceptance condition. The records separately retain the solver's success flag, message and evaluation count. Residual based acceptance must not be silently described as a theorem of existence or as a complete diagnostic of every solver status.

C0 is continued through expression levels 1, 0.9, 0.75, 0.5, 0.25, 0.1, 0.05, 0. Other classes use the converged C0 WT seed for their WT solve and matching C0 expression seeds for the 5% and null solves. This produces 26 class and expression cases. Since all seven null source terms vanish, seven entries solve the same null system. The count is not a count of 26 distinct equilibria of one fixed vector field.

The early inherited checkpoint retained only an aggregate statement about four unsuccessful continuation attempts. Their individual seeds, class labels and error messages are not available in that record. The completed analysis preserves that limitation and supplies a documented successful retry strategy. It does not invent an attempt history and does not reinterpret failure of a particular numerical search as nonexistence of a root.

The initial helper's compact physiological filter is supplemented by the later full inherited gate review. The complete review distinguishes positivity, finite diagnostics, chemical intervals, source bookkeeping, current residuals and architecture checks. Four WT class cases remain outside the physiological gates despite solved stable roots. The supplementary sensitivity verifier explicitly reapplies the complete inherited gates to all 384 nearby cases; a successful nonlinear solve alone is not the acceptance criterion.

Source record: `evidence/task44/core.py::stationary,physiology;evidence/task44/equilibrium_review.py;`

6.5 Jacobians, spectra and boundary differentiation

The dynamic Jacobian is formed in the eleven scaled independent coordinates. The coordinate difference is the relative step multiplied by the larger of the coordinate magnitude and 0.1. The two principal steps are 10^{-5} and 5×10^{-6} . Time is then restored by division by t_0 . Eigenvalues from the two calculations are matched individually, rather than compared only through their largest real parts. The saved maximum matched relative discrepancy is approximately 5.43×10^{-8} .

At a regulatory boundary, the physical side or a smooth extension must be used. A centred difference that crosses the production clamp at $r = 1$ halves that Jacobian column and produces a spurious regulatory mode. The corrected code uses a second order inward derivative. The resulting regulatory eigenvalue is $-1/30$ per second. The chemical modes remain substantially slower. The same issue is relevant when interpreting the derivative at the null expression endpoint: the source uses an inward expression difference where appropriate.

Singular values and condition numbers refer to the declared scaled coordinates. They are numerical conditioning diagnostics and cannot be interpreted as coordinate independent measures of biological fragility. Negative real eigenvalues support local stability of the computed root under the stated smooth physical linearisation. They do not prove global attraction, forward invariance of the physiological set, or that a trajectory beginning at the frozen resting state never leaves a gate.

The parameter sensitivity calculation further checks decay estimates using fourth order differentiation of the ten coordinate chemical block at two steps, together with the exact regulatory eigenvalue. This uses the block structure while retaining the complete eleven mode spectrum. No chemical or volume state is discarded to obtain a faster numerical calculation.

Source record: evidence/task44/core.py::jacobian;evidence/task44/equilibrium_review.py;evidence/task44/sensitivity_analysis.py::resolved_decay.

6.6 Implicit sensitivities and what a decomposition means

At a locally regular equilibrium the state sensitivity is obtained by solving the linear system $F_z z_\mu = -F_\mu$. The output derivative includes both direct parameter dependence and the dependence mediated by the solved state. The implementation validates the result against independently solved neighbouring roots, rather than assuming that a finite difference of the original residual verifies the complete equilibrium response.

For expression, the comparison uses steps 10^{-4} and 5×10^{-5} at expression 1, 0.5, 0.05 and 0, with a second order inward rule at zero. Six quantities are checked: outflow, NKCC cycle flux, AE4 chloride flux, total pump cycles, NHE flux and intracellular pH. All eleven state derivatives are also checked. Part IV prints the four derivative sets and their checks, including the zero denominator qualification for any compensation ratio.

For the sixteen uncertain parameters, the independent neighbouring parameter points are $p \exp(\pm h)$ for $h = 10^{-3}$ and 5×10^{-4} . WT and null roots are resolved independently at every perturbed point. Fixed state derivatives use the smaller steps 10^{-5} and 5×10^{-6} to resolve cancellations. The output includes derivatives and elasticities for WT secretion and pH, the stationary null deficit, NBC and non NBC alkalinity shares, AE4 demand, the conditional NHE and no NBC capacities, compensation and both slow decay times. These are local partial perturbations without a new WT recalibration.

The original `coordinate_attribution.csv` is a chain rule accounting by state coordinate. In the outflow law, explicit dependence occurs through lumen volume, so a coordinate attribution can put the output derivative into that coordinate while all transporters influence it indirectly. Such a

table is not a unique mechanistic assignment of causal percentages to NKCC, pump and other pathways. Part IV retains the table and the modal projection data so this limitation is visible.

An equilibrium time parameter can have zero stationary sensitivity while affecting a time dependent response. Likewise, effective capacity and a full stimulation multiplier can enter as a product and have identical stationary sensitivities without being separately identifiable. The source explicitly reports both examples. None of the chosen finite difference steps is an experimentally justified parameter range.

Source record: `evidence/task44/integrated_context.tex;evidence/task44/sensitivity_results.tex;evidence/task44/output/sensitivity_extended.json;evidence/task44/output/coordinate_attribution.csv.`

6.7 Modal response and nonnormal amplification

The modal review uses the right eigenvector matrix and its inverse left rows, with right vectors normalised to Euclidean length one. For an expression perturbation, it evaluates each input projection, secretion output projection, transfer residue and contribution to the stationary derivative. The propagator norm is sampled over the recorded time grid extending to 10000 s. Its maximum of approximately 1.37 is a sampled result in the chosen state metric, not a proved supremum or a physiological energy gain.

The WT stationary derivative is small partly because modal contributions of opposite signs cancel. The slowest WT mode contributes approximately -5.5754×10^{-5} pL/s per unit expression, while the net derivative is approximately 1.2853×10^{-6} . The linear derivative at 600 s remains approximately 1.0027×10^{-5} , so the local approach to the stationary response is not instantaneous. The 30 s regulatory mode has zero expression input projection in this experiment because its target does not depend on expression. This does not imply that perturbing the regulatory input would have the same response.

Source record: `evidence/task44/equilibrium_results.tex;evidence/task44/output/equilibrium_review.json;evidence/task44/output/local_step_response.csv.`

6.8 Inverse scan, refinement, perturbation and extended stimulation

The inverse calculation retains the one prescribed conductance family and the common conserved WT initial state. The 17 point ordered recruitment scan is followed by bracketed root refinement. Both the adaptive integral and original sampled trapezoidal integral are retained. The latter includes integer seconds together with the distinct initial points at zero and 10^{-6} s. The selected historical point is reproduced for WT, 5% expression and null before a new comparison crossing is discussed.

The refined crossings are checked with two solver settings and against independently integrated production states with quadrature of dense flow. For each of eight effective parameters, the implicit crossing derivative is compared with independently solved neighbouring crossings at two logarithmic perturbation sizes. The WT denominator uses the same parameter perturbation as the null numerator. This is necessary to interpret the derivative of a ratio rather than a change in the null numerator alone.

Physiological gates are checked at accepted solver endpoints and the original observation grid. This is a finite numerical monitoring procedure, not a proof of interval wide invariance between every sampled point. The extended selected and crossing null cases are separate constant stimulus diagnostics through 3600 s. Their first pH crossing times are refined numerically; the late roots are labelled mathematical continuations outside the inherited pH interval. The report preserves the original 600 s observables alongside the later failure.

Source record: `evidence/task44/inverse_analysis.py;evidence/task44/inverse_verify.py;evidence/task44`

4/inverse_detail/protocol.json;evidence/task44/inverse_detail/independent_verification.json.

6.9 What clean replay verifies

The original recorded clean replay applies to numerical checkpoint `f783785a863440df459e9c5530beba4c7eb59110`. It ran eighteen recorded commands after removing generated numerical outputs and inverse caches in a separate disposable checkout. The comparison covers 47 authoritative JSON/CSV files: 46 regenerated products and one unchanged historical record. It reports 331157 bit identical numeric values, with explicitly listed runtime and ancestor validated commit metadata exclusions. The optional 532 trial cache files were regenerated rather than reused. A separate supplement records exact agreement for 7567 additional numeric values and the complete 384 nearby gate checks.

The present report assembly copies these receipts verbatim and verifies their file hashes and internal references. It does not claim to have repeated that entire numerical replay in this report writing session. Report compilation, reference checks, data table coverage, preservation of numerical payloads and visual review are recorded separately from the original scientific verification. The distinction between these two verification levels is retained in the final build record.

Source record: `evidence/task44/output/clean_recomputation_verification.json;evidence/task44/output/clean_sensitivity_gate_verification.json;evidence/task44/recompute_pinned_checkpoint.py`.

Part II

Complete mathematical analysis from the final Task 44 sources

The following chapters retain the complete technical exposition in the final Task 44 report, including every included mathematical fragment and its tables. Their scope statements refer to the original Codex investigation. The report assembly expands the original TeX inputs, preserves the numerical values, and changes only document integration, reference namespaces and file paths. Additional detailed tables appear in Part IV. Source verification here checks the preserved records; it is not a new execution of the original full simulation programme.

Source record: `evidence/task44/report.tex` and `includedTeXfiles`.

7 Scope, checkpoint continuity and interpretation

This report continues the existing branch `analysis/task-44-full-system-mathematics`, recovered at `b7b775a0277d93263c4de339745806e5658ec9c2`. The complete numerical checkpoint independently replayed for this report is `f783785a863440df459e9c5530beba4c7eb59110`. Task 43 remains on the separate, unmerged parameter provenance branch; its final report checkpoint is `bc3e37645b9a7519dee0da9d1e2df55e291dd983`. The companion crosswalk reads numerical results at the exact Task 44 commit, with source file hashes. Neither a fit nor a new production model is introduced.

The archived `Marty.tex` fixes intracellular concentrations other than sodium, potassium and chloride, lumen concentrations, membrane potentials and cell volume. Those assumptions remove precisely the carbon, alkalinity, storage and osmotic responses at issue here. Its reduction is therefore not reused. The following scaling is an invertible coordinate change; every retained degree of freedom remains dynamic or is reconstructed by an exact constraint. The inherited algebraic electrical approximation is identified as an assumption, not justified retrospectively by a nonexistent capacitance parameter.

The report uses four levels of conclusion. A proved identity is conditional on its stated equations and domain. A local numerical result applies at the specified parameter point and branch of roots. A finite time numerical result applies to its stated stimulus, initial state and observation window. Unresolved questions include biological mechanism identity, global uniqueness, global stability, uncertainty ranges and robustness over those ranges.

8 The retained system and exact scaling

The analysis keeps the intracellular and luminal amounts of sodium, potassium, chloride, total inorganic carbon and total alkalinity; both volumes; and the AE4 regulatory fraction. Carbonate speciation and two membrane voltages are recovered algebraically at every evaluation. The older reduction in `Marty.tex` is therefore not used: neither a fixed pH nor a fixed volume can be inserted into the following identities. Alkalinity is a conserved acid–base coordinate, not bicarbonate alone.

Let N, H, E, B, J_4 denote inward NKCC1 cycles, NHE1 sodium uptake with proton extrusion, AE2 chloride uptake with bicarbonate extrusion, NBC cycles and AE4 cycles respectively. A positive NBC cycle imports one sodium and two bicarbonates. Write $P = P_a + P_b$ for the total pump cycles, K_a, K_b for potassium efflux from cell to lumen and bath, and C_a for chloride efflux from cell to lumen. The paracellular flux $j_{p,s}$ is positive from lumen to bath for species s . The carbon

dioxide fluxes D_b, D_a are positive from bath and lumen into the cell. Water fluxes q_b, q_a, q_t, Q are positive from bath to cell, cell to lumen, bath to lumen, and lumen to sink respectively. These are signed fluxes, not their positive parts. The general AE4 source vector is

$$\nu = (\nu_{\text{Na}}, \nu_{\text{K}}, \nu_{\text{Cl}}, \nu_T, \nu_A), \quad S^4 = J_4 \nu.$$

Take $C_0 = [\text{Na}]_e$, $V_0 = V_i(0)$, $L_0 = V_l(0)$ and the initial stimulated WT NKCC1 *cycle* flux J_0 . Define

$$t_0 = \frac{C_0 V_0}{J_0}, \quad \tau = t/t_0, \quad \varepsilon = L_0/V_0, \quad v = V_i/V_0, \quad \ell = V_l/L_0, \quad x_s = \frac{n_{s,i}}{C_0 V_0}, \quad y_s = \frac{n_{s,l}}{C_0 L_0}.$$

Every ionic or carbon flux below is divided by J_0 , and every water flux by J_0/C_0 ; a hat marks these fluxes. Dimensionless concentrations are $c_{s,i} = x_s/v$ and $c_{s,l} = y_s/\ell$. Primes mean $d/d\tau$. The full dimensionless amount and volume equations are

$$\begin{aligned} x'_{\text{Na}} &= \hat{N} + \hat{H} + \hat{B} + \nu_{\text{Na}} \hat{J}_4 - 3\hat{P}, \\ x'_{\text{K}} &= \hat{N} + \nu_{\text{K}} \hat{J}_4 + 2\hat{P} - \hat{K}_a - \hat{K}_b, \\ x'_{\text{Cl}} &= 2\hat{N} + \hat{E} + \nu_{\text{Cl}} \hat{J}_4 - \hat{C}_a, \\ x'_T &= -\hat{E} + 2\hat{B} + \nu_T \hat{J}_4 + \hat{D}_b + \hat{D}_a, \\ x'_A &= \hat{H} - \hat{E} + 2\hat{B} + \nu_A \hat{J}_4, \\ v' &= \hat{q}_b - \hat{q}_a, \end{aligned} \tag{1}$$

$$\begin{aligned} \varepsilon y'_{\text{Na}} &= 3\hat{P}_a - \hat{j}_{p,\text{Na}} - \hat{Q} c_{\text{Na},l}, \\ \varepsilon y'_{\text{K}} &= -2\hat{P}_a + \hat{K}_a - \hat{j}_{p,\text{K}} - \hat{Q} c_{\text{K},l}, \\ \varepsilon y'_{\text{Cl}} &= \hat{C}_a - \hat{j}_{p,\text{Cl}} - \hat{Q} c_{\text{Cl},l}, \\ \varepsilon y'_T &= -\hat{j}_{p,\text{HCO}_3} - \hat{D}_a - \hat{Q} c_{T,l}, \\ \varepsilon y'_A &= -\hat{j}_{p,\text{HCO}_3} - \hat{Q} c_{A,l}, \\ \varepsilon \ell' &= \hat{q}_a + \hat{q}_t - \hat{Q}, \quad \theta_4 r' = \beta - r, \quad \theta_4 = 30/t_0. \end{aligned} \tag{2}$$

The neutral manifold is

$$x_A = x_{\text{Na}} + x_{\text{K}} - x_{\text{Cl}} - a_f, \quad y_A = y_{\text{Na}} + y_{\text{K}} - y_{\text{Cl}}, \quad a_f = \frac{n_{\text{fixed}}}{C_0 V_0}. \tag{3}$$

Eliminating these two amounts leaves eleven independent dynamic coordinates. This is exact: cell charge is conserved; luminal charge obeys $\dot{q}_l = -Q q_l/V_l$ and hence remains zero from zero initial charge. Alkalinity and its effect on pH remain in the reconstructed state.

8.1 Dimensionless constitutive laws

Here h_i and b_i denote the dimensionless hydrogen and bicarbonate concentrations from the carbonate closure below. All half concentrations and dissociation constants in this subsection are divided by C_0 ; numerical rate constants retain their declared unit conversions. For constant full stimulation the frozen NKCC1 and NHE1 multipliers are $m_N = 1.75$ and $m_H = 2.3$. More generally their separate clipped calcium coordinates have the frozen endpoints used by the production objects: the active NKCC1 endpoints are 0.058 and 0.1 μM , whereas NBC and NHE1 use 0.058 and 0.25 μM . This distinction matters away from full stimulation.

Put $X = c_{\text{Na},i} c_{\text{K},i} c_{\text{Cl},i}^2$. The Palk law actually selected in Task 40 Palk et al. [2010a] is

$$\hat{N} = e_N m_N \frac{\alpha_N A_1}{J_0 A_3} \frac{1 - \eta_2 X}{1 + \eta_4 X}, \quad \eta_2 = \frac{A_2 C_0^4}{A_1}, \quad \eta_4 = \frac{A_4 C_0^4}{A_3}. \tag{4}$$

The coefficients are $A_1 = 157.5$, $A_2 = 2.0096 \cdot 10^{-5}$, $A_3 = 1.0306$ and $A_4 = 1.3852 \cdot 10^{-6}$ in the repository's mM convention. The effective amount flux α_N comes from the frozen WT calibration, not an imported membrane density.

For NHE1, choose $k_* = 1000k_{1+}$, converting its printed rate per millisecond into a rate per second. Let barred rates be divided by k_* :

$$\begin{aligned}\bar{a} &= \frac{1000k_{1+}}{k_*} \frac{c_{\text{Na},e}}{c_{\text{Na},e} + K_{\text{Na},e}} \frac{K_{H,e}}{h_e + K_{H,e}}, & \bar{b} &= \frac{1000k_{2+}}{k_*} \frac{h_i}{h_i + K_{H,i}} \frac{K_{\text{Na},i}}{c_{\text{Na},i} + K_{\text{Na},i}}, \\ \bar{c} &= \frac{1000k_{1-}}{k_*} \frac{c_{\text{Na},i}}{c_{\text{Na},i} + K_{\text{Na},i}} \frac{K_{H,i}}{h_i + K_{H,i}}, & \bar{d} &= \frac{1000k_{2-}}{k_*} \frac{h_e}{h_e + K_{H,e}} \frac{K_{\text{Na},e}}{c_{\text{Na},e} + K_{\text{Na},e}}.\end{aligned}$$

Thus, with the exact printed Cha constants selected by production Cha et al. [2009a],

$$M_H = \frac{1}{1 + (1 + h_e/K_{M,e})(K_{M,i}/h_i)^3}, \quad \hat{H} = \frac{n_H k_*}{J_0} e_H m_H M_H \frac{\bar{a}\bar{b} - \bar{c}\bar{d}}{\bar{a} + \bar{b} + \bar{c} + \bar{d}}.$$

The printed $k_{2-} = 183 \text{ ms}^{-1}$ is retained: the small rounding discrepancy in microscopic reversibility is not silently repaired. The other acid-base flux laws are

$$\begin{aligned}\hat{E} &= e_2 \gamma_2 \tanh \left[\frac{\log(c_{\text{Cl},e} b_i / (c_{\text{Cl},i} b_e))}{w_2} \right], & \gamma_2 &= G_2/J_0, \\ \hat{B} &= \gamma_B u \tanh \left[\frac{\log(c_{\text{Na},e} b_e^2 / (c_{\text{Na},i} b_i^2)) + \psi_b}{w_B} \right], & \gamma_B &= G_B/J_0, \\ \hat{D}_b &= \delta_b (c_{\text{CO}_2,e} - c_{\text{CO}_2,i}), & \hat{D}_a &= \delta_a (c_{\text{CO}_2,l} - c_{\text{CO}_2,i}).\end{aligned}$$

Here $\delta_{a,b} = P_{\text{CO}_2,a,b} C_0/J_0$, $\psi_b = V_b/V_T$, $V_T = RT/F$, and u is the frozen NBC calcium coordinate. The tanh widths are dimensionless effective parameters.

For the inherited AE4 shared carrier, write k_4 for its common chloride attempt rate. Define

$$\begin{aligned}L_c &= \log(c_{\text{Cl},e}/c_{\text{Cl},i}), & L_s &= \log(c_{s,i}/c_{s,e}) + 2 \log(b_i/b_e), \quad s \in \{\text{Na}, \text{K}\}, \\ a_c^\pm &= \exp(\pm L_c/2), & a_s^\pm &= (k_s/k_4) \exp(\pm L_s/2), \\ U &= a_c^+ + a_{\text{Na}}^- + a_{\text{K}}^-, & W &= a_c^- + a_{\text{Na}}^+ + a_{\text{K}}^+, \quad \pi_i = U/(U + W), \quad \pi_o = W/(U + W).\end{aligned}$$

The scalar cycle is exactly

$$\hat{J}_4 = e_4 \gamma_4 (1 + 0.25r) \left[(a_{\text{Na}}^+ + a_{\text{K}}^+) \pi_i - (a_{\text{Na}}^- + a_{\text{K}}^-) \pi_o \right], \quad \gamma_4 = n_4 k_4/J_0.$$

The frozen cooperative gate is absent. Task 40 applies equal ensemble sodium and potassium routing to this scalar cycle. The seven Catalán classes change its source vector but do not provide new class specific kinetics. Local detailed balance of the inherited carrier edges does not by itself establish thermodynamic consistency of those substituted vectors.

To specify the electrical laws, scale each conductance G as $g = GV_T 10^{15}/(FJ_0)$. The reversal potential for a valence z_s is $e_{s,uv} = z_s^{-1} \log(c_{s,v}/c_{s,u})$. With $g_{K,a}, g_{K,b}, g_{C,a}$ incorporating the frozen calcium gate a_{Ca} and background conductances, explicitly

$$g_{K,a} = g_{K,\text{total}} a_{\text{Ca}} f_K + g_{a,\text{bg}}, \quad g_{K,b} = g_{K,\text{total}} a_{\text{Ca}} (1 - f_K) + g_{b,\text{bg}}, \quad g_{C,a} = g_{\text{Cl},a} a_{\text{Ca}},$$

the channel and paracellular fluxes are

$$\begin{aligned}\hat{K}_a &= g_{K,a} (\psi_a - e_{K,il}), & \hat{K}_b &= g_{K,b} (\psi_b - e_{K,ie}), \\ \hat{C}_a &= -g_{C,a} (\psi_a - e_{\text{Cl},il}), & \hat{J}_{p,s} &= \frac{g_{p,s}}{z_s} (\psi_b - \psi_a - e_{s,le}).\end{aligned}$$

The apical pump is

$$\hat{P}_a = \gamma_P f_P \frac{c_{\text{Na},i}^3}{c_{\text{Na},i}^3 + K_{P,\text{Na}}^3} \frac{c_{\text{K},l}^2}{c_{\text{K},l}^2 + K_{P,\text{K}}^2}, \quad \gamma_P = G_P/J_0,$$

and the basolateral expression replaces $f_P, c_{\text{K},l}$ by $1 - f_P, c_{\text{K},e}$. These pump rates are voltage independent in the frozen model. The calcium gate is $c_{\text{Ca}}^n/(c_{\text{Ca}}^n + K_{\text{Ca}}^n)$, using the same calcium units in numerator and denominator.

For water define dimensionless osmolarities

$$o_i = c_{\text{Na},i} + c_{\text{K},i} + c_{\text{Cl},i} + c_{T,i} + (\beta_i + \iota_i)/v, \quad o_l = c_{\text{Na},l} + c_{\text{K},l} + c_{\text{Cl},l} + c_{T,l},$$

where $\beta_i = n_{\text{buffer},i}/(C_0 V_0)$ and $\iota_i = n_{\text{other impermeant}}/(C_0 V_0)$. The bath expression includes its declared untracked osmolyte concentration. Exactly as in the production law, free hydrogen and hydroxide are omitted from osmolarity, and the tiny luminal buffer is not added as an osmotic particle. No new approximation is introduced by the present scaling. Then

$$\hat{q}_b = \lambda_b(o_i - o_e), \quad \hat{q}_a = \lambda_a(o_l - o_i), \quad \hat{q}_t = \lambda_t(o_l - o_e), \quad \hat{Q} = \omega \varepsilon (\ell - d_l)_+,$$

with $\lambda_j = L_j C_0^2/J_0$, $\omega = k_{\text{out}} t_0$, and $d_l = V_{\text{dead}}/L_0$. These equations close (1)–(2) without fixing a volume, composition, carbon pool or voltage.

9 Carbonate closure and pH derivatives

Write p for pH, $h = 10^{-p}$ in molar units, $K_1 = 10^{-pK_1}$ and $K_2 = 10^{-pK_2}$. The carbonate fractions are

$$(\alpha_0, \alpha_1, \alpha_2) = \frac{(h^2, K_1 h, K_1 K_2)}{h^2 + K_1 h + K_1 K_2}, \quad f(p) = \alpha_1 + 2\alpha_2, \quad g(p) = \frac{1}{1 + 10^{pK_b - p}}.$$

In concentration units mM let $w(p) = 10^3(10^{p-pK_w} - 10^{-p}) = [\text{OH}] - [\text{H}]$. For either compartment, with fixed finite buffer amount n_b , the closure is the single equation

$$G(p; n_T, n_A, V) = n_T f(p) + n_b g(p) + V w(p) - n_A = 0. \quad (5)$$

Equivalently, for the cell, $x_A = x_T f(p) + \beta_i g(p) + v w(p)/C_0$; the lumen uses y_T, y_A, ℓ and $\beta_l = n_{b,l}/(C_0 L_0)$. Thus dilution of the finite buffer is retained exactly.

Proposition: unique speciation on the stated domain. Suppose $V > 0$, $n_T \geq 0$, $n_b \geq 0$, and all constants are finite and strictly positive in their concentration representation. For each real n_A , equation (5) has exactly one real pH. The solution depends smoothly on the retained amounts and volume on their open physical domain. For the production implementation one additionally requires $n_T > 0$ and

$$G(p_{\min}) \leq 0 \leq G(p_{\max});$$

smoothness within the executable domain is asserted with strict bracket inequalities. This numerical bracket is distinct from a physiological pH acceptance interval.

Proof. Let Z take the values 0, 1, 2 with probabilities $\alpha_0, \alpha_1, \alpha_2$. Differentiation gives

$$f'(p) = \log(10) \text{Var}(Z), \quad g'(p) = \log(10)g(1 - g), \quad w'(p) = \log(10)([\text{H}] + [\text{OH}]).$$

Consequently

$$G_p = \log(10)\{n_T \text{Var}(Z) + n_b g(1 - g) + V([\text{H}] + [\text{OH}])\} > 0. \quad (6)$$

The water term sends G to $-\infty$ as $p \rightarrow -\infty$ and to $+\infty$ as $p \rightarrow +\infty$. Continuity, strict monotonicity and the implicit function theorem prove the claims. The statement for the finite numerical bracket follows by monotonicity. $\square$

At fixed buffer amount the partial derivatives and trajectory derivative are

$$\frac{\partial p}{\partial n_A} = \frac{1}{G_p}, \quad \frac{\partial p}{\partial n_T} = -\frac{f}{G_p}, \quad \frac{\partial p}{\partial V} = -\frac{w}{G_p}, \quad \dot{p} = \frac{\dot{n}_A - f\dot{n}_T - w\dot{V}}{G_p}. \quad (7)$$

If the buffer amount is a parameter, $\partial p / \partial n_b = -g / G_p$. The very small explicit volume derivative at fixed amounts does not permit constant volume: volume changes all transported concentrations, osmolarity, electrical reversals and therefore $\dot{n}_T$ and $\dot{n}_A$. Nor does small free hydrogen concentration justify dropping carbon or buffering.

10 The coupled electrical closure

Use the dimensionless conductances and voltages defined above. Set $g_a = g_{K,a} + g_{C,a}$, $g_p = \sum_s g_{p,s}$, $c_a = -g_{K,a}e_{K,il} - g_{C,a}e_{Cl,il} + \hat{P}_a$ and $c_p = -\sum_s g_{p,s}e_{s,le}$. The apical equation gives

$$\psi_a(\psi_b) = \frac{g_p\psi_b - c_a + c_p}{g_a + g_p}.$$

The remaining current equation, including the electrogenic NBC, is

$$R(\psi_b) = g_{K,b}(\psi_b - e_{K,ie}) + \hat{P}_b + \hat{B}(\psi_b) + g_p(\psi_b - \psi_a(\psi_b)) + c_p = 0. \quad (8)$$

Proposition: unique electrical root. At any chemical state with positive concentrations and finite logarithms, suppose all conductances and NBC capacity are nonnegative, $g_a + g_p > 0$, and $s_0 = g_{K,b} + g_p g_a / (g_a + g_p) > 0$. If $w_B > 0$ and $u \in [0, 1]$, the mathematical current closure has a unique root on the real voltage line, smoothly dependent on the chemical state and parameters. The production routine further requires its physical voltage bracket $[-0.5, 0.5]$ V to contain that root.

Proof. Writing $L_B = \log(c_{Na,e}b_e^2 / (c_{Na,i}b_i^2))$,

$$R'(\psi_b) = s_0 + \frac{\gamma_B u}{w_B} \operatorname{sech}^2((L_B + \psi_b)/w_B) > 0.$$

The NBC term is bounded whereas the passive contribution grows linearly with positive slope s_0 . Thus $R(\psi_b) \rightarrow \pm\infty$ at the two ends of the real line. Monotonicity and the implicit function theorem complete the argument. For any retained variable z , $\partial_z \psi_b = -R_z / R_{\psi_b}$; differentiating the expression for ψ_a gives its corresponding derivative. $\square$

On a domain with positive amounts and volumes, strict pH and voltage brackets and $V_l > V_{\text{dead}}$, these closures make the constant input chemical vector field smooth. At the outflow threshold it is locally Lipschitz but not differentiable. The regulatory law on $0 \leq r \leq 1$ has the smooth analytical extension $\dot{r} = (\beta - r)/30$ and capacity multiplier $1 + 0.25r$; its derivative at $r = 1$ means the derivative from the physical side, or equivalently that extension. The executable trial clamp outside $[0, 1]$ must not be differentiated by a centred difference across the boundary. Doing so gives a spurious regulator eigenvalue $-1/60$ instead of $-1/30 \text{ s}^{-1}$. These regularity statements prove local existence and uniqueness of trajectories in the stated domain. They do not prove global positivity, global equilibrium uniqueness or global stability.

11 Alkalinity balance and conditional capacity obstruction

Proposition: parent alkalinity budget. For the parent equal routing class C0, the intracellular balance is

$$\dot{n}_{A,i} = H + 2B - E - 2J_4. \quad (9)$$

Hence an intracellular stationary state must satisfy $2J_4 = H + 2B - E$. For $J_4 \neq 0$ define

$$\mathcal{A} = \frac{H + 2B - E}{2J_4} = 1 + \frac{\dot{n}_{A,i}}{2J_4}, \quad \mathcal{A}_0 = \frac{H - E}{2J_4}.$$

Here $\mathcal{A}_0$ removes the NBC contribution at the same state; it is not the result of resolving the NBC null system. For $J_4 = 0$ these ratios are undefined, and the relevant statement is the unnormalised balance $\dot{n}_{A,i} = H + 2B - E$.

Proof and finite time form. NHE1 proton extrusion adds one alkalinity equivalent, AE2 bicarbonate export removes one, NBC imports two, and the parent AE4 cycle removes two. Neutral carbon dioxide exchange, water transfer and the included potassium, sodium and chloride pathways have no direct alkalinity source. Integration yields, for every finite valid interval $[0, T]$,

$$2 \int_0^T J_4 dt = \int_0^T (H + 2B - E) dt + n_{A,i}(0) - n_{A,i}(T). \quad (10)$$

Changing volume is already retained because these are amount balances. Equation (7) then gives the pH change without identifying alkalinity change with pH change. $\square$

Corollary: a conditional obstruction without NBC. Suppose a declared admissible domain gives $H \leq H_{\max}$ and $E \geq -E_{\max}$. At an equilibrium with $B = 0$, $J_4 \leq (H_{\max} + E_{\max})/2$. More generally, maintaining a positive mean demand J_d for time T requires

$$2J_d \leq H_{\max} + E_{\max} + \frac{n_{A,i}(0) - n_{A,i}(T)}{T}$$

when NBC is absent. If $n_{A,i}(T) \geq A_{\min}$, a demand greater than the capacity term can persist for at most $[n_{A,i}(0) - A_{\min}]/[2J_d - H_{\max} - E_{\max}]$. This is conditional on the stated domain and demand, not a universal impossibility of secretion without NBC.

For the frozen fixed bath and NHE1 amount one can obtain a rigorous but conservative bound from the inherited pH condition $p_i \geq 6.6$. In the dimensional Cha notation a, d are fixed bath quantities, $b \leq b_0(h) = 1000k_{2+}h/(h + K_{H,i})$ and $c \geq 0$. Both $M_H(h)$ and $ab_0/(a + b_0 + d)$ increase with h . Therefore

$$H \leq n_H m_H M_H(h_{\max}) \frac{ab_0(h_{\max})}{a + b_0(h_{\max}) + d}, \quad h_{\max} = 10^3 10^{-6.6} \text{ mM}.$$

With $m_H \leq 2.3$ this yields $H_{\max} = 0.08202591 \text{ fmol/s}$; the tanh AE2 law gives $E_{\max} = 0.005 \text{ fmol/s}$. Consequently the corresponding NBC absent bound is $J_4 \leq 0.04351295 \text{ fmol/s}$. A *specified* demand of 0.075 fmol/s requires $B \geq 0.03148705 \text{ fmol/s}$ at equilibrium. The chosen demand is an illustrative model loading level, not a measured AE4 requirement. Neither this bound nor the source based kinetic law identifies the effective salivary carrier amounts.

12 Equal routing and general source projections

Proposition: exact chloride identity. For the full system with a constant AE4 source vector define

$$\kappa(\nu) = \nu_{\text{Cl}} - 2\nu_{\text{Na}} + \nu_A.$$

At every valid state,

$$C_a = 6P - H + \kappa(\nu)J_4 + 2\dot{n}_{\text{Na},i} - \dot{n}_{A,i} - \dot{n}_{\text{Cl},i}. \quad (11)$$

For the parent equal routing vector $\kappa = 0$, yielding precisely

$$C_a = 6P - H + 2\dot{n}_{\text{Na},i} - \dot{n}_{A,i} - \dot{n}_{\text{Cl},i}, \quad C_a^* = 6P^* - H^*.$$

Proof. From the three intracellular balances,

$$\begin{aligned} 2\dot{n}_{\text{Na},i} - \dot{n}_{A,i} - \dot{n}_{\text{Cl},i} &= 2(N + H + B + \nu_{\text{Na}}J_4 - 3P) \\ &\quad - (H - E + 2B + \nu_AJ_4) - (2N + E + \nu_{\text{Cl}}J_4 - C_a) \\ &= H - 6P - \kappa(\nu)J_4 + C_a. \end{aligned}$$

Rearrangement gives the claim. Neither carbon, volume nor pH was frozen. $\square$

Only the explicit AE4 term disappears from this stationary balance. AE4 still changes the chemical state, pump, NHE1, NKCC1, pH, volumes, membrane voltages and hence secretion. The proposition does not prove zero AE4 sensitivity, a small null phenotype for every admissible parameterisation, or observational equivalence of mechanisms. Away from equilibrium the three storage terms cannot be omitted. Integrating (11) gives the analogous finite time chloride budget by replacing each derivative integral with its endpoint amount difference.

The five relevant linear projections of a source vector are

$$\Pi_{\text{Cl}}\nu = \nu_{\text{Cl}}, \quad \Pi_T\nu = \nu_T, \quad \Pi_A\nu = \nu_A, \quad q\nu = \nu_{\text{Na}} + \nu_{\text{K}} - \nu_{\text{Cl}} - \nu_A, \quad \kappa\nu = -2\nu_{\text{Na}} + \nu_{\text{Cl}} + \nu_A.$$

The charge source is $(q\nu)J_4$. Thus a class with nonzero $q\nu$ would require its additional transported current in the electrical closure; merely changing amount sources would not preserve the neutral manifold. All seven frozen classes have $q\nu = 0$, so their explicit AE4 current is zero. On that neutral subspace, $\kappa\nu = \nu_{\text{K}} - \nu_{\text{Na}}$: the cancellation tests equal sodium and potassium source coefficients.

Class	ν_{Na}	ν_{K}	ν_{Cl}	ν_T	ν_A	κ
C0	-1/2	-1/2	1	-2	-2	0
C1	1	-1	1	-1	-1	-2
C2	1	-2	1	-1	-2	-3
C3a	0	-1	1	-2	-2	-1
C3b	-1	0	1	-2	-2	1
C4a	0	-1	1	-1	-2	-1
C4b	-1	0	1	-1	-2	1

Each individual balance annihilates the kernel of its displayed linear functional. For example a change in ν_T alone is invisible to $\Pi_{\text{Cl}}, \Pi_A, q, \kappa$ but not to carbon balance. This is exactly why C3a/C4a and C3b/C4b have the same listed stationary chloride coefficient but different carbon signatures. Jointly the five projections have rank five: if chloride, carbon and alkalinity projections vanish, $q = 0$ gives $\nu_{\text{K}} = -\nu_{\text{Na}}$, and $\kappa = 0$ then gives $\nu_{\text{Na}} = \nu_{\text{K}} = 0$. No nonzero direction is lost by all five together. Equality in one projected balance is not equality of molecular mechanisms or of the whole dynamical response.

12.1 Independent verification and scaling implications

The analysis script `verify_exact.py` independently assembles carbonate chemistry, all selected kinetic laws, the nonlinear electrical solve, finite lumen transport, changing volumes and the thirteen amount, volume and regulatory derivatives. It compares these with production at 84

random valid states spanning all seven source classes and varied AE4 expression. It also verifies the pH derivative formula at two perturbation sizes, the positive electrical derivative, the exact charge and chloride identities, and the correct inward regulatory derivative at $r = 1$. Rational source coefficients are retained in `output/exact_verification.json`. These implementation checks support the algebra but do not turn random samples into a global theorem.

Numerically $\varepsilon = 0.072515$ and $\theta_4 = 0.031977$; $\gamma_B = 0.515137$, $\gamma_2 = 0.022260$ and $\gamma_4 = 0.668363$. The water coefficients are $\lambda_a = 404.37$, $\lambda_b = 4820.57$ and $\lambda_t = 24.34$; $\omega = 187.64$. These suggest some rapidly adjusting variables but are insufficient to justify freezing either volume: their driving osmotic differences and coupling to amounts matter. The verified dimensional recovery retains them all. The small dissociation concentrations, luminal buffer amount and inherited carrier relaxation ratio are recorded separately in the machine readable groups. A small buffer amount is a numerical convention here, not a measured physiological asymptotic limit. There is no membrane capacitance or differential voltage equation in the frozen system, so no electrical relaxation parameter can be inferred from its conductance groups. The algebraic voltage law is an inherited modelling assumption, not a new singular perturbation theorem proved by this analysis.

13 Numerical scales and their interpretation

Quantity	Value	Units
C_0	145	mM
V_0	1.45333439	pL
L_0	0.10538841	pL
J_0	0.22461808	fmol/s
t_0	938.18578	s
ϵ_L	0.072515	1
θ_4	0.031977	1
G_B/J_0	0.515137	1
G_2/J_0	0.022260	1
$L_a C_0^2/J_0$	404.366	1
$L_b C_0^2/J_0$	4820.571	1
$L_t C_0^2/J_0$	24.337	1

Here the table's ϵ_L is the ε in the equations. The induced cell and lumen amount scales are 210.733487 and 15.2813191 fmol; the induced water flux scale is $J_0/C_0 = 0.00154909$ pL/s. For dimensional recovery multiply cell amounts by $C_0 V_0$, lumen amounts by $C_0 L_0$, the two volumes by their distinct scales, time by t_0 , ionic fluxes by J_0 and water fluxes by J_0/C_0 . pH is already dimensionless. The thermal voltage is 26.726659 mV.

Group	Meaning	Value
$n_{b,i}/(C_0V_0)$	Cell buffer amount	0.14236
$n_{b,l}/(C_0L_0)$	Lumen buffer amount	6.54394e-11
$n_{\text{fixed}}/(C_0V_0)$	Fixed cell charge	0.370429
$n_{\text{other}}/(C_0V_0)$	Other cell osmoles	0.59174
K_1/C_0	First carbonate dissociation	5.47813e-06
K_2/C_0	Second carbonate dissociation	3.45646e-10
K_w/C_0^2	Water dissociation	4.75624e-13
$K_{b,i}/C_0$	Cell buffer dissociation	6.89655e-07
$K_{b,l}/C_0$	Lumen buffer dissociation	6.89655e-07
$P_{\text{CO}_2,b}C_0/J_0$	Basolateral carbon dioxide exchange	12.9108
$P_{\text{CO}_2,a}C_0/J_0$	Apical carbon dioxide exchange	6.4554
$J_4(0^+)/J_0$	Initial AE4 to NKCC cycle ratio	0.0219659
$J_4(0^+)/(2J_0)$	Initial chloride loading ratio	0.0109829
t_{carrier}/t_0	Inherited carrier relaxation ratio	0.000139211

The carbonate constants in this table use mM before division by C_0 ; K_w uses mM squared. These units differ from the molar representation used to define pH in the proof, with the conversion shown there. The small initial AE4 loading ratio is evaluated at the inherited WT state under immediate full stimulation. It is not a sustained loading fraction. The reference carrier capacity ratio is $\gamma_4 = 0.668363$, the pump capacity ratio is 0.356160 and the outflow group is 187.637155. The Palk concentration groups are $\eta_2 = 56.402853$ and $\eta_4 = 594.147609$. Dimensionless maximal channel conductances are 38.722939 for CaCC and 17.265005 for potassium; the paracellular sodium, potassium, chloride and bicarbonate values are respectively 0.369964, 0.061661, 0.308304 and 0.061661. Stimulus gates multiply the channel capacities as specified in the constitutive laws. Small dissociation concentrations do not imply a dispensable alkalinity coordinate. The cell buffer amount is appreciable relative to the amount scale. The small lumen buffer is an inherited numerical convention, whereas finite lumen carbon remains dynamical. Large water conductances allow small osmotic imbalances to move appreciable water. No uniform error bound or normally attracting limiting manifold has been established that would permit freezing either volume or lumen composition. The present results therefore make no singular perturbation reduction.

14 Verification, equilibrium framework and compensation

14.1 Production identity and dimensional recovery

The analysis loads the actual frozen Task 40 objects and WT conserved initial state. It checks all six stimulated parameter-object hashes and the resting state hash against the frozen manifest, rather than rebuilding from dataclass defaults. The source audit compares 150 entries with the Task 40 parent manifest: 149 match; the sole difference is `AGENTS.md`, an instructional metadata change. All listed production Python sources match their recorded Git blobs. The recovery audit also found no differences in the protected scientific tree relative to the main head it inspected. The scientific base is `c4d4f207f702d8eee09ab94d2d90ae90552dd641`; the recovered Task 44 checkpoint is `b7b775a0277d93263c4de339745806e5658ec9c2`. The complete audit and object hashes are retained in `output/source_audit.json`. For an unambiguous initial-state anchor, the frozen thirteen-state SHA256 is

`a0b96ef16e5967de4dfe2bdabb882f2a28d478e211f920bc19cd735a5a143acb`

Dimensional recovery uses $n_{s,i} = C_0V_0x_s$, $n_{s,l} = C_0L_0y_s$, $V_i = V_0v$, $V_l = L_0\ell$ and $t = t_0\tau$. If $\hat{I}$ is the integrated dimensionless water observable scaled by V_0 , then $I = V_0\hat{I}$ and $Q = (J_0/C_0)\hat{Q}$. Separate dimensional and dimensionless integrations through 600 s differ by at most 7.524×10^{-10}

in a state component divided by its declared scale, over the saved comparison times. A separate check recovers cumulative water to 1.443×10^{-15} pL. The reproduced original sampled WT output is 0.992524545 pL, compared with frozen 0.992524544 pL; its adaptive integral is 0.992544443 pL. These two observables have different numerical definitions. The WT, 5% expression and null reference checks reproduce all 34 diagnostic columns at eleven frozen display times, with a largest cumulative-output discrepancy of 6.74×10^{-10} pL. These are numerical equivalence checks on the retained equations, not a reduced-model fit.

14.2 Executable domain and inherited acceptance gates

The carbonate executable bracket is $[3, 11]$ for pH in both compartments. For the smooth local analysis, require $G(3) < 0 < G(11)$ for the amount residual in (5), positive transported concentrations and positive volumes. The NBC electrical solver brackets the basolateral voltage on $[-0.5, 0.5]$ V; strict bracketing of its residual defines the interior domain used for differentiation. The apical voltage is then recovered from the coupled current equation, without imposing a separate apical bracket. The numerical solvers can accept an exact endpoint root; that fact does not extend the stated smooth interior domain. Require $V_i > V_{\text{dead}}$ to remain on the differentiable, active-outflow branch. The physical regulatory range is $0 \leq r \leq 1$; its boundary derivative uses the inward rule or smooth extension already specified above.

The inherited physiological checks are more restrictive than numerical speciation. They require all twelve amount/volume components to be strictly positive and all states, derivatives and diagnostics to be finite, together with the following intracellular bounds:

Quantity	Accepted domain
Sodium	$[\text{Na}]_i < 40$ mM
Potassium	$50 \leq [\text{K}]_i \leq 200$ mM
Chloride	$30 \leq [\text{Cl}]_i \leq 80$ mM
pH	$6.6 \leq \text{pH}_i \leq 7.3$
Bicarbonate	$0 < [\text{HCO}_3]_i < 100$ mM
Total inorganic carbon	$[T]_i > 0$
Cell volume	$0 < V_i < 3$ pL

The same diagnostics check charge, carbon and buffer accounting to 10^{-10} fmol/s, water accounting to 10^{-12} pL/s, each membrane current residual to 10^{-20} A, and compartmental speciation alkalinity to 10^{-9} mM. The declared diagnostic-key set must match, and NBC's signed charge and carbon sources must equal $-B$ and $2B$ exactly in the diagnostic representation. These last two sources are not zero residuals. No additional physiological luminal pH interval has been introduced.

The supplemental full-gate verification independently re-solves 128 parameter-perturbed WT/null roots and 256 nearby expression roots used for compensation validation. All 384 pass these inherited checks; the largest conservation residual divided by its own tolerance is 4.086×10^{-5} . The largest dimensionless independent equilibrium residual is 6.880×10^{-12} . These results are retained in `output/sensitivity_full_gate_verification.json`; gate acceptance does not remove the separate thermodynamic qualifications.

14.3 An eleven-coordinate implicit-function problem

Let

$$z = (x_{\text{Na}}, x_{\text{K}}, x_{\text{Cl}}, x_{\text{T}}, v, y_{\text{Na}}, y_{\text{K}}, y_{\text{Cl}}, y_{\text{T}}, \ell, r) \in \mathbb{R}^{11}, \quad \frac{dz}{d\tau} = F(z, \mu).$$

The two alkalinity coordinates are reconstructed by (3); both carbon amounts, both volumes and the regulatory state remain dynamic. The parameter μ denotes expression or one specified active

model setting, with the scaling fixed at the reference values. Carbonate and voltage closures are evaluated inside F . Constant full stimulation defines the equilibrium problem $F(z^*, \mu) = 0$. The numerical solver finds ten positive chemical/volume coordinates in logarithmic variables and sets the *equilibrium* condition $r^* = 1$. All eleven equations are subsequently checked, and all eleven coordinates are retained in the dynamic Jacobian.

Proposition: local equilibrium sensitivity. Suppose the stated interior closure conditions hold, the vector field and observable $Q(z, \mu)$ are continuously differentiable locally, and $F_z(z^*, \mu)$ is nonsingular. Then a unique differentiable equilibrium branch exists in a neighbourhood of this root and parameter. Its derivatives are

$$\frac{dz^*}{d\mu} = -F_z^{-1} F_\mu, \quad (12)$$

$$\frac{dQ^*}{d\mu} = Q_\mu - Q_z F_z^{-1} F_\mu. \quad (13)$$

At a physical boundary the claim uses the smooth extension and is restricted to inward admissible perturbations. The result follows by differentiating $F(z^*(\mu), \mu) = 0$ and applying the implicit function theorem. It establishes local uniqueness of the branch, not global uniqueness or global existence. The dimensional dynamic Jacobian is $J = F_z/t_0$; this constant time scaling does not change (12). $\square$

The terms in (13) separate explicit parameter dependence from the response mediated by the complete state. For AE4 expression the outflow law has $Q_{e_4} = 0$. This mathematical zero does not mean that AE4 has no biological effect on secretion. Likewise, Q_z has only a lumen-volume component on the active-outflow branch, but every retained network process can influence that volume.

The expression derivatives were checked against independently solved nearby equilibria at steps 10^{-4} and 5×10^{-5} , centred at $e_4 = 1, 0.5, 0.05$ and second-order inward at $e_4 = 0$. Validation covers all eleven coordinates and $Q, N, J_4, P, H, \text{pH}_i$. Across these checks, the largest observable component relative discrepancy is 1.034×10^{-7} ; the largest state-derivative relative discrepancy is 8.070×10^{-8} . The maximum nearby-root residual is 5.683×10^{-12} in the dimensionless independent equations. This validates local derivatives at the sampled points, without assigning those difference steps the meaning of uncertainty intervals.

14.4 Three distinct measures of NKCC compensation

Because N is an NKCC1 *cycle* flux carrying two chlorides, the local stationary chloride compensation gain is

$$C_N(e_4) = -\frac{2 dN^*/de_4}{dJ_4^*/de_4}, \quad dJ_4^*/de_4 \neq 0. \quad (14)$$

Here the parent J_4 carries one chloride per cycle. At a zero denominator the ratio is undefined and the two unnormalised derivatives must be given separately. The null entry below is an inward derivative and has a nonzero denominator even though $J_4^* = 0$ there.

e_4	dN^*/de_4	dJ_4^*/de_4	C_N
1	-0.0117368	0.0252674	0.929011
0.5	-0.0283978	0.0620609	0.915159
0.05	-0.0719960	0.160300	0.898266
0	-0.0795571	0.177417	0.896838

Flux derivatives in the table are in fmol/s per unit expression. Equation (14) describes incremental replacement at equilibrium; it does not quantify the complete knockout.

For the original finite-time experiment let $\mathcal{I} = [60, 600]$ s and superscripts W and 0 denote WT and AE4 null. The relative integrated NKCC increase and the integrated chloride replacement fraction are instead

$$R_N = \frac{\int_{\mathcal{I}} 2(N^0 - N^W) dt}{\int_{\mathcal{I}} 2N^W dt} = 0.2316345,$$

$$C_{N,\mathcal{I}} = \frac{\int_{\mathcal{I}} 2(N^0 - N^W) dt}{\int_{\mathcal{I}} (J_4^W - J_4^0) dt} = 0.8850704.$$

These signed integrals use the frozen one-second flux records, with nonzero denominators. NKCC supplies an additional 36.33701 fmol of chloride against 41.05550 fmol of missing AE4 chloride. Thus 23.16% is an increase relative to WT NKCC loading, whereas 88.51% is replacement of lost AE4 chloride over that specified window. Neither is the local gain C_N . The window also differs from the $[0, 600]$ s cumulative secretion comparison. The constant-stimulus stationary null secretion deficit is 0.94617%, compared with the frozen finite-time 3.8575%; their difference retains transient storage and state evolution.

14.5 Counting local cases and interpreting slow coordinates

The 26 equilibrium entries are class/expression *cases*: eight C0 expression values and three values for each of the other six source classes. At zero expression the AE4 source vanishes, so all seven null entries solve the same system. The count must not be read as 26 distinct equilibria at one parameter point. Twenty two entries pass the inherited physiological gates. Across all cases the smallest singular value of the dimensional Jacobian is at least $9.202 \times 10^{-4} \text{ s}^{-1}$ and its condition number ranges from 1.882×10^4 to 2.931×10^4 in the declared scaled coordinates. Nonsingularity supports the local framework; the conditioning and numerical step checks qualify its implementation.

The slow eigenvectors involve volume and ionic storage together. For the slowest WT mode (345.82 s), intracellular potassium, cell volume and intracellular chloride account for 28.82%, 20.65% and 18.35% of the sum of absolute components of the scaled right eigenvector. For the slowest null mode (459.92 s), the corresponding shares are 31.04%, 24.68% and 16.55%. These are coordinate-dependent descriptive shares, not fractions of physiological energy, transported chloride or causal control. They supplement the secretion input/output residues and nonnormality diagnostics reported below. In particular, they give no justification for removing either volume or declaring the full chemical system instantaneous.

15 NBC removal: capacity obstruction versus secretion

At the parent WT root, NBC supplies $J_B/J_4 = 0.954815$ of the AE4 alkalinity demand. The conditional no NBC capacity bound covers 57.9613% of that particular demand. These are state and capacity comparisons; the total support ratio at equilibrium is identically one. Resolving the full system as NBC capacity decreases gives:

NBC capacity fraction	J_4^* (fmol/s)	J_B^* (fmol/s)	pH_i^*
1.00	0.075072	0.071680	7.0504
0.75	0.062485	0.058576	7.0331
0.50	0.047042	0.042486	7.0135
0.25	0.028351	0.022995	6.9918
0.00	0.006327	0.000000	6.9683

At zero NBC, a root satisfying the inherited physiological gates has $Q = 0.00153598288$ pL/s and $\text{pH}_i = 6.96833154$. The reference WT value is 0.00158406711 pL/s, so secretion falls by approximately 3.035%. AE4 loading falls from 0.07507245 to 0.00632651 fmol/s. The conditional obstruction is respected because the network selects a much lower AE4 demand. Additional alkalinity support is needed to sustain the larger stated loading level under the capacity assumptions; universal NBC necessity for all secretion is disproved for this frozen mathematical system by the valid counterexample root. Existence of this root does not establish a unique global attractor or identify the biological NBC isoform.

16 Independent equilibrium and spectral review

The full inherited physiological and conservation gates were applied to all 26 roots. Twenty two roots pass; the four excluded roots are retained in the table below. All 26 Jacobians are nonsingular and have negative real spectral parts on the 11 dimensional charge manifold. This establishes numerical local stability in the inward regulatory linearisation, not global uniqueness or global physiological validity.

Class	e_4	pH_i	$[\text{Na}]_i$	τ_{slow} (s)	Failed gate
C1	1	7.0139	45.694	564.4	na_i_mM
C2	1	6.5211	57.460	500.5	na_i_mM, ph_i
C4a	1	6.5214	31.108	455.7	ph_i
C4b	1	6.5792	15.581	313.6	ph_i

The first inherited checkpoint recorded 22 roots and four unsuccessful large continuation steps, without individual seeds, class labels or error strings. That aggregate failure record is preserved verbatim in the review JSON; missing details have not been reconstructed. The completed analysis follows C0 through eight expression levels. Other classes use the converged C0 WT as the WT seed and matching C0 roots for 5 percent expression and the null. This documented retry strategy obtains all 26 roots, including the four outside the physiological gates.

Eigenvalues were matched individually between relative coordinate difference steps 10^{-5} and 5×10^{-6} . The greatest matched relative eigenvalue discrepancy is $5.43\text{e-}08$. At full regulatory activation the production evaluator clips values beyond one. A centred derivative across this boundary halves the regulatory Jacobian column. The corrected second order inward derivative gives the regulatory eigenvalue $-1/30 \text{ s}^{-1}$. The much slower chemical modes remain, with decay times 345.82 s in WT and 459.92 s in the null.

16.1 Thermodynamics at the local roots

Class	e_4	$A_4/(RT)$	Signed flux compatibility
C0	1	-0.9589	opposed
C1	1	3.8925	supported
C2	1	3.8589	supported
C3a	1	1.5135	supported
C3b	1	-3.7440	opposed
C3b	0.05	-2.2240	opposed
C4a	1	0.7093	supported
C4b	1	-4.3271	opposed
C4b	0.05	-2.3400	opposed

Positive forward affinity supports the proposed positive chloride cycle direction. The product of signed cycle flux and affinity separately checks the actual implemented flux. These diagnostics

do not replace the inherited scalar cycle law. Every null has exactly zero AE4 flux, regardless of the hypothetical forward affinity. Thermodynamic opposition is retained and is not repaired by refitting.

16.2 Modal input and output projection

For $\dot{z} = Jz + B\delta e_4$ and $\delta Q = Cz$, let $J = V\Lambda V^{-1}$. The residue of mode k is $(CV)_k(V^{-1}B)_k$, and its stationary sensitivity contribution is this residue divided by $-\lambda_k$. The right vectors, inverse left rows and both projections are reported for every WT and null mode. The Euclidean norm uses the declared dimensionless coordinates and is not a physiological energy norm.

State	τ_{slow}	$\kappa(V)$	$\max_t \ e^{Jt}\ _2$	dQ^*/de_4
WT	345.82	22.02	1.37	1.28527e-06
Null	459.92	21.59	1.37	5.06256e-05

The propagator maxima are sampled numerical diagnostics over 0 to 10000 s, not proved suprema. State amplification, expression input projection and secretion output projection must be considered together. A small stationary secretion sensitivity can result from cancellation of modal contributions even when slow state modes are present. In WT, the 345.82 s and 223.48 s modes contribute -5.5754×10^{-5} and $+1.6013 \times 10^{-5}$ pL s⁻¹ per unit expression to the stationary secretion derivative. These contributions partially cancel faster positive contributions, leaving the much smaller net derivative $+1.2853 \times 10^{-6}$ pL s⁻¹. At the null, the 459.92 s mode contributes -3.5795×10^{-4} pL s⁻¹, against a net derivative of $+5.0626 \times 10^{-5}$ pL s⁻¹. The slow modes therefore do project onto expression driven secretion and substantially shape its transient approach; their effects are not absent merely because the final WT derivative is small. For example, at 600 s the WT linear step derivative is 1.0027×10^{-5} pL s⁻¹, still above its eventual value. The 30 s regulatory mode has zero expression input projection at both endpoints, because its target does not depend on expression. Its WT secretion output projection is nonzero, so a perturbation that actually excites regulation is a different experiment. There is therefore no basis for claiming that this model lacks a slow mode. The linear expression response at either endpoint is not a validated approximation to the complete WT to null change.

16.3 Independent reference and sensitivity verification

All 34 diagnostic columns at all 11 frozen display times were reproduced for WT, 5 percent expression and the null using the original Task 40 stimulus. The largest relative diagnostic discrepancy was 2.26e-07; the largest cumulative secretion discrepancy was 6.73e-10 pL. The original stimulus is unstimulated at the exact initial instant; the separate constant stimulus equilibrium experiment is fully stimulated there. This distinction matters for instantaneous transporter fluxes, even though initial outflow is state determined.

Implicit expression sensitivities for all independent coordinates and six observables were checked against separately solved nearby roots at 10^{-4} and 5×10^{-5} . Centred differences were used at expression 1, 0.5 and 0.05, and second order inward differences at the null. All component discrepancies and root residuals are provided in the review JSON. Chloride compensation retains the factor two for NKCC cycles.

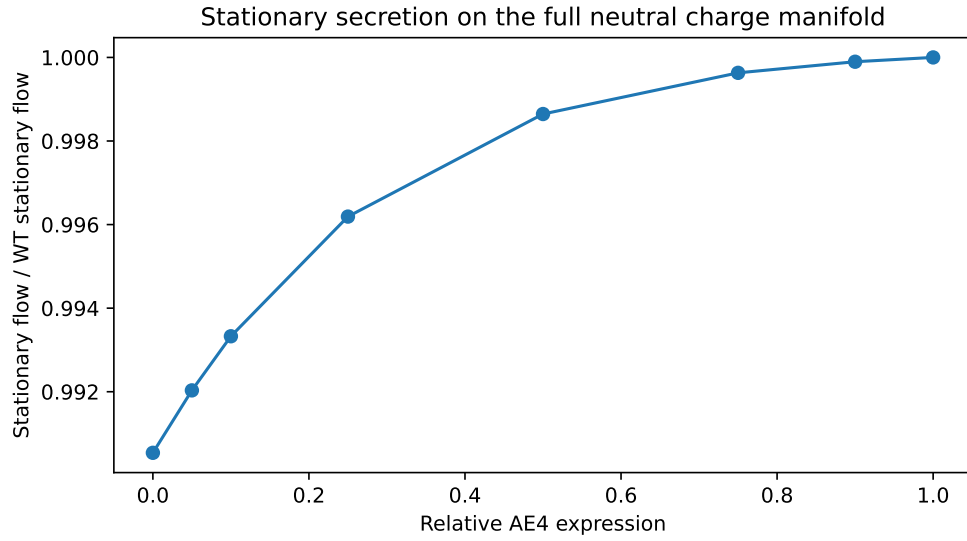


Figure 1: Constant stimulus C0 equilibrium secretion normalised by WT. The vertical axis deliberately resolves the small change around one; the complete null deficit is 0.94617%, not the 600 s cumulative deficit. Each point is a solved local root, not evidence of global uniqueness.

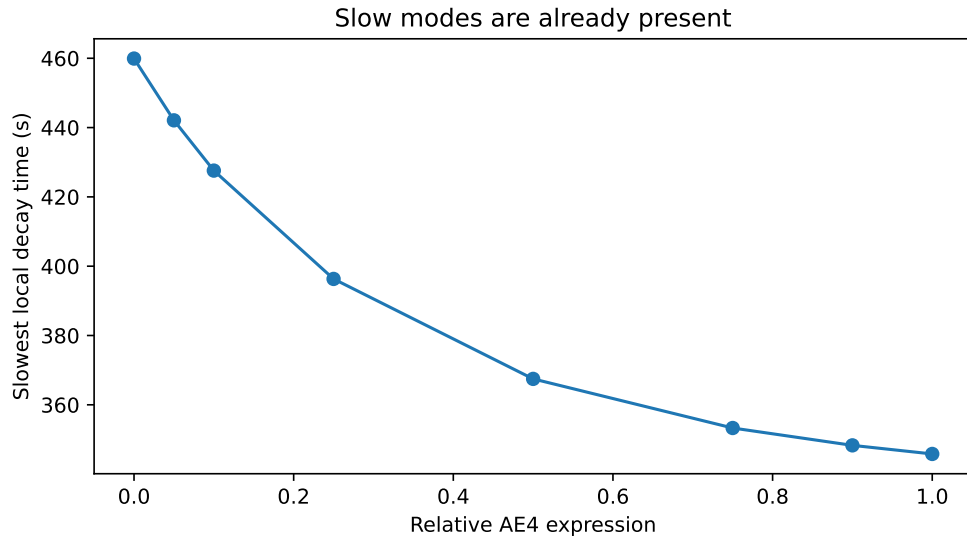


Figure 2: Dominant local decay times on the eleven dimensional neutral manifold during expression continuation. Carbon, alkalinity, both volumes, finite lumen composition and regulation are retained. These local timescales do not alone establish the shape of genotype divergence.

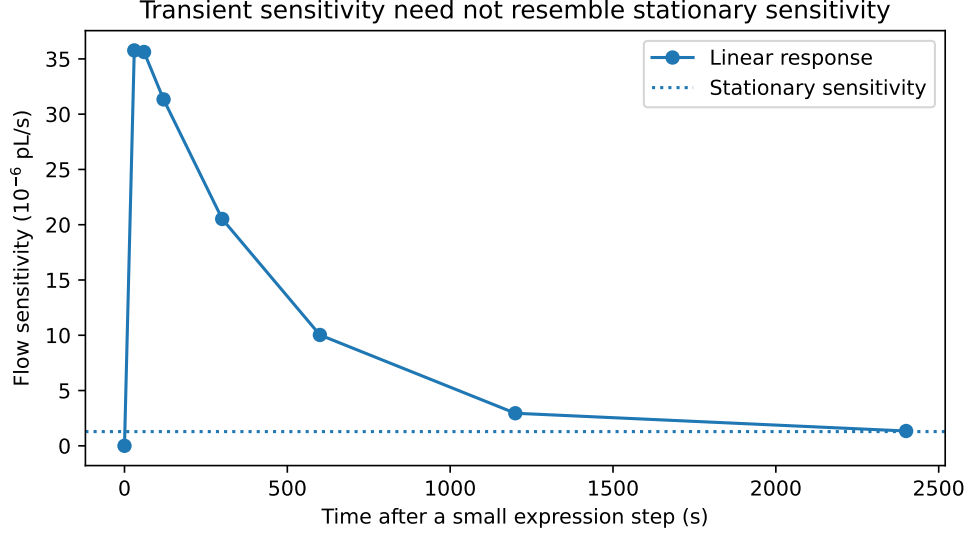


Figure 3: Local expression step response at the WT equilibrium. This linear response retains input and secretion output projection. It is a derivative experiment, not a replacement for a finite knockout trajectory from the inherited resting state.

17 All frozen source classes

The exact projections were derived above without class specific refitting. For completeness, the full expression root of every class is tabulated here. Sodium concentrations are in mM. Physiological acceptance and thermodynamic compatibility are separate tests; passing this table’s gates does not erase the opposed fluxes reported in the spectral review.

Class	pH _i [*]	[Na] _i [*]	t _{slow} (s)	Failed criteria
C0	7.0504	18.083	345.82	None
C1	7.0139	45.694	564.38	sodium
C2	6.5211	57.460	500.52	sodium, pH
C3a	7.0220	24.508	431.19	None
C3b	7.0734	14.206	314.61	None
C4a	6.5214	31.108	455.72	pH
C4b	6.5792	15.581	313.64	pH

In particular C0 is a physiological mathematical root with thermodynamic opposition under the substituted ensemble source convention. Edge kinetics inherited from the carrier are insufficient to guarantee consistency of the whole substituted molecular cycle. The source vector analysis identifies the conserved signatures that require measurement; it does not validate the scalar kinetics for each molecular hypothesis.

18 The prescribed Task 41 inverse family

The Task 41 construction is target selected. Its added hypothesis changes stimulated CaCC conductance before the full electrical and NBC closure:

$$g_{\text{CaCC}}(e, a; \rho) = g_{\text{parent}}\{1 - a\rho(1 - e)\}, \quad \rho = 1 - b, \quad 0 \leq \rho < 1. \quad (15)$$

Here e is AE4 expression and a is the protocol normalised recruitment fraction used by NBC:

$$a = \min\left\{1, \max\left\{0, \frac{[\text{Ca}]_i - 0.058 \mu\text{M}}{(0.25 - 0.058) \mu\text{M}}\right\}\right\}.$$

The parent CaCC Hill gate remains a distinct retained factor. The full conserved system, carbonate closure, electrical closure, both volumes and AE4 regulation remain active. WT nests the parent exactly for every state and every ρ . At full activation, the null retains fraction $b = 1 - \rho$ of its parent CaCC conductance.

Every 600 s trajectory uses the identical frozen WT initial conserved state and original stimulus protocol. No genotype resting solve, new mechanism or parameter fit is introduced. All inherited positivity, concentration, pH, volume, source and conservation gates are checked at accepted solver endpoints and at the original observation times. The original WT stimulation requirement is also retained; it is not imposed retrospectively on mutants.

For $X \in \{A, S\}$ define

$$D_X(\rho, p) = 1 - \frac{I_{0,X}(\rho, p)}{I_{1,X}(p)}. \quad (16)$$

$I_{e,A}$ is the adaptive time integral of outflow and $I_{e,S}$ is the original trapezoidal observable on integer seconds, including $t = 0$ and 10^{-6} s. These are distinct numerical observations. The prescribed comparison threshold is $D_X = 0.303$: it is one reported standard error below the experimental mean of 35%, whose reported standard error is 4.7 percentage points. It is not a confidence limit, a measured biological range or an admissibility gate.

18.1 Reproduction and numerical crossing

The selected Task 41 parameter is $b = 0.10511872843288446$, or $\rho = 0.8948812715671155$. The three selected trajectories reproduce the frozen sampled cumulative outputs to within 6.7e-10 pL.

Case	Sampled output (pL)	Adaptive output (pL)	Sampled deficit (%)
WT	0.992524545	0.992544443	0
5% AE4	0.762622459	0.762624862	23.163365
AE4 null	0.692175197	0.692173709	30.261151

In particular, the frozen selected null deficit is slightly below 30.3%. It must not be described as exactly attaining this new comparison threshold.

An ordered scan of 17 recruitment fractions, followed by bracketed numerical root refinement in the prescribed family, gives:

Observable	Crossing ρ	AE4 dependent recruitment (%)	Residual b
Adaptive	0.8950726882	89.5072688	0.1049273118
Original sampled	0.8950806285	89.5080628	0.1049193715

Both crossings pass every inherited 600 s gate. At the adaptive crossing, the original sampled deficit is 30.2984507%; at the sampled crossing, the adaptive deficit is 30.3015492%. Refinement at relative solver tolerances 10^{-8} and 2×10^{-9} , with respective maximum steps 2 s and 1 s, changes the two crossing fractions by $1.4\text{e-}13$ and $1.7\text{e-}11$, respectively. Independent integration of the 13 production states, followed by adaptive quadrature of dense flow, also reproduces the selected cases and crossing.

The deficit increases at the sampled points, but monotonicity between them and exclusion of all earlier unsampled crossings have not been proved. These are the first numerical crossings found in this prescribed family, not global lower bounds, universal recruitment requirements or independent validation of the coupling hypothesis.

18.2 The separate extension beyond 600 s

Continuing constant full stimulation to 3600 s gives the following null diagnostics. Each run passes all inherited gates through 600 s. Intracellular pH is the only later gate failure; all other

monitored gates continue to pass.

Case	First pH = 7.3 time (s)	pH at 3600 s	Limiting local root pH
Selected Task 41	1486.30957	7.3117946	7.3120147
Adaptive crossing	1484.74517	7.3118437	7.3120644

The limiting roots found from these late states are mathematical continuations outside the inherited pH domain. They do not establish globally unique equilibria or physiological long term validity. These separate diagnostics do not change any frozen 600 s Task 41 result.

18.3 Local uncertainty of the crossing

For a locally regular numerical crossing with $\partial_\rho D_X \neq 0$,

$$\frac{d\rho_X}{d \log p} = -\frac{\partial_{\log p} D_X}{\partial_\rho D_X}, \quad \frac{d \log b_X}{d \log p} = -\frac{1}{1 - \rho_X} \frac{d\rho_X}{d \log p}. \quad (17)$$

Eight effective or architecture derived quantities were selected using the Task 43 provenance classification. A perturbed WT denominator is used with the same perturbed parameter in the null. The conserved initial state is always held at the inherited WT value. Thus changing the buffer amount also changes the algebraic initial pH; it does not trigger a new resting solve.

The following elasticities refer to the adaptive crossing. The companion sampled results are retained in the machine readable crosswalk.

Parameter	$d \log \rho / d \log p$	$d \log b / d \log p$
NBC capacity	-0.0059550	+0.0507982
AE4 carrier amount	-0.0050569	+0.0431371
NKCC effective scale	-0.0021926	+0.0187041
NHE1 carrier amount	+0.0008018	-0.0068396
Pump capacity	-0.0355825	+0.3035333
CaCC conductance	+0.0983320	-0.8388120
Cell buffer amount	+0.0071557	-0.0610411
Apical water permeability	-0.0002975	+0.0025380

The derivative calculation is checked against independently solved nearby crossings at logarithmic perturbation sizes 10^{-3} and 5×10^{-4} . Every perturbed WT and null trajectory passes its inherited gates. The maximum relative discrepancy between implicit crossing derivatives and nearby root differences is 1.7e-06. The larger relative sensitivity of residual recruitment b is relevant when interpreting a conductance assay. None of these eight quantities has an established experimental uncertainty interval in the audited configuration. Accordingly, these calculations establish local sensitivity only; they do not prove robustness over a finite range or identify a finite parameter change that overturns the conclusion.

18.4 A discriminating experiment and scope

At the numerical comparison crossing the null must retain approximately 10.49% of stimulated CaCC conductance in this family. A direct test would compare stimulated CaCC conductance and channel recruitment in WT and AE4 loss cells while matching calcium activation, voltage, intracellular chloride and pH, so that a changed driving force is distinguished from a changed conductance. Surface channel abundance and channel open probability would help distinguish recruitment from gating. Concurrent measurements of pH, chloride, cell volume and secretion over the first ten minutes and after approximately 25 minutes would test the predicted coupled response and its later pH failure. A secretion measurement alone does not identify this specific dependency.

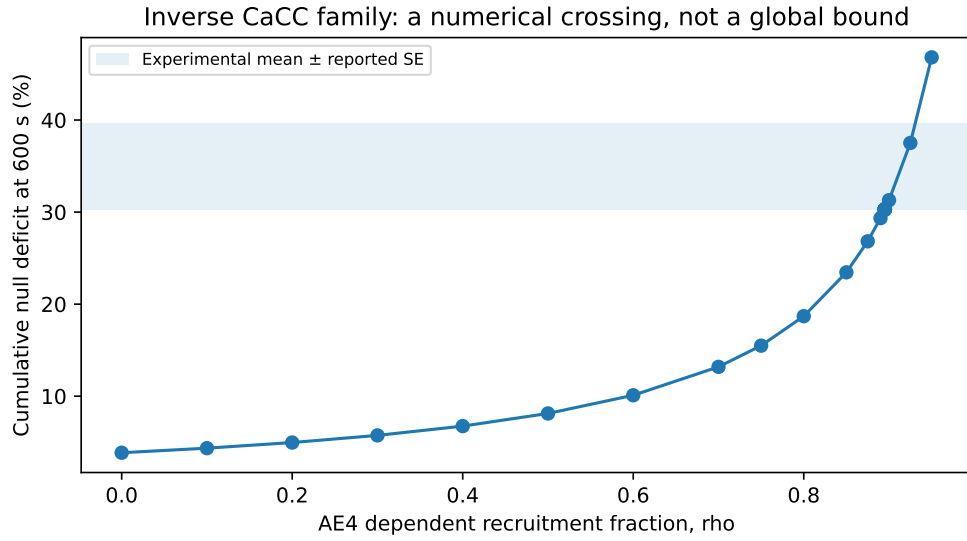


Figure 4: Ordered numerical scan within the one prescribed Task 41 family. The experimental band represents the reported mean plus or minus one standard error, not a confidence interval. The refined adaptive and sampled crossings are recorded separately. Neither a global lower bound nor global monotonicity follows from the plotted points.

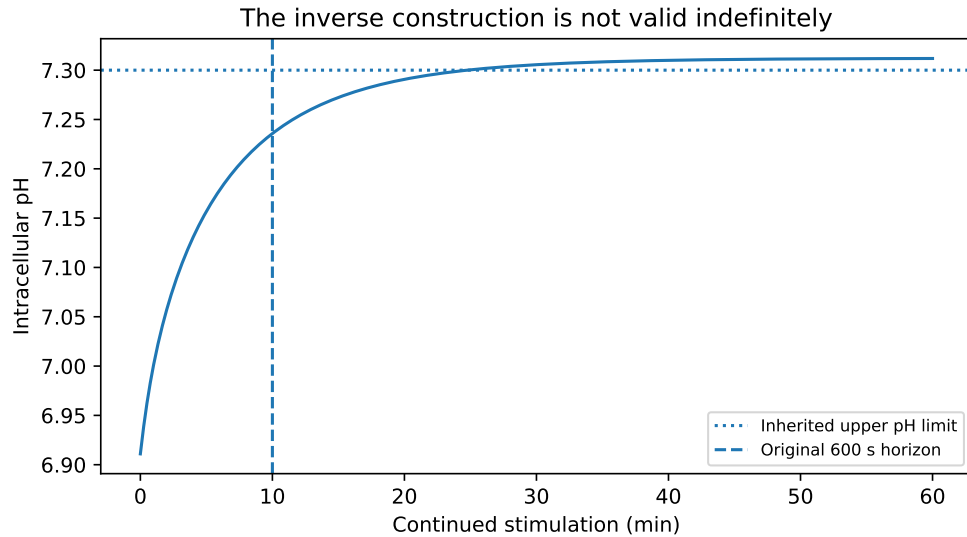


Figure 5: Separate extension of the adaptive crossing null beyond the declared 600 s Task 41 window. It first crosses the inherited upper pH gate near 1485 s; the selected case has a nearly identical late crossing. The later diagnostic does not revise the frozen 600 s observable; it limits claims of sustained physiological validity.

19 Dependence on uncertain parameters

19.1 Sensitivity crosswalk with Task 43

The sixteen selected parameters are uncertain effective settings or inherited calibrated capacities. Each is perturbed independently by $p \exp(\pm h)$, with $h = 10^{-3}$ and 5×10^{-4} ; both genotypes are solved afresh at each perturbed point. These steps are numerical checks and do not define uncertainty intervals. No defensible physiological interval is available from the provenance audit, so no global robustness claim is made.

Parameter	E_Q	$\partial \text{pH} / \partial \log p$	E_D	E_{C_N}	$E_{\tau_{WT}}$
NBC capacity	0.01992	0.06483	2.085	-0.06438	0.107
AE4 amount	0.0008114	-0.07246	0.08494	0.02339	-0.06397
NKCC scale	0.05536	-0.001558	-1.494	0.06404	0.1901
NHE amount	-0.003261	0.07084	-0.2293	0.007679	-0.01473
Pump capacity	0.5259	0.1206	4.634	-0.2734	-1.522
CaCC conductance	0.03247	-0.005524	-0.2212	0.01105	0.008274
Cell buffer amount	-0.001575	0.001396	-0.2063	0.009504	0.3086
Apical water permeability	0.002129	-8.317e-05	-0.003558	0.0002239	-3.74e-05
AE4 activation time	0	0	0	0	0
AE4 activation gain	0.0001623	-0.01449	0.01699	0.004678	-0.01279
Apical pump fraction	0.04956	-0.003571	-0.05983	0.003151	0.04517
NHE stimulation gain	-0.003261	0.07084	-0.2293	0.007678	-0.01473
Basolateral CO ₂ permeability	-0.004448	-0.08319	-0.4338	0.05062	-0.06818
AE2 capacity	-7.335e-05	0.000363	-0.03056	-0.03383	7.508e-05
NBC saturation width	-0.01435	-0.0467	-1.502	0.01777	-0.07263
NKCC stimulation gain	0.05536	-0.001558	-1.494	0.06404	0.1901

Here $E_Y = \partial \log Y / \partial \log p$ and $D = 1 - Q_{\text{null}} / Q_{\text{WT}}$ is the stationary deficit. Its small baseline value makes E_D potentially large; the machine readable table also supplies percentage point derivatives. pH is already logarithmic, so its unnormalised derivative is shown; the hydrogen concentration elasticity is $-\log(10) \partial \text{pH} / \partial \log p$. All derivatives are partial sensitivities at the inherited parameter point, without recalibration. Pump capacity has the largest local effect on WT secretion and on the stationary null deficit. The latter is also sensitive to NBC capacity and its assumed saturation width, and to effective NKCC capacity. These are priorities for experimental constraint. The calculations do not show that a finite parameter change overturns a conclusion, and cannot establish that the small null phenotype persists throughout an unknown uncertainty range. The exact balance identities survive these capacity changes within the specified architecture; their numerical consequences need not. The equilibrium total alkalinity support ratio is identically one and cannot demonstrate robustness. The saved results instead give the NBC share J_B / J_4 , the observed non NBC share $(J_H - J_2) / (2J_4)$, and a conditional capacity ratio $(H_{\text{max}} + J_{2,\text{max}}) / (2J_4)$. The bound assumes the fixed bath, positive sodium and $\text{pH}_i \geq 6.6$. It permits AE2 to supply alkalinity at its maximal reverse capacity. Its local sensitivity is useful, but it does not prove the available capacity over an unknown parameter interval. The AE4 activation time has no local effect on constant stimulus equilibria. Its eigenvalue is $-1/\tau_4$ and is separated from the slower chemical mode at this point; a zero local slow mode sensitivity does not establish that activation timing is irrelevant to finite time secretion. The buffer calculation holds the fixed charge and impermeant osmole pools fixed while changing only buffer amount. NKCC capacity and its stimulated multiplier produce the same stationary perturbation because only their product enters at constant full stimulation. Independent root perturbations validate the IFT derivatives at both steps, and expression perturbations validate the chloride compensation gain at both steps for every nearby parameter point. Fixed state parameter derivatives use the smaller steps 10^{-5} and 5×10^{-6} to resolve cancellation in the buffer response. Decay sensitivities additionally use fourth order differentiation of the chemical block at two steps, together with the exact regulatory eigenvalue; the block triangular structure retains all eleven dynamical eigenvalues. The supplied files retain

root residuals, physiological gate results, Jacobian step checks and all derivative comparisons. The separate inverse family calculation supplies the Task 41 crossing sensitivity.

The independent complete gate supplement resolves all 128 perturbed WT and null parameter roots and 256 additional expression roots used to check compensation. Every numerical solve succeeds and every inherited gate passes. Its largest conservation residual divided by its own tolerance is 4.086×10^{-5} ; the largest independent RHS residual is 6.880×10^{-12} . This supplements the original derivative checks without weakening a gate. Exact reproduction of this supplementary calculation from the clean numerical checkout is recorded separately.

20 Comparison with experimental evidence

The gland phenotype and the comparison threshold. In the 2015 mouse submandibular gland study, stimulation used $0.3 \mu\text{M}$ carbachol and $5 \mu\text{M}$ isoproterenol. AE4 deletion reduced total secretion over ten minutes by $35 \pm 4.7\%$ (mean $\pm$ standard error; six glands per genotype). Initial flow over two to three minutes was comparable, with the deficit developing later. AE2 deletion had no detectable flow effect against its own control strain. AE4 deficient acinar cells had lower resting chloride and slower chloride reuptake; initial chloride exit and resting pH were not detectably changed. These are distinct measurements, rather than interchangeable manifestations of a single steady flux Peña Münzenmayer et al. [2015a]. The prescribed $30.3\% = 35\% - 4.7\%$ inverse target is therefore a comparison value one reported standard error below the mean. It is not a confidence bound, an uncertainty interval for a model parameter, or a universal biological minimum.

The Task 40 cumulative null deficit of 3.8575% establishes a small effect for the frozen model and protocol. The equal routing identity does not establish that small magnitude for every admissible parameter choice. Nor does agreement at 600 s establish the observed time course. The present simulations initialise every genotype at the inherited WT resting state, whereas the experiment used established knockout animals. That difference should remain explicit when comparing the initial response or stored intracellular chloride.

Compensation and pH. The 2015 isolated NKCC1 assay used bicarbonate free solution, $30 \mu\text{M}$ ethoxzolamide and $50 \mu\text{M}$ T16Ainh-A01; bumetanide identified the NKCC1 dependent component. Initial chloride uptake showed no detectable genotype difference. Stimulation induced NHE dependent alkalinisation also showed no genotype difference Peña Münzenmayer et al. [2015a]. Neither finding measures the complete stimulated network integral. Consequently, the approximately 23.16% rise in integrated NKCC1 loading in the frozen Task 40 null is an unresolved experimental tension. It must not be described as observed compensation. The local equilibrium compensation derivative, the relative integrated NKCC1 increase and the fraction of missing AE4 chloride replaced by NKCC1 are three separate quantities throughout this report. Unchanged assay alkalinisation does not establish equal NHE1 net flux at different ionic states, and unchanged resting pH does not identify individual alkalinity sources. A discriminating experiment would measure chloride loading, pH, sodium and volume through the same combined stimulus, with pathway specific perturbations and parallel controls.

What the 2018 model actually established. The published model predicted an approximately 24% flow reduction after AE4 deletion. Its reduced flow settled within about two minutes; the authors explicitly contrasted this with experimental development over roughly ten minutes. It predicted little NKCC1 activation following either exchanger deletion. It included carbon dioxide hydration, NHE1, AE2 and AE4, without another proton or bicarbonate transporter, and held lumen volume constant Vera-Sigüenza et al. [2018a]. Those are historical model properties, not experimental determinations of the full network. The present retained carbon, alkalinity and

volume coordinates make the storage terms explicit. Numerical slow modes and their projection onto secretion must be examined before attributing a temporal discrepancy to absent slow regulation. A slow eigenvalue alone does not reproduce delayed genotype divergence.

Regulation and the inverse CaCC family. The 2021 study found that beta adrenergic stimulation increased native AE4 activity, that H89 prevented the response, and that constitutively active PKA increased AE4 activity in CHO cells. The S173A mutant lost PKA activation, whereas S273A retained it. The authors described phosphorylation of S173 as a probable mechanism; H89 is a nonselective inhibitor Peña Münzenmayer et al. [2021a]. This supports regulated AE4 activity, but does not measure the model's AE4 regulatory gain, relaxation time or an AE4 to CaCC recruitment law. The Task 41 inverse crossing quantifies a requirement within one family selected using the phenotype. It supplies no independent validation of that family. Its direct experimental test is the fraction of stimulated apical chloride conductance lost upon AE4 perturbation at matched calcium, membrane potential, chloride, pH and volume, resolved from the initial response into sustained secretion. This distinguishes altered channel recruitment from changed driving force and from intracellular chloride storage.

Molecular source classes. Catalán and colleagues studied human AE4 expressed in HEK293 cells, combining mutagenesis, transport assays and molecular dynamics. The T756A–T448I mutant lost transport with extracellular sodium while retaining activity with potassium. Their discussion proposes both bicarbonate and carbonate cycles, including exchanged sodium and potassium, and explicitly calls for further experiments to resolve the alternatives Catalán et al. [2025a]. Thus the seven frozen classes are specified source hypotheses, not seven experimentally established full cycles. Their conserved projections and equilibrium tests separate chloride, carbon, alkalinity and charge consequences under inherited scalar kinetics. Matching one projected signature cannot establish identical molecular mechanisms or identical whole cell responses. An unsuccessful root search cannot establish nonexistence. A thermodynamically opposed or nonphysiological case must retain that qualification. Joint measurements of cation and anion flux ratios, reversal conditions and current would constrain the source vector; rates measured across those gradients would constrain the separate kinetic law.

The most urgent missing measurements. The evidence checked here does not establish the identity, abundance or capacity of the assumed inward $1\text{Na} : 2\text{HCO}_3$ pathway. The alkalinity proposition gives a conditional requirement for additional support when the admissible NHE1 and AE2 contribution is insufficient. It does not prove that a particular NBC isoform is universally necessary. The assumed loading allocation used to construct NBC capacity is not a measured transporter partition. Measurements of sodium and bicarbonate influx with current and pH during combined stimulation are therefore a priority, alongside native NHE1 and NKCC1 capacity and state dependence. The sensitivity crosswalk identifies which effective capacities, conductances and buffer settings matter most at the analysed point. Until independent ranges constrain them, local elasticities do not establish robustness over biological uncertainty.

Scope of source verification. The comparison uses the 2015 primary full text and Figures 1 and 2, the repository copy of the 2018 published paper, the 2021 primary abstract and figure captions, and the 2025 primary author manuscript including its discussion. These four references were checked to that extent. No systematic literature survey, new experimental measurement or complete source audit of every inherited kinetic coefficient is claimed here; parameter provenance is recorded separately in Task 43.

21 What is established and what remains unresolved

Status	Conclusion and scope
Proved or exact	Unique mathematical carbonate and electrical roots under the stated positivity and slope assumptions, smooth dependence inside strict executable brackets, local well posedness, exact retained nondimensionalisation, alkalinity storage budget, conditional capacity obstruction and equal routing chloride identity. Joint source projections have rank five.
Local numerical	Twenty six class/expression cases yield nonsingular stable local roots; twenty two pass the physiological gates. WT/null slow times are 345.82/459.92 s. The stationary null deficit is 0.94617%; the WT local chloride compensation is 92.901%. Local IFT and parameter derivatives are independently verified. A physiological NBC absent equilibrium exists.
Finite time numerical	Frozen Task 40 observables reproduce. The 60–600 s integrated NKCC increase is 23.163% and missing AE4 chloride compensation 88.507%. First numerical Task 41 crossings need about 89.51% AE4 dependent stimulated CaCC recruitment. The 600 s gates pass; selected continued null cases later violate the pH gate.
Unresolved	Global uniqueness, stability and positivity; a justified reduced model; global exclusion of earlier inverse crossings; robustness over unknown parameter ranges; native carrier abundance and pathway identity; molecular source stoichiometry and compatible kinetics; the isolated NKCC assay tension; independent validation of the Task 41 CaCC coupling.

Several earlier interpretations require explicit correction. Stationary cancellation does not imply zero total sensitivity or a uniformly small phenotype. The system already contains slow chemical modes; the corrected regulatory derivative gives 30 s, not the spurious 60 s obtained by crossing the production clamp. The initial unsuccessful continuation record is not evidence of nonexistence. NBC is not universally necessary for secretion in this frozen system, though it supplies most of the larger AE4 alkalinity demand. The old selected Task 41 point gives a sampled deficit of 30.26115%, slightly below 30.3%, and the refined crossing depends on the declared integral. The comparison value is mean minus one standard error. The construction remains target selected and its late pH failure must accompany any claim about sustained stimulation.

The immediate experimental priorities are functional pump capacity and apical distribution; NBC equivalent sodium/bicarbonate flux, current and its saturation curve; ion conditioned NKCC and NHE capacity; simultaneous TIC, buffering, pH and volume measurements; and stimulated CaCC conductance and recruitment under acute AE4 perturbation at matched calcium and driving forces. Source stoichiometry requires combined cation, chloride, carbon and current measurements. These measurements would constrain the effective parameters that dominate local sensitivities and separate a transport pathway from a fitted phenomenological coupling. An accepted physiological point and small numerical perturbations do not supply those missing ranges.

22 Reproduction, verification and deliverables

All new material is under `analysis/44_full_system_mathematics/`. The frozen scientific base is `c4d4f207f702d8eee09ab94d2d90ae90552dd641`. The original source manifest, active parameter objects, source hashes and recovery record distinguish scientific files from historical metadata changes. The complete numerical checkpoint was cloned independently, checked out detached, and regenerated with empty inverse trial caches before this standalone report was written. All eighteen

commands passed. Forty six regenerated authoritative JSON/CSV files plus one unchanged historical record were compared: all 331157 numeric values matched bit for bit. Only elapsed runtime and an ancestor validated verification commit field were excluded. All 532 inverse trial files were regenerated; the optional cache is not part of the published scientific deliverables. All 1751 tracked files outside Task 44 were unchanged in that clean replay.

File or group	Content
README.md, RECOVERY.md	Reproduction commands, recovered history and derivative correction.
core.py, run_analysis.py	Exact coordinates, production adapter, scaling, continuation, local derivatives and baseline reproduction.
verify_exact.py	Independent full RHS reconstruction, pH and electrical closure checks, exact source and balance identities at valid states.
equilibrium_review.py	Full physiological gates, matched eigenvalues, modal projection and independent nearby roots.
sensitivity_analysis.py	Sixteen uncertain parameter sensitivities and derivative verification; Task 43 crosswalk source.
verify_sensitivity_gates.py	All 384 nearby roots tested against the complete inherited gates, with solver failures retained separately.
inverse_analysis.py, inverse_verify.py	Original family, both crossing observables, long diagnostics, eight parameter sensitivities and independent thirteen state integration/quadrature.
output/, inverse_detail/	Machine readable source audit, scales, trajectories, roots, spectra, derivatives, projections, crossing scan, refinement, gates and verification.
output/clean_recomputation_verification.json	Exact numerical checkpoint, commands, file comparisons, exclusions and protected file audit.
report.tex, build.sh, sources.bib	Standalone report source, bibliography and clean build command.
FINAL_STATUS.md	Final verified checkpoint, findings, failures, caveats, artifact paths and checksums.

The PDF build resolves every citation and reference. Figures are generated from saved verified results; axis ranges and the distinct observables are stated in captions. The report, TeX fragments, vector figures and complete machine readable outputs are retained for scientific review. No model source, frozen output, Task 1–42 analysis or manuscript text was changed. The two analysis branches remain separate and unmerged.

Part III

Complete parameter provenance from the final Task 43 sources

The following chapters retain the completed Task 43 audit and its complete inventory table. The additional parameter records in Part IV expose the detailed justification fields that were only supplied electronically in that source report. The original source bibliography is preserved separately; duplicate reference keys in this combined document are namespaced rather than silently reconciled.

Source record: `evidence/task43/report.tex`; `evidence/task43/output/parameter_inventory.json`.

23 Source of the audit

The audit uses repository <https://github.com/esig626/ae4-salivary-transport-control> at main revision `c4d4f207f702d8eee09ab94d2d90ae90552dd641`. It loads the actual nested Task 40 model through its historical validation helper and checks the complete active parameter hashes against `results/40_ae4_equal_cation_routing/frozen_inputs.json`. It does not overwrite the helper or weaken its historical numerical checks. The canonical initial condition is the separately frozen WT resting state, not the constructor default tuple.

The model lineage includes the original AE4 reconstruction [Vera-Sigüenza et al., 2018b], the Palk and Benjamin NKCC1 law [Palk et al., 2010b, Benjamin and Johnson, 1997], and the Cha NHE kinetic model [Cha et al., 2009b]. The experimental evidence for AE4 transport and regulation constrains mechanism and direction, but does not automatically identify every abundance, gain or timescale in this implementation [Peña Münzenmayer et al., 2015b, 2016, 2021b, Catalán et al., 2025b]. This report preserves that distinction.

24 What counts as a justification

A published numerical coefficient has a source, but transfer from another tissue or model still needs an applicability argument. A value determined by a balance equation is justified conditional on the concentrations, fluxes and structural assumptions used in that balance. A value selected to keep a reference trajectory within physiological limits is an effective modelling choice; acceptance of that trajectory does not make the value measured, unique, or known within an uncertainty interval. A numerical inversion bracket is not a physiological observation. A parameter selected using the knockout phenotype must remain labelled as an inverse design choice.

The inventory records value, units, original metadata, active role, classification, exact code locations, the equation role and source, mathematical justification, default value and whether the active value differs from that default. Separate fields state what constrains each quantity, what does not justify it, its logical domain, its uncertainty status, its relevance to manuscript conclusions, and an experiment that could constrain it. Classification is conservative: inherited metadata identifies a lineage, not a newly verified measurement of salivary transporter abundance. Each physiological uncertainty interval remains explicitly null in the machine readable inventory. Logical positivity or fractional bounds are not substituted for experimental uncertainty.

The current categories include physical constants; published kinetic coefficients; inherited model conversions or settings; reference and WT constraint derived values; WT architecture derived values; effective assumptions; numerical brackets and regularisation; protocol and law selections;

inactive legacy values; constructor seeds; and the target selected Task 41 parameter. These categories should not be collapsed into a single “from the literature” column.

25 Parameters that can be explained mathematically

25.1 NKCC1 scale: derived from a modelled resting flux

The production law has the form

$$J_N = \alpha_N m_N(t) \psi([Na]_i, [K]_i, [Cl]_i), \quad \psi = \frac{A_1 - A_2[Na]_i[K]_i[Cl]_i^2}{A_3 + A_4[Na]_i[K]_i[Cl]_i^2}.$$

At the accepted resting state, the scale is determined by

$$\alpha_N = \frac{J_{N,rest}}{\psi(c_{rest})} = 0.017334746894096052 \text{ fmol/s.} \quad (18)$$

Equation (18) makes the choice reproducible and avoids an independent extra fit. However, $J_{N,rest}$ is itself an inherited model flux. The scale is not a directly measured salivary carrier amount. The much larger legacy field `homeostasis.nkcc1_capacity_fmols` is inactive for this selected law and must not appear in a publication table as its active capacity.

The four Palk coefficients contain fourth order concentration factors. Their conversion to mM is essential. They are effective coefficients for the adopted fixed bath, so an uncertainty study that changes bath composition cannot automatically reuse them without rechecking the underlying reduction.

25.2 NHE1 carrier amount: a WT resting pH calibration

The active NHE1 law is the eight state Cha reduction with its Mod2 term. Printed rate and binding constants are retained, including the documented rounding discrepancy in the fourth reverse rate. The salivary carrier amount is

$$N_H = 2.339370005697548 \times 10^{-5} \text{ fmol.}$$

This was determined against a resting pH target of 6.91 on the selected R09 background. The derivation fixes one scale conditional on that background and target. It does not identify a cardiac carrier count as a salivary measurement, nor does it prove unique joint identification of the carrier amount, buffer pool and other pH pathways.

The completion audit also records the fixed Mod2 exponents $n_H = 3$ and $m_H = 1$, which were absent from the earlier inventory because they are module constants rather than dataclass fields. Both are part of the selected kinetic law. Their inclusion corrects an inventory omission without changing the model.

Legacy tanh NHE capacity and older NHE kinetic fields remain in the dataclass for other models, but are inactive in production. The active stimulus multiplier is 2.3. Calling it fixed before the test documents absence of retuning; it does not establish that its transfer to every tissue and stimulus is exact. The mathematical companion report shows explicitly how the alkalinity bound depends on this multiplier and N_H .

25.3 NBC capacity: derived conditional on an assumed partition

The NBC source imports one sodium and two bicarbonate equivalents per inward cycle. Its capacity is

$$G_B = 0.11570913197464398 \text{ fmol/s.}$$

The construction begins from the assumed fraction $s = 0.70$ of positive WT chloride loading assigned to NKCC1 at the Task 31 reference scale. If that NKCC1 chloride flux is L_N , the corresponding AE4 demand is

$$J_{4,req} = \frac{1-s}{s} L_N. \quad (19)$$

Using $L_N = 0.2562444047732706$ fmol/s and the selected stimulated NHE flux, the approximate reference requirement is $B_{req} = J_{4,req} - H_{stim}/2$, before evaluating the full coupled electrical closure. The capacity is then chosen so the NBC law supplies that cycle rate at the reference chemical state with its self consistent voltage. The derivation is recorded in the unchanged Task 36 design and NBC module.

This is a conditional structural construction, not a knockout fit and not a measured transporter density. In particular, equation (19) does not turn the 70/30 partition into experimental evidence. The subsequently evolved trajectory need not retain that partition exactly. The manuscript should state the assumed partition, how it sets the capacity, and what changes when that assumption changes. A broad uncertainty interval for s has not been established by this audit.

The NBC increment is zero at the declared resting calcium and fully recruited at $0.25 \mu\text{M}$. Its width parameter is effective. Its molecular identity is deliberately not assigned to a particular salivary SLC4 isoform.

25.4 Fixed charge and osmotic pools

Reference electroneutrality determines the fixed intracellular anion equivalent amount from

$$X = n_{Na,i} + n_{K,i} - n_{Cl,i} - n_{A,i}.$$

The active value is 78.06176220594485 fmol equivalents. Its status is reference constraint derived, conditional on the reference tuple and carbonate buffer model. It is not a directly counted molecular pool.

The other impermeant osmotic pool is 124.69936937124379 fmol on the selected background, rather than the smaller dataclass default. It is an inherited effective osmotic setting. The separate finite intracellular buffer amount is 30 fmol and is counted exactly once in the osmolarity. Effective osmotic closure and charge closure are different roles and should not be conflated.

26 Parameters that remain assumptions or transferred settings

The AE4 carrier amount and transition attempt rates are inherited effective model settings. Their thermodynamic construction constrains rate ratios and reversal properties of the parent cycle; it does not supply independent measurements of the absolute carrier number or all attempt rates. Moreover, fixed source substitution in the later mechanism panel does not itself establish thermodynamic consistency of the inherited scalar kinetics with each substituted cycle.

Channel conductances, pump capacity and apical fractions determine current closure and the distribution of secretion. Their active values are explicit in the inventory, but physiological feasibility at one reference point is not evidence that each value is unique. The same applies to the effective outlet rate and dead volume. The code itself describes the outlet law as a transparent compliance closure rather than a calibrated secretion law.

The water coefficients are inherited model conversions with stated units. They should be described as transferred model parameters, not newly measured hydraulic permeabilities of the present preparation. The effective beta activation time of 30 s and AE4 gain of 0.25 likewise need to remain explicit assumptions. Experimental activation of a pathway does not identify all intermediate time constants [Peña Münzenmayer et al., 2021b].

The NKCC1 algebraic recruitment reaches its endpoint at $0.10\ \mu\text{M}$ calcium, while NBC reaches its endpoint at $0.25\ \mu\text{M}$. The production input fully recruits both. This difference is preserved and recorded, not silently reconciled. Any analysis of submaximal calcium would need to account for it.

The numerical pH bracket from 3 to 11 only enables inversion; the actual inherited physiological criterion is pH 6.6 to 7.3. The tiny positive luminal buffer amount $10^{-9}\ \text{fmol}$ is numerical regularisation approximating no fixed luminal buffer, not an abundance measurement. Constructor concentrations and volumes do not replace the frozen solved initial state used by the production experiments.

Finally, the Task 41 residual CaCC recruitment coefficient $b = 0.10511872843288446$ is target selected. It is not an active parameter of the parent Task 40 model and is not reclassified as literature supported merely because the resulting trajectory has an appropriate secretion magnitude.

The CCh and IPR dose fields, 0.3 and $5\ \mu\text{M}$ respectively, are experiment labels in this implementation. The selected input callable returns prescribed calcium and beta activation from the input arm and time window; it does not map those dose values through a dose response law. They are therefore retained as protocol metadata but marked inactive for the selected right hand side. The previous inventory incorrectly marked both active. Changing the nominal dose labels alone cannot change the model trajectory.

The frozen initial state is now recorded separately, with all twelve conserved amount and volume coordinates and the AE4 regulatory fraction. The twelve constructor reference fields remain distinct. The upstream NBC design quantities, selected transport laws, R1 regulatory family and WT construct are also explicit. These additions explain the larger record count; they add no fitted degrees of freedom.

The externally passed WT genotype object and the three prescribed AE4 expression values, 1, 0.05 and 0, are also recorded. The other transporter expression multipliers remain one in those comparisons. These values define interventions, not an uncertainty interval for protein abundance. The prescribed Task 41 family requires $0 < b \leq 1$, equivalently $0 \leq \rho < 1$; the endpoint $b = 0$ is outside its implemented domain.

The additional file `output/numerical_controls.json` records the production integration specification, conservation tolerances, pH inversion tolerances and voltage bracket. In particular, $[-0.5, 0.5]\ \text{V}$ is the implemented voltage search bracket and not a measured physiological voltage interval. Integration begins at $10^{-6}\ \text{s}$ with the shared frozen state; the distinct resting observable at exactly $t = 0$ is preserved. Exact unit conversions are stated separately from physiological settings.

27 What the mathematical analysis adds to justification

The separate Task 44 report uses the unchanged reference parameter set and retains carbon, alkalinity, volume, lumen and electrical feedback. It supplies exact structural identities and a conditional NHE supply bound, then evaluates local stationary sensitivities. These calculations help identify which assumptions affect which conclusions. They do not estimate experimental uncertainty intervals.

An important distinction is between statements independent of capacities, such as the cancellation of the explicit equal routing AE4 term in the chloride balance, and numerical conclusions that depend on the adopted kinetics and scales, such as large NKCC1 replacement. Even an exact balance identity does not determine the magnitude of the knockout flow deficit. The dependence of pump and NHE fluxes on state remains part of the mechanism.

For later manuscript development, every nonliterature value should therefore be accompanied

Setting	E_Q	$dpH/d \log p$	E_D	E_τ
NBC capacity	0.019923	0.064832	2.0848	0.10701
AE4 carrier	0.00081137	-0.072461	0.084942	-0.063968
NKCC scale	0.055361	-0.001558	-1.4937	0.19015
NHE carrier	-0.0032611	0.070838	-0.22928	-0.014728
pump capacity	0.52593	0.12061	4.6337	-1.5222
CaCC conductance	0.03247	-0.0055239	-0.22116	0.0082736
buffer pool	-0.0015749	0.0013957	-0.20631	0.30856
water apical	0.0021295	-8.3167e-05	-0.003558	-3.7398e-05
AE4 tau	0	0	0	0
AE4 gain	0.00016227	-0.014492	0.016988	-0.012794
apical pump fraction	0.049564	-0.0035713	-0.059826	0.045169
NHE stimulated multiplier	-0.0032611	0.070838	-0.22928	-0.014728
CO2 basolateral permeability	-0.004448	-0.083186	-0.43378	-0.068179
AE2 capacity	-7.3346e-05	0.00036297	-0.030565	7.5082e-05
NBC log width	-0.014352	-0.046703	-1.5016	-0.072634
NKCC stimulated multiplier	0.055361	-0.001558	-1.4937	0.19015

Table 3: Local stationary sensitivities imported from Task 44. No measured uncertainty intervals are implied.

by either its conditional derivation or a plainly stated effective role. Where several values were selected together on the same WT background, their correlation should be retained in future uncertainty work. Independently perturbing every value as though each had its own measured error bar would create an unsupported uncertainty model.

The remaining parameter work is substantive rather than typographical: independently justify or measure ranges for the effective settings; define a WT feasible set using evidence not reused from the knockout comparison; and assess which conclusions survive across that set. A local derivative or a collection of physiologically passing simulations does not prove robustness over all plausible parameterisations. This audit has performed no new fitting and established no calibrated uncertainty intervals.

27.1 Completed sensitivity crosswalk

The completed companion calculation examines sixteen uncertain or effective settings. All 128 nearby WT and AE4 null equilibrium solves pass the inherited physiological gates. Parameter differences of 10^{-3} and 5×10^{-4} in log coordinates are numerical verification steps, not uncertainty ranges. The machine readable crosswalk reads the verified results directly from immutable Task 44 numerical source commit `f783785a863440df459e9c5530beba4c7eb59110` and records SHA256 hashes of all seven imported source files. Later report-only commits need not have that same head SHA. It maps every companion parameter name back to the actual inventory value and classification.

At the adopted constant full stimulus equilibrium, WT secretion is 0.0015840671 pL/s and intracellular pH is 7.0503863. The AE4 null stationary secretion deficit is 0.94617%. This stationary deficit is distinct from the frozen Task 40 cumulative deficit of 3.8575% at 600 s. Intracellular storage, changing volumes, finite lumen composition and regulation remain active in the companion analysis.

In the table, $E_Q = d \log Q_{WT} / d \log p$, $E_D = d \log D / d \log p$ for the positive stationary null deficit D , and $E_\tau = d \log \tau_{slow, WT} / d \log p$. The pH column uses $dpH / d \log p$ because pH is already logarithmic. A large elasticity of a small deficit is not a large absolute percentage point change.

The largest local sensitivities of the stationary null deficit are to pump capacity ($E_D = 4.634$), NBC capacity (2.085), NBC saturation width (-1.502), and NKCC effective capacity (-1.494). These

values identify priorities for experimental constraint. They do not establish a finite parameter change that recovers the experimental phenotype. The common stationary effects of NHE carrier amount and its stimulated multiplier reflect their product under full stimulation; separate resting and stimulated measurements are needed to distinguish them.

At the WT equilibrium, NBC supplies 95.48% of the AE4 alkalinity demand. Under the stated bath composition and inherited intracellular pH lower gate of 6.6, the conditional NHE/AE2 capacity calculation covers at most 57.96% of that current demand without NBC. This comparison holds the required AE4 demand at the analysed level; it does not rule out a lower supported flux, another admissible state or an additional pathway. The capacity ratio has local elasticity 0.931 to NHE carrier amount and to the stimulated NHE multiplier. Direct measurements of maximal acid extrusion are therefore essential before treating the conditional obstruction as a biological conclusion.

The companion continuation also finds a physiological equilibrium with zero NBC capacity: pH 6.96833 and secretion 0.00153598 pL/s, only 3.035% below the full NBC equilibrium. Its AE4 cycle flux falls from 0.0750724 to 0.00632651 fmol/s. Consequently, any earlier broad claim that NBC is structurally necessary for secretion must be qualified: the proved obstruction concerns sustaining the specified AE4 loading level. It does not imply that all physiological secretion fails without NBC.

The WT and null slowest local decay times are approximately 346 and 460 s. The WT decay time is particularly sensitive to pump capacity ($E_\tau = -1.522$) and the finite intracellular buffer pool (0.309). The model therefore already contains slow modes. The effective AE4 regulatory time has zero local stationary influence in the selected R1 family while the slower chemical mode remains dominant; this is not evidence that time dependent regulation is irrelevant. Acute pH, TIC and volume measurements alongside secretion would constrain those chemical modes and their observable effects.

The separate inverse calculation stays inside the one prescribed Task 41 family. Its first numerical 600 s crossing requires an AE4 dependent stimulated CaCC fraction $\rho \simeq 0.89507$, approximately 89.51%. The original selected value $\rho = 1 - b = 0.89488127$ gives the frozen 30.2612% deficit, slightly below the exact 30.3% comparison threshold. The threshold is 35% minus the reported standard error of 4.7 percentage points in the 2015 comparison; it is not a confidence bound. Adaptive integration and the original sampled trapezoidal observable are retained separately in the companion files. This is the first crossing found in the ordered numerical scan. Monotonicity between samples and exclusion of earlier unsampled crossings are unproved; it is neither a global lower bound nor a universal recruitment requirement.

Inverse sensitivities hold the inherited conserved WT initial state fixed, perturb the WT denominator and null system together, and perform no new resting solve. A buffer perturbation therefore changes the initial algebraic pH. Within that family, the adaptive crossing has local log elasticity 0.09833 to CaCC conductance and -0.03558 to pump capacity, larger in magnitude than its elasticities to the other six tested settings. The relative sensitivity of the residual independent recruitment $1 - \rho$ is also recorded, because a small change in ρ can be a larger relative change in the remaining approximately 10.49%. Under continued full stimulation, both the selected and crossing null cases eventually violate the inherited upper pH gate at about 1485 s. This diagnostic does not change their declared 600 s results, but prevents an unsupported long duration interpretation.

The most urgent experiments consequently differ by claim. Functional pump turnover and membrane distribution, NBC capacity and saturation shape, and ion conditioned NKCC uptake constrain secretion and compensation. Maximal NHE acid extrusion constrains the conditional alkalinity obstruction. Buffer titration with TIC and volume time series constrains slow relaxation. Matched CaCC current measurements during acute AE4 perturbation would directly test the very large recruitment dependence required by the inverse family. Strong modelled NKCC compensation remains in tension with the isolated 2015 NKCC assay; changing the assay context in the model

does not erase that discrepancy.

28 Reproducibility and use in the manuscript

Run `python analysis/43_parameter_provenance/audit.py` from the repository root. It verifies the production parameter hashes and writes both CSV and JSON inventories under this task’s `output` directory. Each record contains source paths and justification notes beyond the abbreviated table below. `prepare_report.py` typesets the table from those records. The report and the full inventory belong on the Task 43 branch, separate from the mathematical analysis and from the main manuscript.

The inventory has 137 records and 109 active flags. This is an inventory of configuration and roles, not a count of independent degrees of freedom: it includes physical constants, protocol settings, law selections, derived values and other nonfit quantities. Five legacy fields and twelve constructor seed fields are explicitly separated. The Task 41 inverse coefficient is also outside the parent model.

Run `analysis/43_parameter_provenance/verify.py` to compare the saved CSV/JSON products with a fresh reconstruction, check every nested production field, verify source hashes, confirm the fixed Cha exponents and demonstrate that changing dose labels leaves the selected input unchanged. The audit performs no fresh sensitivity simulations. With the companion commit available in the fetched Git history, `import_crosswalk.py--check` independently reconstructs the saved crosswalk from that exact commit; no adjacent Task 44 working directory is required. The report tables are also checked against their machine readable sources. The source manifest contains SHA256 values for the production modules and frozen inputs; verification also checks that protected paths are unchanged from recovered Task 43 checkpoint `693f44c0b05e92faec737aa7793c3a99ec76445c`.

No model source, historical analysis, frozen trajectory or main manuscript is modified by this audit. The exact original metadata is retained in the machine readable inventory, while the new classification states what that metadata does and does not justify.

29 Complete active object and legacy inventory

The “Used” column denotes use by the selected production configuration as classified by the audit. “No” for constructor seeds means they are not the production initial state; the constructor still uses that tuple for reference consistency checks. Full units, source locations, defaults and detailed justifications are in `parameter_inventory.csv` and `parameter_inventory.json`.

Setting	Value	Used	Classification
<code>constants.temperature_K</code>	310.15	Yes	effective assumption
<code>constants.gas_constant_J_mol_K</code>	8.31446262	Yes	physical constant
<code>constants.faraday_C_mol</code>	96485.3321	Yes	physical constant
<code>acid_base.carbon_pka1</code>	6.1	Yes	inherited model setting
<code>acid_base.carbon_pka2</code>	10.3	Yes	inherited model setting
<code>acid_base.water_pkw</code>	14	Yes	inherited model setting
<code>acid_base.cell_buffer_pka</code>	7	Yes	effective assumption
<code>acid_base.lumen_buffer_pka</code>	7	Yes	effective assumption
<code>acid_base.ph_lower</code>	3	Yes	numerical bracket
<code>acid_base.ph_upper</code>	11	Yes	numerical bracket
<code>bath.na_mM</code>	145	Yes	effective assumption
<code>bath.k_mM</code>	5	Yes	effective assumption

Setting	Value	Used	Classification
bath.cl_mM	126.161593	Yes	reference constraint derived
bath.tic_mM	25	Yes	effective assumption
bath.ph	7.4	Yes	effective assumption
bath.buffer_total_mM	0	Yes	effective assumption
bath.untracked_osmolyte_mM	0	Yes	effective assumption
geometry.cell_buffer_total_fmole	30	Yes	effective assumption
geometry.lumen_buffer_total_fmole	1e-09	Yes	numerical regularisation
geometry.fixed_cell_anion_equivalents_fmole	78.0617622	Yes	reference constraint derived
geometry.cell_impermeant_osmoles_fmole	124.699369	Yes	inherited effective setting
homeostasis.nkcc1_capacity_fmole_s	0.32	No	inactive legacy
homeostasis.nhe1_capacity_fmole_s	0.00744098059	No	inactive legacy
homeostasis.nhe1_model	cha_2009_eight_state_mod2	Yes	law selection
homeostasis.nhe1_published_gfmole_s	0.0305	No	inactive legacy
homeostasis.nhe1_published_k_h_mM	0.00045	No	inactive legacy
homeostasis.nhe1_published_k_na_mM	15	No	inactive legacy
homeostasis.nhe1_cha_carrier_amount_fmole	2.33937001e-05	Yes	WT constraint derived
homeostasis.ae2_capacity_fmole_s	0.005	Yes	effective assumption
homeostasis.thermodynamic_saturation_log_width	2	Yes	effective assumption
homeostasis.co2_basolateral_permeability_fmole_s_mM	0.02	Yes	effective assumption
homeostasis.co2_apical_permeability_fmole_s_mM	0.01	Yes	effective assumption
membranes.nak_capacity_fmole_s	0.08	Yes	effective assumption
membranes.nak_na_half_mM	10	Yes	inherited model setting
membranes.nak_k_half_mM	1.5	Yes	inherited model setting
membranes.apical_pump_fraction	0.075075	Yes	effective assumption
membranes.apical_k_fraction	0.3	Yes	effective assumption
membranes.g_k_total_S	1.4e-08	Yes	effective assumption
membranes.g_cl_apical_S	3.14e-08	Yes	effective assumption
membranes.calcium_half_uM	0.26	Yes	effective assumption
membranes.calcium_hill	1.46	Yes	effective assumption
membranes.g_para_na_S	3e-10	Yes	inherited model setting
membranes.g_para_k_S	5e-11	Yes	inherited model setting
membranes.g_para_cl_S	2.5e-10	Yes	effective assumption
membranes.g_para_hco3_S	5e-11	Yes	effective assumption
membranes.g_apical_background_S	0	Yes	effective assumption
membranes.g_basolateral_background_S	0	Yes	effective assumption
water.apical_hydraulic_pL_s_mOsm	0.00432	Yes	published model conversion
water.basolateral_hydraulic_pL_s_mOsm	0.0515	Yes	published model conversion
water.paracellular_hydraulic_pL_s_mOsm	0.00026	Yes	published model conversion
water.lumen_dead_volume_pL	0.1	Yes	effective assumption
water.outflow_rate_s	0.2	Yes	effective assumption
initial.cell_na_mM	15	No	constructor seed
initial.cell_k_mM	140	No	constructor seed
initial.cell_cl_mM	40	No	constructor seed

Setting	Value	Used	Classification
initial.cell_tic_mM	20	No	constructor seed
initial.cell_ph	7.2	No	constructor seed
initial.cell_volume_pL	1	No	constructor seed
initial.lumen_na_mM	145	No	constructor seed
initial.lumen_k_mM	5	No	constructor seed
initial.lumen_cl_mM	126.161593	No	constructor seed
initial.lumen_tic_mM	25	No	constructor seed
initial.lumen_ph	7.4	No	constructor seed
initial.lumen_volume_pL	0.1	No	constructor seed
ae4.carrier_amount_fmole	0.150126427	Yes	inherited effective setting
ae4.common_cl_attempt_rate_s	1	Yes	inherited effective setting
ae4.na_loaded_attempt_rate_s	0.1	Yes	inherited effective setting
ae4.k_loaded_attempt_rate_s	1.9	Yes	inherited effective setting
ae4.cooperative_gate	None	No	inherited effective setting
nbc.capacity_fmole_s	0.115709132	Yes	WT architecture derived
nbc.log_width	2	Yes	effective assumption
nbc.resting_calcium_uM	0.058	Yes	effective assumption
nbc.fully_recruited_calcium_uM	0.25	Yes	effective assumption
nbc.nhe1_stimulated_multiplier	2.3	Yes	effective assumption
nkcc.alpha_eff_fmole_s	0.0173347469	Yes	WT constraint derived
nkcc.A1	157.5	Yes	published kinetic coefficient
nkcc.A2	2.0096e-05	Yes	published kinetic coefficient
nkcc.A3	1.0306	Yes	published kinetic coefficient
nkcc.A4	1.3852e-06	Yes	published kinetic coefficient
cha.k1_plus_per_ms	10.5	Yes	published kinetic coefficient
cha.k1_minus_per_ms	0.201	Yes	published kinetic coefficient
cha.k2_plus_per_ms	15.8	Yes	published kinetic coefficient
cha.k_na_o_mM	195	Yes	published kinetic coefficient
cha.k_na_i_mM	16.2	Yes	published kinetic coefficient
cha.k_h_o_mM	0.00162	Yes	published kinetic coefficient
cha.k_h_i_mM	0.000605	Yes	published kinetic coefficient
cha.modifier_k_i_mM	3.07e-05	Yes	published kinetic coefficient
cha.modifier_k_o_mM	4.8e-07	Yes	published kinetic coefficient
cha.k2_minus_table_per_ms	183	Yes	published kinetic coefficient
regulation.tau_ae4_s	30	Yes	effective assumption
regulation.basal_capacity_multiplier	1	Yes	effective assumption
regulation.fully_activated_increment	0.25	Yes	effective assumption
regulation.coupling_scale	1	Yes	effective assumption
nkcc_activation.fully_activated_multiplier	1.75	Yes	inherited effective setting

Setting	Value	Used	Classification
nkcc_activation.resting_calcium_uM	0.058	Yes	inherited effective setting
nkcc_activation.stimulated_calcium_uM	0.1	Yes	inherited effective setting
protocol.arm	CCH_IPR	Yes	protocol setting
protocol.onset_s	0	Yes	protocol setting
protocol.duration_s	600	Yes	protocol setting
protocol.cch_dose_uM	0.3	No	protocol setting
protocol.ipr_dose_uM	5	No	protocol setting
protocol.resting_calcium_uM	0.058	Yes	protocol setting
protocol.stimulated_calcium_uM	0.25	Yes	protocol setting
protocol.beta_occupancy_on	1	Yes	protocol setting
Task41.b	0.105118728	No	target selected
cha.intracellular_modifier_hill	3	Yes	published kinetic coefficient
cha.extracellular_modifier_hill	1	Yes	published kinetic coefficient
nkcc.law	palk_benjamin_2010_eq17	Yes	law selection
ae4.law	equal_routing_adapter	Yes	law selection
regulation.family	R1	Yes	law selection
regulation.construct	WT	Yes	protocol setting
nkcc_activation.family	N1_CCH_ALGEBRA_IC	Yes	law selection
nbc_design.TARGET_NKCC1_POSITIVE_CL_SHARE	0.7	No	effective assumption
nbc_design.TASK31_R09_NKCC1_CL_F_MOL_S	0.256244405	No	WT architecture derived
nbc_design.TASK31_R09_NHE1_FMOL_S	0.00789058049	No	WT architecture derived
nbc_design.DERIVED_AE4_CL_FMOL_S	0.109819031	No	WT architecture derived
nbc_design.DERIVED_STIMULATED_NHE1_FMOL_S	0.0181483351	No	WT architecture derived
nbc_design.DERIVED_REQUIRED_NBC_CYCLE_FMOL_S	0.100744863	No	WT architecture derived
frozen_initial.na_i_fmol	16.7203587	Yes	WT constraint derived
frozen_initial.k_i_fmol	170.002009	Yes	WT constraint derived
frozen_initial.cl_i_fmol	88.031802	Yes	WT constraint derived
frozen_initial.tic_i_fmol	8.26351181	Yes	WT constraint derived
frozen_initial.alk_i_fmol	20.6288031	Yes	WT constraint derived
frozen_initial.volume_i_pL	1.45333439	Yes	WT constraint derived
frozen_initial.na_l_fmol	13.1277687	Yes	WT constraint derived
frozen_initial.k_l_fmol	2.71897261	Yes	WT constraint derived
frozen_initial.cl_l_fmol	15.3786879	Yes	WT constraint derived
frozen_initial.tic_l_fmol	0.540262316	Yes	WT constraint derived
frozen_initial.alk_l_fmol	0.468053423	Yes	WT constraint derived
frozen_initial.volume_l_pL	0.105388407	Yes	WT constraint derived
frozen_initial.ae4_effective_activation_fraction	0	Yes	WT constraint derived
genotype.name	WT	No	protocol setting
genotype.ae4_expression	1	Yes	protocol setting
genotype.ae2_expression	1	Yes	protocol setting
genotype.nkcc1_expression	1	Yes	protocol setting
genotype.nhe1_expression	1	Yes	protocol setting
protocol.ae4_expression_cases	[1.0, 0.05, 0.0]	Yes	protocol setting

Part IV

Expanded numerical and provenance records

30 Complete equilibrium case tables

All 26 cases are printed, including the four physiological exclusions and the repeated null systems. Numeric entries use standard e notation where needed: 1.2e-5 means 1.2×10^{-5} . Tables round for reading; every full precision value remains in the unchanged electronic source. The equilibrium evaluator time field is not an equilibration time.

Table 5: Stationary flow and intracellular ions. Flow is in pL/s; concentrations are in mM.

Class	e_4	Q	$[Na]_i$	$[K]_i$	$[Cl]_i$
C0	1	0.00158407	18.0826	110.237	53.2985
C0	0.9	0.0015839	18.0351	110.398	53.2649
C0	0.75	0.00158348	17.9502	110.691	53.2007
C0	0.5	0.00158192	17.7592	111.381	53.0352
C0	0.25	0.00157803	17.4794	112.498	52.7302
C0	0.1	0.0015735	17.2567	113.518	52.4206
C0	0.05	0.00157145	17.1728	113.944	52.2851
C0	0	0.00156908	17.0844	114.424	52.1287
C1	1	0.00108167	45.6937	79.5546	41.9556
C1	0.05	0.00153903	18.2226	112.696	51.2039
C1	0	0.00156908	17.0844	114.424	52.1287
C2	1	0.00093557	57.4596	63.0419	42.2884
C2	0.05	0.00151952	18.8973	110.84	50.798
C2	0	0.00156908	17.0844	114.424	52.1287
C3a	1	0.00141723	24.5079	102.228	48.2718
C3a	0.05	0.00155578	17.6797	113.245	51.7658
C3a	0	0.00156908	17.0844	114.424	52.1287
C3b	1	0.00171753	14.2063	115.557	58.0397
C3b	0.05	0.00158668	16.6943	114.61	52.7984
C3b	0	0.00156908	17.0844	114.424	52.1287
C4a	1	0.00128961	31.1075	90.6207	46.3804
C4a	0.05	0.00155274	17.7687	112.368	51.8636
C4a	0	0.00156908	17.0844	114.424	52.1287
C4b	1	0.00167764	15.5809	110.052	57.4878
C4b	0.05	0.00158409	16.7677	113.757	52.9047
C4b	0	0.00156908	17.0844	114.424	52.1287

Table 6: Intracellular carbon, alkalinity, bicarbonate and finite volumes. Concentrations are in mM and volumes in pL.

Class	e_4	T_i	A_i	$[HCO_3]_i$	pH_i	V_i	V_l
C0	1	8.268	18.857	7.4308	7.0504	1.3899	0.10792
C0	0.9	8.5578	19.19	7.7051	7.0582	1.3945	0.10792
C0	0.75	9.0922	19.805	8.2126	7.0726	1.4031	0.10792
C0	0.5	10.388	21.292	9.4523	7.1073	1.4241	0.10791
C0	0.25	12.615	23.826	11.609	7.1657	1.4613	0.10789
C0	0.1	14.822	26.292	13.769	7.2214	1.4994	0.10787
C0	0.05	15.801	27.368	14.734	7.2452	1.5168	0.10786
C0	0	16.955	28.618	15.873	7.2723	1.5378	0.10785
C1	1	8.6202	20.037	7.6798	7.0139	1.2341	0.10541
C1	0.05	15.957	27.686	14.885	7.2477	1.5003	0.1077
C1	0	16.955	28.618	15.873	7.2723	1.5378	0.10785
C2	1	9.8336	13.344	7.1287	6.5211	1.2034	0.10468
C2	0.05	15.688	25.972	14.325	7.1246	1.4738	0.1076
C2	0	16.955	28.618	15.873	7.2723	1.5378	0.10785

Class	e_4	T_i	A_i	$[HCO_3]_i$	pH_i	V_i	V_l
C3a	1	7.7661	18.713	6.9328	7.022	1.3064	0.10709
C3a	0.05	15.621	27.247	14.555	7.2403	1.5037	0.10778
C3a	0	16.955	28.618	15.873	7.2723	1.5378	0.10785
C3b	1	8.773	18.934	7.9256	7.0734	1.4787	0.10859
C3b	0.05	15.98	27.487	14.911	7.25	1.53	0.10793
C3b	0	16.955	28.618	15.873	7.2723	1.5378	0.10785
C4a	1	10.224	13.356	7.413	6.5214	1.2592	0.10645
C4a	0.05	16.061	26.234	14.691	7.1334	1.5001	0.10776
C4a	0	16.955	28.618	15.873	7.2723	1.5378	0.10785
C4b	1	11.739	14.489	8.8135	6.5792	1.4549	0.10839
C4b	0.05	16.427	26.487	15.05	7.1419	1.5266	0.10792
C4b	0	16.955	28.618	15.873	7.2723	1.5378	0.10785

Table 7: Lumen composition and membrane potentials. Chloride is in mM, osmolarity in mOsm, and voltage in V.

Class	e_4	$[Cl]_l$	pH_l	O_l	V_a	V_b
C0	1	147.677	6.67928	301.535	-0.0295982	-0.0791034
C0	0.9	147.669	6.67169	301.535	-0.0296128	-0.0790928
C0	0.75	147.653	6.659	301.535	-0.0296409	-0.0790718
C0	0.5	147.619	6.63427	301.534	-0.0297138	-0.0790152
C0	0.25	147.568	6.60658	301.533	-0.0298506	-0.0789105
C0	0.1	147.528	6.59123	301.532	-0.0299919	-0.0788118
C0	0.05	147.512	6.58692	301.532	-0.0300546	-0.0787723
C0	0	147.496	6.58321	301.531	-0.0301275	-0.07873
C1	1	145.973	6.80765	301.416	-0.0349864	-0.0716211
C1	0.05	147.431	6.5952	301.524	-0.0305548	-0.0785273
C1	0	147.496	6.58321	301.531	-0.0301275	-0.07873
C2	1	144.346	6.41508	301.382	-0.0342485	-0.0661334
C2	0.05	147.254	6.49501	301.519	-0.0307075	-0.0782226
C2	0	147.496	6.58321	301.531	-0.0301275	-0.07873
C3a	1	147.258	6.73356	301.495	-0.0319461	-0.077626
C3a	0.05	147.476	6.59201	301.528	-0.0302942	-0.0786693
C3a	0	147.496	6.58321	301.531	-0.0301275	-0.07873
C3b	1	147.953	6.63582	301.566	-0.0275453	-0.0798805
C3b	0.05	147.547	6.58201	301.535	-0.0298199	-0.0788679
C3b	0	147.496	6.58321	301.531	-0.0301275	-0.07873
C4a	1	145.95	6.2529	301.465	-0.0325839	-0.0746932
C4a	0.05	147.332	6.48275	301.527	-0.0302108	-0.0784617
C4a	0	147.496	6.58321	301.531	-0.0301275	-0.07873
C4b	1	146.951	6.11227	301.557	-0.0275475	-0.0786201
C4b	0.05	147.402	6.47139	301.535	-0.0297337	-0.0786676
C4b	0	147.496	6.58321	301.531	-0.0301275	-0.07873

Table 8: Signed stationary fluxes in fmol/s. NKCC and pump columns are cycle fluxes.

Class	e_4	N	H	E	B	J_4	P	C_a	$j_{p,Cl}$
C0	1	0.14939	0.0060362	-0.00074806	0.07168	0.075072	0.06319	0.37311	0.13918
C0	0.9	0.15066	0.0057091	-0.00065769	0.06915	0.072334	0.063118	0.373	0.13911
C0	0.75	0.15301	0.0051453	-0.00049737	0.064444	0.067265	0.062987	0.37278	0.13897
C0	0.5	0.15864	0.0039755	-0.00013973	0.052944	0.055001	0.062686	0.37214	0.13862
C0	0.25	0.16791	0.002519	0.00038759	0.033563	0.034628	0.062226	0.37084	0.13797
C0	0.1	0.17629	0.0015978	0.00082224	0.015665	0.016052	0.061844	0.36946	0.13733
C0	0.05	0.17972	0.001309	0.00099221	0.0082788	0.0084372	0.061696	0.36887	0.13706
C0	0	0.1835	0.0010393	0.0011767	6.8683e-05	0	0.061537	0.36819	0.13675
C1	1	0.088636	0.004194	-7.307e-05	0.040726	0.085719	0.073091	0.26292	0.10502
C1	0.05	0.17617	0.0012374	0.0010667	0.0042075	0.0085858	0.0634	0.36199	0.13509
C1	0	0.1835	0.0010393	0.0011767	6.8683e-05	0	0.061537	0.36819	0.13675
C2	1	0.082846	0.029022	-0.00027869	0.046924	0.061574	0.073455	0.22699	0.091941
C2	0.05	0.1741	0.0033544	0.00099391	0.0071942	0.0083744	0.064342	0.35758	0.13382

Class	e_4	N	H	E	B	J_4	P	C_a	$j_{p,Cl}$
C2	0	0.1835	0.0010393	0.0011767	6.8683e-05	0	0.061537	0.36819	0.13675
C3a	1	0.13369	0.0062131	-0.00067532	0.067623	0.071067	0.069174	0.33777	0.12907
C3a	0.05	0.1781	0.0013424	0.00098686	0.0082313	0.0084091	0.062558	0.36559	0.13615
C3a	0	0.1835	0.0010393	0.0011767	6.8683e-05	0	0.061537	0.36819	0.13675
C3b	1	0.16185	0.0057461	-0.0007987	0.074573	0.077845	0.054775	0.40075	0.14664
C3b	0.05	0.18129	0.0012765	0.00099744	0.008324	0.0084635	0.060809	0.37204	0.13793
C3b	0	0.1835	0.0010393	0.0011767	6.8683e-05	0	0.061537	0.36819	0.13675
C4a	1	0.11464	0.042992	-0.00041119	0.056852	0.078554	0.071493	0.30742	0.1192
C4a	0.05	0.17755	0.0032376	0.0010046	0.0073198	0.0084363	0.062701	0.36453	0.13576
C4a	0	0.1835	0.0010393	0.0011767	6.8683e-05	0	0.061537	0.36819	0.13675
C4b	1	0.14901	0.053092	-0.00051442	0.065128	0.091931	0.058431	0.38943	0.1429
C4b	0.05	0.18078	0.0031227	0.0010148	0.0074359	0.0084898	0.06095	0.37106	0.13757
C4b	0	0.1835	0.0010393	0.0011767	6.8683e-05	0	0.061537	0.36819	0.13675

Table 9: Separate physiological and thermodynamic classifications at every root.

Class	e_4	Physiology	Failed gate	$A_4/(RT)$	Flux compatibility
C0	1	Pass	None	-0.958921	opposed
C0	0.9	Pass	None	-0.886387	opposed
C0	0.75	Pass	None	-0.758624	opposed
C0	0.5	Pass	None	-0.476574	opposed
C0	0.25	Pass	None	-0.0627855	opposed
C0	0.1	Pass	None	0.282549	supported
C0	0.05	Pass	None	0.420015	supported
C0	0	Pass	None	0.571436	zero_flux
C1	1	Fail	na_i_mM	3.89255	supported
C1	0.05	Pass	None	5.62265	supported
C1	0	Pass	None	5.74871	zero_flux
C2	1	Fail	na_i_mM, ph_i	3.85893	supported
C2	0.05	Pass	None	8.00389	supported
C2	0	Pass	None	8.58518	zero_flux
C3a	1	Pass	None	1.51348	supported
C3a	0.05	Pass	None	3.02927	supported
C3a	0	Pass	None	3.20596	zero_flux
C3b	1	Pass	None	-3.74397	opposed
C3b	0.05	Pass	None	-2.22395	opposed
C3b	0	Pass	None	-2.06309	zero_flux
C4a	1	Fail	ph_i	0.709318	supported
C4a	0.05	Pass	None	2.90591	supported
C4a	0	Pass	None	3.31614	zero_flux
C4b	1	Fail	ph_i	-4.3271	opposed
C4b	0.05	Pass	None	-2.34	opposed
C4b	0	Pass	None	-1.95291	zero_flux

Table 10: Root, state charge and current residual diagnostics. Independent residuals use the dimensionless vector field.

Class	e_4	$\ F\ _\infty$	$\kappa(J)$	Charge (fmol)	Current (A)	Amount RHS	Volume RHS
C0	1	4.3571e-11	24849	1.4211e-14	3.8774e-26	1.7708e-14	6.7495e-14
C0	0.9	6.9845e-10	24857	1.4211e-14	1.1632e-25	4.8475e-13	1.082e-12
C0	0.75	1.4406e-10	24903	1.4211e-14	2.5849e-26	9.9226e-14	2.074e-13
C0	0.5	4.2564e-12	25173	1.4211e-14	3.2312e-26	3.3445e-15	5.6888e-15
C0	0.25	1.8392e-10	26074	1.4211e-14	1.551e-25	1.0159e-13	2.849e-13
C0	0.1	9.4522e-11	27458	2.2204e-15	1.1299e-23	1.6197e-13	1.0409e-13
C0	0.05	2.2513e-10	28241	2.8422e-14	1.6479e-24	3.645e-14	3.0522e-13
C0	0	1.2427e-11	29314	1.4211e-14	1.1309e-25	3.0431e-15	1.925e-14
C1	1	1.3937e-12	22777	2.8422e-14	1.8095e-25	1.7764e-15	1.5656e-16
C1	0.05	5.3949e-11	28199	1.4211e-14	2.8434e-25	1.7125e-14	8.3572e-14
C1	0	1.8975e-12	29314	1.4211e-14	5.8161e-26	1.491e-15	2.1315e-16
C2	1	3.7703e-09	19380	1.4211e-14	1.8418e-25	1.9159e-12	4.6477e-12
C2	0.05	7.7569e-11	25851	1.4211e-14	1.0857e-24	2.4652e-14	8.7135e-15
C2	0	1.8975e-12	29314	1.4211e-14	5.8161e-26	1.491e-15	2.1315e-16
C3a	1	1.7863e-10	26347	1.4211e-14	2.0033e-25	8.846e-14	2.7672e-13
C3a	0.05	7.2033e-12	27980	1.4211e-14	1.6414e-24	1.6556e-14	1.1159e-14
C3a	0	1.8975e-12	29314	1.4211e-14	5.8161e-26	1.491e-15	2.1315e-16

Class	e_4	$\ F\ _\infty$	$\kappa(J)$	Charge (fmol)	Current (A)	Amount RHS	Volume RHS
C3b	1	4.8111e-11	24602	1.4211e-14	3.8774e-26	1.1262e-13	7.4529e-14
C3b	0.05	9.0095e-11	28516	0	1.6867e-24	2.0005e-14	1.3957e-13
C3b	0	1.8975e-12	29314	1.4211e-14	5.8161e-26	1.491e-15	2.1315e-16
C4a	1	1.4568e-11	19319	1.4211e-14	3.2312e-26	3.8997e-15	1.6365e-15
C4a	0.05	5.2386e-10	26074	1.1102e-16	1.1988e-24	1.4613e-13	5.8846e-14
C4a	0	1.8975e-12	29314	1.4211e-14	5.8161e-26	1.491e-15	2.1315e-16
C4b	1	6.5386e-10	18817	1.1102e-16	1.9387e-26	1.9672e-13	1.7423e-13
C4b	0.05	2.891e-12	26375	1.4211e-14	1.1891e-24	1.645e-14	4.4784e-15
C4b	0	1.8975e-12	29314	1.4211e-14	5.8161e-26	1.491e-15	2.1315e-16

Source record: `evidence/task44/output/equilibrium_summary.csv`.

Source record: `evidence/task44/output/equilibrium_review.json`.

31 All state vectors, Jacobians and eigenvalues

The following matrices are the recorded dimensionless dynamic Jacobians F_z , in the eleven coordinate order $(x_{Na}, x_K, x_{Cl}, x_T, v, y_{Na}, y_K, y_{Cl}, y_T, \ell, r)$. The physical eigenvalues are in s^{-1} after division by t_0 . The thirteen state entries are dimensional amounts and volumes followed by the regulatory fraction, in the production order specified above. Full precision matrices and solver messages are retained in `equilibria.json`.

C0, expression 1.0

State: 25.13268, 153.2164, 74.07855, 11.49151, 26.20874, 1.38988, 12.71138, 3.520701, 15.93738, 0.3722667, 0.2947073, 0.1079203, 1.

Solver flag: True; evaluations: 54; residual: 4.35706e-11; physiological: True.

Row/column	1	2	3	4	5	6	7	8	9	10	11
1	-27.851	-7.2169	-9.0535	-8.2564	11.055	0.061709	0.073671	-0.080312	0.00015615	0.015089	-0.033422
2	-19.449	-8.807	-19.054	-9.6279	15.983	-2.0056	7.1965	4.7377	-0.0050748	-4.815	-0.033422
3	-36.772	-8.0917	-36.833	10.98	24.438	-2.0673	7.1203	4.818	-0.005231	-4.8295	0.066845
4	-5.8392	-3.3635	4.1567	-35.015	2.7024	-6.169	-6.1425	6.1318	6.2963	-0.0016594	-0.13369
5	5463.5	5463.5	5463.5	5463.5	-11349	-394.88	-394.88	-394.88	-394.88	821.17	0
6	2.5678	4.5608	11.505	-3.697	-8.0708	-22.449	-0.45952	-1.0034	-0.0064381	-133.06	3.1554e-25
7	-2.3484	41.679	86.021	0.32885	-63.008	-4.2373	-154.99	-24.543	-0.010722	21.094	0
8	0.074759	47	99.443	-3.9843	-72.353	30.629	-97.793	-83.452	0.077501	-108.99	0
9	-11.232	-12.137	9.4596	20.231	0.73896	43.23	42.885	-42.639	-100.69	-3.4073	0
10	-5830.9	-5830.9	-5830.9	-5830.9	12112	5773.2	5773.2	5773.2	5773.2	-12193	0
11	0	0	0	0	0	0	0	0	0	0	-31.273

Eigenvalues: -0.0028916897; -0.0044747554; -0.032669813; -0.033333333; -0.0413017; -0.061790606; -0.10806272; -0.13646588; -0.2; -9.1014622; -15.905022.

C0, expression 0.9

State: 25.1501, 153.9511, 74.27844, 11.93393, 26.761, 1.394509, 12.70767, 3.523573, 15.93632, 0.3739098, 0.2949219, 0.1079195, 1.

Solver flag: True; evaluations: 38; residual: 6.98445e-10; physiological: True.

Row/column	1	2	3	4	5	6	7	8	9	10	11
1	-27.739	-7.024	-9.2537	-7.6403	11.028	0.064257	0.076781	-0.083647	0.00015775	0.015712	-0.032203
2	-19.404	-8.6907	-19.137	-8.9273	15.934	-2.0016	7.1931	4.7333	-0.0049141	-4.8145	-0.032203
3	-37.05	-8.2755	-36.525	9.5681	24.462	-2.0659	7.1138	4.8169	-0.0050718	-4.8296	0.064406
4	-5.4636	-2.9226	3.6171	-32.208	2.6044	-6.1642	-6.1366	6.1254	6.2966	-0.00098697	-0.12881
5	5445.3	5445.3	5445.3	5445.3	-11311	-394.88	-394.88	-394.88	-394.88	821.18	2.068e-20
6	2.5343	4.5172	11.49	-3.6989	-8.0277	-22.445	-0.45733	-1.0081	-0.0062367	-133.02	-2.5244e-24
7	-2.3559	41.482	85.788	0.32902	-62.799	-4.234	-154.88	-24.548	-0.010395	21.062	0
8	0.025367	46.752	99.193	-3.9863	-72.093	30.606	-97.706	-83.433	0.075139	-108.98	6.3109e-25
9	-11.432	-12.338	9.6697	20.256	0.72359	43.263	42.918	-42.671	-100.69	-3.4177	0
10	-5811.5	-5811.5	-5811.5	-5811.5	12072	5773.3	5773.3	5773.3	5773.3	-12193	-1.034e-20
11	0	0	0	0	0	0	0	0	0	0	-31.273

Eigenvalues: -0.0028711031; -0.0044077049; -0.032032096; -0.033333333; -0.038615936; -0.061750647; -0.10801877; -0.13632315; -0.2; -9.0846545; -15.88169.

C0, expression 0.75

State: 25.18567, 155.3092, 74.64529, 12.75718, 27.78786, 1.403088, 12.70107, 3.528616, 15.93435, 0.3767946, 0.2953438, 0.1079174, 1.

Solver flag: True; evaluations: 43; residual: 1.44058e-10; physiological: True.

Row/column	1	2	3	4	5	6	7	8	9	10	11
1	-27.57	-6.7103	-9.5731	-6.7581	10.988	0.068718	0.08223	-0.089494	0.00016045	0.016809	-0.029947
2	-19.352	-8.5107	-19.251	-7.8983	15.852	-1.9943	7.1875	4.7252	-0.0046565	-4.8136	-0.029947
3	-37.473	-8.5385	-36.05	7.5272	24.483	-2.063	7.1028	4.8146	-0.004817	-4.8298	0.059893
4	-4.9221	-2.2574	2.8011	-28.115	2.4655	-6.1558	-6.1263	6.1142	6.297	0.00021677	-0.11979
5	5412	5412	5412	5412	-11242	-394.89	-394.89	-394.89	-394.89	821.19	0
6	2.4779	4.4412	11.46	-3.6867	-7.9501	-22.437	-0.45299	-1.0169	-0.0059141	-132.95	1.2622e-24
7	-2.3693	41.123	85.364	0.32794	-62.415	-4.2275	-154.67	-24.558	-0.0098709	21.006	0
8	-0.058816	46.304	98.734	-3.9732	-71.618	30.56	-97.557	-83.391	0.071356	-108.96	-2.5244e-24
9	-11.781	-12.688	10.038	20.291	0.69692	43.326	42.983	-42.735	-100.69	-3.4364	-2.0195e-23
10	-5776	-5776	-5776	-5776	11998	5773.4	5773.4	5773.4	5773.4	-12194	-2.5849e-21
11	0	0	0	0	0	0	0	0	0	0	-31.273

Eigenvalues: -0.0028304045; -0.004290714; -0.029772028; -0.033333333; -0.03602262; -0.061676038; -0.10795563; -0.13607934; -0.2; -9.0535326; -15.839225.

C0, expression 0.5

State: 25.29149, 158.6214, 75.52913, 14.79375, 30.32204, 1.424133, 12.68636, 3.539572, 15.92946, 0.383044, 0.2964781, 0.1079096, 1.

Solver flag: True; evaluations: 62; residual: 4.25642e-12; physiological: True.

Row/column	1	2	3	4	5	6	7	8	9	10	11
1	-27.276	-6.1015	-10.17	-5.4143	10.914	0.078118	0.093743	-0.10186	0.0001656	0.019151	-0.024487
2	-19.31	-8.184	-19.395	-6.246	15.683	-1.9768	7.177	4.7061	-0.0041905	-4.8118	-0.024487
3	-38.186	-8.9174	-35.212	4.4294	24.45	-2.0549	7.0809	4.808	-0.0043561	-4.8304	0.048973
4	-4.0915	-1.1388	1.4177	-21.711	2.2643	-6.1382	-6.1045	6.0907	6.2981	0.0028436	-0.097946
5	5332.1	5332.1	5332.1	5332.1	-11076	-394.92	-394.92	-394.92	-394.92	821.25	0
6	2.3711	4.2802	11.372	-3.5866	-7.7735	-22.414	-0.44149	-1.0381	-0.0053304	-132.8	-3.1554e-25
7	-2.3995	40.27	84.361	0.31904	-61.492	-4.2095	-154.22	-24.584	-0.0089235	20.882	-1.5777e-25
8	-0.2244	45.264	97.628	-3.8654	-70.489	30.433	-97.265	-83.271	0.064514	-108.93	1.2622e-24
9	-12.541	-13.45	10.841	20.331	0.64082	43.499	43.156	-42.907	-100.68	-3.4783	0
10	-5690.7	-5690.7	-5690.7	-5690.7	11821	5773.8	5773.8	5773.8	5773.8	-12195	0
11	0	0	0	0	0	0	0	0	0	0	-31.273

Eigenvalues: -0.0027212505; -0.0040421657; -0.023337525; -0.033333333; -0.034848527; -0.061481273; -0.10785004; -0.13556203; -0.2; -8.9772761; -15.739084.

C0, expression 0.25

State: 25.54177, 164.3879, 77.052, 18.4336, 34.81592, 1.461251, 12.66495, 3.554817, 15.92113, 0.3915857, 0.2986369, 0.1078902, 1.

Solver flag: True; evaluations: 80; residual: 1.83918e-10; physiological: True.

Row/column	1	2	3	4	5	6	7	8	9	10	11
1	-26.9	-5.3438	-10.855	-4.1987	10.802	0.089305	0.10754	-0.11671	0.00017022	0.022038	-0.015417
2	-19.278	-7.7789	-19.445	-4.6442	15.416	-1.9495	7.1655	4.6773	-0.0037157	-4.8101	-0.015417
3	-38.82	-9.1408	-34.264	1.948	24.238	-2.0388	7.0556	4.794	-0.0038859	-4.8316	0.030833
4	-3.3209	0.0041049	-0.02268	-15.988	2.1063	-6.1177	-6.0788	6.0629	6.2999	0.0060414	-0.061666
5	5196.6	5196.6	5196.6	5196.6	-10794	-394.99	-394.99	-394.99	-394.99	821.4	5.1699e-21
6	2.282	4.0702	11.178	-3.2578	-7.5081	-22.363	-0.42117	-1.0722	-0.0047355	-132.61	0
7	-2.4427	38.863	82.691	0.28979	-59.927	-4.1747	-153.58	-24.633	-0.007957	20.709	-7.8886e-26
8	-0.3853	43.611	95.732	-3.511	-68.614	30.185	-96.933	-83.021	0.057534	-108.88	-3.1554e-25
9	-13.624	-14.527	11.986	20.309	0.56915	43.827	43.486	-43.235	-100.66	-3.5394	2.0195e-23
10	-5546.1	-5546.1	-5546.1	-5546.1	11520	5774.8	5774.8	5774.8	5774.8	-12197	-5.1699e-21
11	0	0	0	0	0	0	0	0	0	0	-31.273

Eigenvalues: -0.0025231865; -0.0037184213; -0.016704518; -0.033333333; -0.034634893; -0.06111238; -0.10773076; -0.13483578; -0.2; -8.8430546; -15.575308.

C0, expression 0.1

State: 25.87486, 170.2099, 78.60007, 22.22366, 39.42291, 1.49941, 12.64779, 3.566471, 15.91344, 0.3978245, 0.3008224, 0.1078675, 1.

Solver flag: True; evaluations: 90; residual: 9.45216e-11; physiological: True.

Row/column	1	2	3	4	5	6	7	8	9	10	11
1	-26.522	-4.8001	-11.278	-3.4447	10.662	0.094668	0.11427	-0.124	0.00017025	0.023546	-0.0071466
2	-19.154	-7.4443	-19.39	-3.6526	15.136	-1.9272	7.1576	4.655	-0.003466	-4.8097	-0.0071466
3	-39.018	-9.1236	-33.566	0.89382	23.917	-2.0219	7.041	4.779	-0.0036362	-4.8328	0.014293
4	-2.7725	0.73205	-0.95506	-12.895	2.0121	-6.1087	-6.0671	6.05	6.3015	0.0074147	-0.028586
5	5064.4	5064.4	5064.4	5064.4	-10520	-395.07	-395.07	-395.07	-395.07	821.57	-1.2925e-21
6	2.2885	3.9239	10.946	-2.8419	-7.2873	-22.309	-0.4032	-1.1008	-0.0044217	-132.48	-1.5777e-25
7	-2.473	37.534	81.063	0.25279	-58.398	-4.1393	-153.06	-24.681	-0.0074441	20.576	3.1554e-25
8	-0.41605	42.112	93.833	-3.0628	-66.827	29.932	-96.748	-82.755	0.053829	-108.86	-6.3109e-25
9	-14.507	-15.392	12.914	20.249	0.52208	44.158	43.818	-43.564	-100.65	-3.588	-1.0097e-23
10	-5405	-5405	-5405	-5405	11227	5776.1	5776.1	5776.1	5776.1	-12199	0
11	0	0	0	0	0	0	0	0	0	0	-31.273

Eigenvalues: -0.0023387148; -0.0034983584; -0.013004551; -0.033333333; -0.034512761; -0.060739148; -0.10764851; -0.13425348; -0.2; -8.7055549; -15.422196.

C0, expression 0.05

State: 26.04863, 172.8354, 79.3086, 23.9684, 41.51367, 1.51685, 12.64123, 3.570821, 15.91029, 0.4000405, 0.3017623, 0.1078572, 1.

Solver flag: True; evaluations: 62; residual: 2.25131e-10; physiological: True.

Row/column	1	2	3	4	5	6	7	8	9	10	11
1	-26.335	-4.6025	-11.407	-3.1677	10.588	0.095541	0.11542	-0.12526	0.00016905	0.023848	-0.0037563
2	-19.061	-7.3045	-19.345	-3.3048	15.002	-1.919	7.1544	4.6471	-0.0033955	-4.8099	-0.0037563
3	-39.011	-9.077	-33.286	0.62796	23.75	-2.0146	7.0367	4.7724	-0.0035646	-4.8332	0.0075125
4	-2.5367	0.99207	-1.2882	-11.909	1.9719	-6.1077	-6.0656	6.0482	6.3022	0.0075004	-0.015025
5	5006.2	5006.2	5006.2	5006.2	-10399	-395.11	-395.11	-395.11	-395.11	821.64	2.5849e-21
6	2.3129	3.8724	10.834	-2.6506	-7.2001	-22.285	-0.39632	-1.1117	-0.0043327	-132.43	-6.3109e-25
7	-2.4823	36.963	80.34	0.23578	-57.724	-4.1241	-152.87	-24.702	-0.0072972	20.525	-3.1554e-25
8	-0.39886	41.481	92.98	-2.8566	-66.051	29.822	-96.694	-82.639	0.052767	-108.85	3.1554e-25
9	-14.857	-15.732	13.281	20.227	0.50679	44.299	43.959	-43.705	-100.64	-3.6063	2.0195e-23
10	-5342.8	-5342.8	-5342.8	-5342.8	11098	5776.6	5776.6	5776.6	5776.6	-12200	0
11	0	0	0	0	0	0	0	0	0	0	-31.273

Eigenvalues: -0.0022619877; -0.0034253228; -0.011809039; -0.033333333; -0.034431621; -0.060579032; -0.10761986; -0.13402661; -0.2; -8.642952; -15.356859.

C0, expression 0.0

State: 26.27248, 175.9622, 80.16363, 26.07319, 44.0093, 1.537802, 12.6342, 3.575436, 15.90681, 0.4022997, 0.3028266, 0.1078454, 1.

Solver flag: True; evaluations: 66; residual: 1.24269e-11; physiological: True.

Row/column	1	2	3	4	5	6	7	8	9	10	11
1	-26.098	-4.396	-11.523	-2.8703	10.491	0.095614	0.11561	-0.12549	0.00016683	0.023966	0
2	-18.924	-7.1443	-19.277	-2.9448	14.838	-1.9106	7.1506	4.6393	-0.0033337	-4.8104	-3.9443e-26
3	-38.942	-9.003	-32.966	0.40619	23.539	-2.0062	7.0327	4.7648	-0.0035005	-4.8338	0
4	-2.2542	1.2664	-1.638	-10.941	1.9222	-6.1083	-6.066	6.0485	6.3029	0.0072486	-3.1554e-25
5	4937.9	4937.9	4937.9	4937.9	-10257	-395.15	-395.15	-395.15	-395.15	821.73	0
6	2.3531	3.819	10.698	-2.4291	-7.1041	-22.259	-0.38918	-1.123	-0.0042541	-132.38	-6.3109e-25
7	-2.49	36.305	79.485	0.21607	-56.935	-4.107	-152.66	-24.725	-0.0071658	20.472	1.5777e-25
8	-0.36073	40.761	91.966	-2.6179	-65.15	29.698	-96.645	-82.506	0.051818	-108.84	6.3109e-25
9	-15.246	-16.106	13.687	20.211	0.49211	44.458	44.119	-43.864	-100.63	-3.6256	0
10	-5270	-5270	-5270	-5270	10947	5777.2	5777.2	5777.2	5777.2	-12202	0
11	0	0	0	0	0	0	0	0	0	0	-31.273

Eigenvalues: -0.0021743099; -0.0033552801; -0.010624601; -0.033333333; -0.034309271; -0.060398218; -0.10759125; -0.13377908; -0.2; -8.5679955; -15.281905.

C1, expression 1.0

State: 56.38905, 98.17567, 51.77597, 10.63786, 24.72699, 1.234067, 13.70917, 2.132253, 15.38673, 0.5436435, 0.4546967, 0.1054083, 1.

Solver flag: True; evaluations: 388; residual: 1.39371e-12; physiological: False.

Row/column	1	2	3	4	5	6	7	8	9	10	11
1	-9.6225	-5.2232	-17.414	4.9034	10.181	0.052284	0.066621	-0.077882	0.00023194	0.022206	0.076324
2	-11.409	-14.186	-24.282	-18.05	18.035	-1.1934	14.244	3.9813	-0.005294	-4.9246	-0.076324
3	-13.678	-12.43	-48.676	12.197	25.464	-1.2457	14.165	4.0592	-0.0055259	-4.9451	0.076324
4	-3.0969	-2.7811	2.7834	-31.565	2.8808	-6.3329	-6.2919	6.2817	6.4424	0.005064	-0.076324
5	6153.3	6153.3	6153.3	6153.3	-12781	-404.29	-404.29	-404.29	-404.29	840.41	0
6	-1.3159	7.0523	16.529	-5.6312	-8.3525	-16.834	2.2819	-2.205	-0.0067429	-151.3	-3.1554e-25
7	0.038924	65.052	123.07	0.5009	-71.302	-2.551	-237.91	-27.334	-0.011317	36.818	0
8	-1.5174	73.28	142.35	-6.0689	-81.04	18.445	-197.17	-67.989	0.081823	-109.49	-2.5244e-24
9	-12.42	-13.835	9.9052	23.618	0.7084	60.572	59.953	-59.951	-98.563	-5.5088	9.8608e-27
10	-6567.1	-6567.1	-6567.1	-6567.1	13641	5910.8	5910.8	5910.8	5910.8	-12475	2.0195e-23
11	0	0	0	0	0	0	0	0	0	0	-31.273

Eigenvalues: -0.0017718692; -0.0037623636; -0.024428354; -0.033333333; -0.039145477; -0.042851348; -0.10646676; -0.2; -0.22910833; -9.7789807; -17.053204.

C1, expression 0.05

State: 27.34018, 169.0834, 76.82347, 23.94153, 41.53835, 1.500344, 12.71311, 3.473817, 15.87763, 0.4081329, 0.3093046, 0.1076952, 1.

Solver flag: True; evaluations: 69; residual: 5.39494e-11; physiological: True.

Row/column	1	2	3	4	5	6	7	8	9	10	11
1	-24.594	-4.5518	-11.927	-2.9379	10.59	0.093248	0.11312	-0.12302	0.00016983	0.023993	0.0076448
2	-18.479	-7.4636	-19.88	-3.4145	15.175	-1.8658	7.472	4.5969	-0.0033982	-4.8171	-0.0076448
3	-36.958	-9.252	-34.227	0.62964	23.916	-1.9591	7.3562	4.72	-0.003568	-4.8405	0.0076448
4	-2.2707	1.0557	-1.3989	-11.796	1.9825	-6.1216	-6.0792	6.0621	6.3116	0.0074018	-0.0076448
5	5061.2	5061.2	5061.2	5061.2	-10513	-395.71	-395.71	-395.71	-395.71	822.86	-6.4623e-22
6	1.6834	3.9706	11.166	-2.6656	-7.2157	-21.921	-0.25606	-1.19	-0.0043363	-133.65	1.5777e-25
7	-2.0666	37.782	82.941	0.2371	-58.397	-4.0105	-156.24	-24.88	-0.0073041	21.579	0
8	-0.62041	42.415	95.967	-2.8727	-66.75	29	-101.22	-81.597	0.052817	-108.91	-1.0097e-23
9	-14.958	-15.856	13.334	20.367	0.50762	45.466	45.115	-44.871	-100.51	-3.7048	6.3109e-25
10	-5401.6	-5401.6	-5401.6	-5401.6	11220	5785.3	5785.3	5785.3	5785.3	-12218	6.4623e-22
11	0	0	0	0	0	0	0	0	0	0	-31.273

Eigenvalues: -0.0021929936; -0.0033742164; -0.011373241; -0.033333333; -0.033747215; -0.059222843; -0.10757387; -0.13779196; -0.2; -8.7071946; -15.433508.

C1, expression 0.0

State: 26.27248, 175.9622, 80.16363, 26.07319, 44.0093, 1.537802, 12.6342, 3.575436, 15.90681, 0.4022997, 0.3028266, 0.1078454, 1.

Solver flag: True; evaluations: 14; residual: 1.89753e-12; physiological: True.

Row/column	1	2	3	4	5	6	7	8	9	10	11
1	-26.098	-4.396	-11.523	-2.8703	10.491	0.095614	0.11561	-0.12549	0.00016683	0.023966	0
2	-18.924	-7.1443	-19.277	-2.9448	14.838	-1.9106	7.1506	4.6393	-0.0033337	-4.8104	9.8608e-27
3	-38.942	-9.003	-32.966	0.40619	23.539	-2.0062	7.0327	4.7648	-0.0035005	-4.8338	0
4	-2.2542	1.2664	-1.638	-10.941	1.9222	-6.1083	-6.066	6.0485	6.3029	0.0072486	3.9443e-26
5	4937.9	4937.9	4937.9	4937.9	-10257	-395.15	-395.15	-395.15	-395.15	821.73	0
6	2.3531	3.819	10.698	-2.4291	-7.1041	-22.259	-0.38918	-1.123	-0.0042541	-132.38	1.5777e-25
7	-2.49	36.305	79.485	0.21607	-56.935	-4.107	-152.66	-24.725	-0.0071659	20.472	0
8	-0.36073	40.761	91.966	-2.6179	-65.15	29.698	-96.645	-82.506	0.051818	-108.84	0
9	-15.246	-16.106	13.687	20.211	0.49211	44.458	44.119	-43.864	-100.63	-3.6256	1.5777e-25
10	-5270	-5270	-5270	-5270	10947	5777.2	5777.2	5777.2	5777.2	-12202	0
11	0	0	0	0	0	0	0	0	0	0	-31.273

Eigenvalues: -0.0021743099; -0.0033552801; -0.010624601; -0.033333333; -0.034309271; -0.060398218; -0.10759125; -0.13377908; -0.2; -8.5679955; -15.281905.

C2, expression 1.0

State: 69.1451, 75.86267, 50.88861, 11.83347, 16.0574, 1.20337, 13.8998, 1.744818, 15.10988, 0.7935139, 0.5347388, 0.1046778, 1.

Solver flag: True; evaluations: 778; residual: 3.77027e-09; physiological: False.

Row/column	1	2	3	4	5	6	7	8	9	10	11
1	-8.7457	-7.0546	-16.864	4.7368	10.39	0.040935	0.050812	-0.065433	3.7518e-05	0.021809	0.054825
2	-18.828	-27.199	-14.529	-20.281	19.142	-0.97993	17.958	3.8104	-0.00089815	-4.9601	-0.10965
3	-7.6652	-13.508	-52.104	8.4699	25.936	-1.0209	17.885	3.8758	-0.00093567	-4.9794	0.054825
4	-5.0978	-5.5436	5.1181	-29.84	3.1331	-6.4123	-6.3703	6.3633	6.4965	-0.069779	-0.054825
5	6310.3	6310.3	6310.3	6310.3	-13107	-407.11	-407.11	-407.11	-407.11	846.18	8.2718e-20
6	-2.6492	8.0202	18.178	-3.8707	-8.5748	-15.241	3.5366	-2.5572	-0.0011433	-155.74	8.0779e-23
7	0.20336	84.284	125.09	0.34431	-73.133	-2.0909	-286.21	-28.362	-0.0019164	41.48	4.039e-23
8	-2.8961	93.64	146.3	-4.1715	-83.133	15.117	-249.45	-64.003	0.013856	-108.25	3.2312e-22
9	-31.358	-33.145	28.779	53.272	-0.5791	65.492	64.72	-64.858	-97.989	-7.5366	3.2312e-22
10	-6734.6	-6734.6	-6734.6	-6734.6	13988	5952.1	5952.1	5952.1	5952.1	-12559	8.2718e-20
11	0	0	0	0	0	0	0	0	0	0	-31.273

Eigenvalues: -0.0019979069; -0.01003652; -0.025845605; -0.033333333; -0.035377417; -0.042900643; -0.10756814; -0.2; -0.28247149; -9.9289579; -17.341529.

C2, expression 0.05

State: 27.85071, 163.3548, 74.86581, 23.12077, 38.27794, 1.473796, 12.7553, 3.402883, 15.84418, 0.4404082, 0.3140022, 0.1075976, 1.

Solver flag: True; evaluations: 104; residual: 7.75692e-11; physiological: True.

Row/column	1	2	3	4	5	6	7	8	9	10	11
1	-24.747	-5.4665	-11.422	-2.3523	10.727	0.091498	0.11128	-0.12117	0.0001157	0.023973	0.0074566
2	-18.509	-7.9068	-20.122	-3.6234	15.464	-1.8355	7.7047	4.5707	-0.0023209	-4.8215	-0.014913
3	-36.132	-9.5336	-35.068	0.63855	24.289	-1.927	7.5906	4.6919	-0.0024366	-4.8449	0.0074566
4	-2.0311	1.1913	-1.5486	-12.613	2.0531	-6.1328	-6.0904	6.0734	6.3187	-0.0039155	-0.0074566
5	5152.4	5152.4	5152.4	5152.4	-10702	-396.06	-396.06	-396.06	-396.06	823.59	4.039e-23
6	1.2495	4.0137	11.559	-2.6764	-7.3112	-21.704	-0.16214	-1.2385	-0.0029616	-134.37	0
7	-1.8488	39.116	85.101	0.23807	-59.47	-3.9448	-158.9	-25.011	-0.0049881	22.356	-2.5244e-24
8	-0.86046	43.799	98.586	-2.8844	-67.939	28.526	-104.52	-81.02	0.03607	-108.8	-2.0195e-23
9	-17.167	-18.097	15.501	23.955	0.33794	46.171	45.811	-45.575	-100.43	-3.9185	2.0195e-23
10	-5498.9	-5498.9	-5498.9	-5498.9	11422	5790.5	5790.5	5790.5	5790.5	-12229	1.2925e-21
11	0	0	0	0	0	0	0	0	0	0	-31.273

Eigenvalues: -0.0022991984; -0.0044018205; -0.01180514; -0.033333333; -0.033658696; -0.05841514; -0.10760509; -0.14095712; -0.2; -8.8066197; -15.54779.

C2, expression 0.0

State: 26.27248, 175.9622, 80.16363, 26.07319, 44.0093, 1.537802, 12.6342, 3.575436, 15.90681, 0.4022997, 0.3028266, 0.1078454, 1.

Solver flag: True; evaluations: 14; residual: 1.89753e-12; physiological: True.

Row/column	1	2	3	4	5	6	7	8	9	10	11
1	-26.098	-4.396	-11.523	-2.8703	10.491	0.095614	0.11561	-0.12549	0.00016683	0.023966	0
2	-18.924	-7.1443	-19.277	-2.9448	14.838	-1.9106	7.1506	4.6393	-0.0033337	-4.8104	9.8608e-27
3	-38.942	-9.003	-32.966	0.40619	23.539	-2.0062	7.0327	4.7648	-0.0035005	-4.8338	0
4	-2.2542	1.2664	-1.638	-10.941	1.9222	-6.1083	-6.066	6.0485	6.3029	0.0072486	3.9443e-26
5	4937.9	4937.9	4937.9	4937.9	-10257	-395.15	-395.15	-395.15	-395.15	821.73	0
6	2.3531	3.819	10.698	-2.4291	-7.1041	-22.259	-0.38918	-1.123	-0.0042541	-132.38	1.5777e-25
7	-2.49	36.305	79.485	0.21607	-56.935	-4.107	-152.66	-24.725	-0.0071659	20.472	0
8	-0.36073	40.761	91.966	-2.6179	-65.15	29.698	-96.645	-82.506	0.051818	-108.84	0
9	-15.246	-16.106	13.687	20.211	0.49211	44.458	44.119	-43.864	-100.63	-3.6256	1.5777e-25
10	-5270	-5270	-5270	-5270	10947	5777.2	5777.2	5777.2	5777.2	-12202	0
11	0	0	0	0	0	0	0	0	0	0	-31.273

Eigenvalues: -0.0021743099; -0.0033552801; -0.010624601; -0.033333333; -0.034309271; -0.060398218; -0.10759125; -0.13377908; -0.2; -8.5679955; -15.281905.

C3a, expression 1.0

State: 32.01803, 133.5549, 63.06404, 10.14596, 24.44714, 1.306437, 13.08013, 3.023995, 15.76924, 0.4126252, 0.3348781, 0.1070862, 1.

Solver flag: True; evaluations: 189; residual: 1.78632e-10; physiological: True.

Row/column	1	2	3	4	5	6	7	8	9	10	11
1	-20.267	-6.7935	-12.101	-4.2029	10.997	0.05692	0.069151	-0.076562	0.0001794	0.016334	6.3109e-25
2	-17.611	-11.03	-20.788	-17.246	17.374	-1.7309	9.0465	4.4784	-0.0054553	-4.8517	-0.063278
3	-27.632	-9.2267	-42.342	12.608	25.49	-1.7878	8.9731	4.5549	-0.0056347	-4.8672	0.063278
4	-5.5612	-4.0236	4.8792	-40.445	2.9848	-6.2258	-6.1971	6.1865	6.3439	-9.402e-06	-0.12656
5	5812.4	5812.4	5812.4	5812.4	-12073	-397.96	-397.96	-397.96	-397.96	827.46	-5.1699e-21
6	0.093869	5.226	13.497	-4.5063	-8.2717	-20.574	0.3192	-1.4052	-0.0069253	-139.37	-5.0487e-24
7	-0.80888	47.816	101.05	0.40084	-67.207	-3.6639	-175.44	-25.45	-0.011548	26.558	2.0195e-23
8	-0.9156	53.913	116.79	-4.8565	-76.82	26.486	-123.98	-78.179	0.083475	-109.33	-8.0779e-23
9	-11.125	-12.197	9.0763	21.451	0.77545	49.118	48.704	-48.518	-99.998	-3.9173	0
10	-6203.3	-6203.3	-6203.3	-6203.3	12885	5818.2	5818.2	5818.2	5818.2	-12285	0
11	0	0	0	0	0	0	0	0	0	0	-31.273

Eigenvalues: -0.0023191439; -0.0044009787; -0.029600937; -0.033333333; -0.047709669; -0.055058198; -0.10778138; -0.15957077; -0.2; -9.4383413; -16.439217.

C3a, expression 0.05

State: 26.5854, 170.2896, 77.8413, 23.49024, 40.97197, 1.503722, 12.67872, 3.521334, 15.89483, 0.4034743, 0.305228, 0.1077789, 1.

Solver flag: True; evaluations: 62; residual: 7.20333e-12; physiological: True.

Row/column	1	2	3	4	5	6	7	8	9	10	11
1	-25.594	-4.6376	-11.646	-3.1461	10.628	0.094324	0.11416	-0.12401	0.00016992	0.023878	-7.8886e-26
2	-18.859	-7.4317	-19.638	-3.4754	15.142	-1.8945	7.312	4.624	-0.0034129	-4.8134	-0.0074875
3	-38.122	-9.1993	-33.853	0.64413	23.914	-1.9888	7.1954	4.748	-0.0035828	-4.8368	0.0074875
4	-2.4762	0.95707	-1.2573	-12.1	1.9894	-6.1146	-6.0724	6.0552	6.3067	0.007492	-0.014975
5	5049.9	5049.9	5049.9	5049.9	-10489	-395.4	-395.4	-395.4	-395.4	822.23	1.6156e-22
6	1.999	3.9331	11.033	-2.7108	-7.2326	-22.109	-0.3285	-1.1481	-0.0043548	-133.05	-3.9443e-26
7	-2.2794	37.516	81.855	0.24113	-58.246	-4.0714	-154.58	-24.786	-0.0073344	21.063	-1.5777e-25
8	-0.51491	42.104	94.727	-2.9215	-66.615	29.441	-98.938	-82.149	0.053036	-108.88	-1.5777e-25
9	-14.817	-15.707	13.213	20.304	0.51099	44.844	44.499	-44.25	-100.57	-3.6502	3.1554e-25
10	-5389.5	-5389.5	-5389.5	-5389.5	11195	5780.8	5780.8	5780.8	5780.8	-12209	-8.0779e-23
11	0	0	0	0	0	0	0	0	0	0	-31.273

Eigenvalues: -0.0022511308; -0.0034199841; -0.011897649; -0.033333333; -0.034183751; -0.059944004; -0.10760606; -0.13593533; -0.2; -8.6925696; -15.413258.

C3a, expression 0.0

State: 26.27248, 175.9622, 80.16363, 26.07319, 44.0093, 1.537802, 12.6342, 3.575436, 15.90681, 0.4022997, 0.3028266, 0.1078454, 1.

Solver flag: True; evaluations: 14; residual: 1.89753e-12; physiological: True.

Row/column	1	2	3	4	5	6	7	8	9	10	11
1	-26.098	-4.396	-11.523	-2.8703	10.491	0.095614	0.11561	-0.12549	0.00016683	0.023966	0
2	-18.924	-7.1443	-19.277	-2.9448	14.838	-1.9106	7.1506	4.6393	-0.0033337	-4.8104	9.8608e-27
3	-38.942	-9.003	-32.966	0.40619	23.539	-2.0062	7.0327	4.7648	-0.0035005	-4.8338	0
4	-2.2542	1.2664	-1.638	-10.941	1.9222	-6.1083	-6.066	6.0485	6.3029	0.0072486	3.9443e-26
5	4937.9	4937.9	4937.9	4937.9	-10257	-395.15	-395.15	-395.15	-395.15	821.73	0
6	2.3531	3.819	10.698	-2.4291	-7.1041	-22.259	-0.38918	-1.123	-0.0042541	-132.38	1.5777e-25
7	-2.49	36.305	79.485	0.21607	-56.935	-4.107	-152.66	-24.725	-0.0071659	20.472	0
8	-0.36073	40.761	91.966	-2.6179	-65.15	29.698	-96.645	-82.506	0.051818	-108.84	0
9	-15.246	-16.106	13.687	20.211	0.49211	44.458	44.119	-43.864	-100.63	-3.6256	1.5777e-25
10	-5270	-5270	-5270	-5270	10947	5777.2	5777.2	5777.2	5777.2	-12202	0
11	0	0	0	0	0	0	0	0	0	0	-31.273

Eigenvalues: -0.0021743099; -0.0033552801; -0.010624601; -0.033333333; -0.034309271; -0.060398218; -0.10759125; -0.13377908; -0.2; -8.5679955; -15.281905.

C3b, expression 1.0

State: 21.00716, 170.8769, 85.82469, 12.97284, 27.99761, 1.478723, 12.39015, 3.943997, 16.06584, 0.346365, 0.2683033, 0.1085876, 1.

Solver flag: True; evaluations: 163; residual: 4.81113e-11; physiological: True.

Row/column	1	2	3	4	5	6	7	8	9	10	11
1	-35.826	-7.5925	-6.5779	-11.287	11.029	0.065096	0.076786	-0.082947	0.00014006	0.014174	-0.069313
2	-20.293	-7.0706	-17.699	-3.7324	14.661	-2.2304	5.9399	4.9525	-0.0047989	-4.786	0
3	-45.358	-7.3001	-32.351	9.5362	23.331	-2.2955	5.8615	5.0355	-0.004939	-4.7998	0.069313
4	-6.121	-2.8419	3.5534	-30.473	2.4579	-6.1246	-6.0996	6.0889	6.2585	-0.0031286	-0.13863
5	5135.2	5135.2	5135.2	5135.2	-10667	-392.45	-392.45	-392.45	-392.45	816.21	-6.4623e-22
6	5.9976	4.0945	9.9492	-3.0788	-7.8718	-23.96	-1.0272	-0.67649	-0.0060852	-127.85	0
7	-4.4651	37.371	74.246	0.27386	-59.061	-4.7055	-141.92	-23.822	-0.010124	16.557	0
8	1.4702	42.148	85.853	-3.3181	-68.14	34.012	-79.967	-87.76	0.073181	-108.65	-1.6156e-22
9	-11.429	-12.173	9.8328	19.311	0.69944	38.416	38.118	-37.832	-101.23	-3.0767	1.0097e-23
10	-5480.6	-5480.6	-5480.6	-5480.6	11384	5737.8	5737.8	5737.8	5737.8	-12121	0
11	0	0	0	0	0	0	0	0	0	0	-31.273

Eigenvalues: -0.0031785847; -0.0044616681; -0.033333333; -0.036673046 + 0.0028277923i; -0.036673046 - 0.0028277923i; -0.067347536; -0.10790701; -0.12204811; -0.2; -8.7581719; -15.443567.

C3b, expression 0.05

State: 25.54313, 175.3595, 80.78407, 24.44984, 42.05678, 1.530048, 12.60442, 3.619287, 15.92525, 0.3967677, 0.2984586, 0.1079334, 1.

Solver flag: True; evaluations: 60; residual: 9.00951e-11; physiological: True.

Row/column	1	2	3	4	5	6	7	8	9	10	11
1	-27.074	-4.5682	-11.176	-3.186	10.547	0.096726	0.11665	-0.12648	0.00016824	0.02382	-0.0075359
2	-19.251	-7.1819	-19.06	-3.1427	14.864	-1.943	7.0037	4.6697	-0.0033795	-4.8065	0
3	-39.885	-8.959	-32.734	0.61238	23.587	-2.0397	6.8849	4.7962	-0.0035478	-4.8298	0.0075359
4	-2.5964	1.026	-1.3184	-11.724	1.9546	-6.1009	-6.0589	6.0414	6.2978	0.0075086	-0.015072
5	4963	4963	4963	4963	-10309	-394.83	-394.83	-394.83	-394.83	821.07	2.5849e-21
6	2.6339	3.814	10.641	-2.5924	-7.1676	-22.457	-0.46164	-1.0762	-0.0043123	-131.83	-1.5777e-25
7	-2.6893	36.431	78.872	0.2306	-57.208	-4.1755	-151.24	-24.621	-0.0072628	20	-3.1554e-25
8	-0.27956	40.881	91.287	-2.7939	-65.494	30.194	-94.547	-83.116	0.052518	-108.81	5.0487e-24
9	-14.897	-15.756	13.347	20.153	0.50266	43.767	43.433	-43.174	-100.7	-3.5645	-1.0097e-23
10	-5296.7	-5296.7	-5296.7	-5296.7	11002	5772.5	5772.5	5772.5	5772.5	-12192	-1.2925e-21
11	0	0	0	0	0	0	0	0	0	0	-31.273

Eigenvalues: -0.0022696463; -0.0034289782; -0.011722463; -0.033333333; -0.034700399; -0.0611975; -0.10762728; -0.13221825; -0.2; -8.5933834; -15.301834.

C3b, expression 0.0

State: 26.27248, 175.9622, 80.16363, 26.07319, 44.0093, 1.537802, 12.6342, 3.575436, 15.90681, 0.4022997, 0.3028266, 0.1078454, 1.

Solver flag: True; evaluations: 14; residual: 1.89753e-12; physiological: True.

Row/column	1	2	3	4	5	6	7	8	9	10	11
1	-26.098	-4.396	-11.523	-2.8703	10.491	0.095614	0.11561	-0.12549	0.00016683	0.023966	0
2	-18.924	-7.1443	-19.277	-2.9448	14.838	-1.9106	7.1506	4.6393	-0.0033337	-4.8104	9.8608e-27
3	-38.942	-9.003	-32.966	0.40619	23.539	-2.0062	7.0327	4.7648	-0.0035005	-4.8338	0
4	-2.2542	1.2664	-1.638	-10.941	1.9222	-6.1083	-6.066	6.0485	6.3029	0.0072486	3.9443e-26
5	4937.9	4937.9	4937.9	4937.9	-10257	-395.15	-395.15	-395.15	-395.15	821.73	0
6	2.3531	3.819	10.698	-2.4291	-7.1041	-22.259	-0.38918	-1.123	-0.0042541	-132.38	1.5777e-25
7	-2.49	36.305	79.485	0.21607	-56.935	-4.107	-152.66	-24.725	-0.0071659	20.472	0
8	-0.36073	40.761	91.966	-2.6179	-65.15	29.698	-96.645	-82.506	0.051818	-108.84	0
9	-15.246	-16.106	13.687	20.211	0.49211	44.458	44.119	-43.864	-100.63	-3.6256	1.5777e-25
10	-5270	-5270	-5270	-5270	10947	5777.2	5777.2	5777.2	5777.2	-12202	0
11	0	0	0	0	0	0	0	0	0	0	-31.273

Eigenvalues: -0.0021743099; -0.0033552801; -0.010624601; -0.033333333; -0.034309271; -0.060398218; -0.10759125; -0.13377908; -0.2; -8.5679955; -15.281905.

C4a, expression 1.0

State: 39.17136, 114.112, 58.4034, 12.87387, 16.81816, 1.259225, 13.27539, 2.637351, 15.53605, 0.6415983, 0.376686, 0.106448, 1.

Solver flag: True; evaluations: 352; residual: 1.45684e-11; physiological: False.

Row/column	1	2	3	4	5	6	7	8	9	10	11
1	-20.598	-11.047	-9.1342	-1.3757	11.217	0.057093	0.070798	-0.079622	3.351e-05	0.01894	0
2	-19.459	-16.869	-17.701	-11.022	17.907	-1.5061	10.905	4.2828	-0.00088397	-4.8788	-0.069944
3	-18.474	-7.7806	-47.697	7.7553	25.822	-1.5632	10.827	4.3625	-0.00091748	-4.8966	0.069944
4	-5.2542	-4.1058	4.2444	-27.076	2.8547	-6.2725	-6.2385	6.2274	6.3889	-0.072945	-0.069944
5	6030.4	6030.4	6030.4	6030.4	-12526	-400.34	-400.34	-400.34	-400.34	832.34	-5.0487e-24
6	-2.0835	4.8572	15.868	-3.1583	-8.3729	-19.154	1.0482	-1.7558	-0.0011239	-143.32	7.8886e-26
7	-0.15623	56.075	108.99	0.28093	-69.812	-3.2025	-197.19	-26.368	-0.0018797	30.928	0
8	-2.6792	61.741	127.51	-3.4037	-79.561	23.152	-150.15	-74.244	0.013589	-108.39	1.5777e-25
9	-31.775	-33.024	29.57	52.126	-0.61504	53.931	53.445	-53.319	-99.487	-5.5131	2.5244e-24
10	-6435.9	-6435.9	-6435.9	-6435.9	13368	5853.1	5853.1	5853.1	5853.1	-12357	0
11	0	0	0	0	0	0	0	0	0	0	-31.273

Eigenvalues: -0.0021943472; -0.010820533; -0.026367862; -0.033333333; -0.038991851; -0.04951702; -0.10850239; -0.18454082; -0.2; -9.6386489; -16.797767.

C4a, expression 0.05

State: 26.65423, 168.5598, 77.79886, 24.0927, 39.35344, 1.500067, 12.6772, 3.506215, 15.87704, 0.4332536, 0.3063804, 0.1077637, 1.

Solver flag: True; evaluations: 62; residual: 5.23861e-10; physiological: True.

Row/column	1	2	3	4	5	6	7	8	9	10	11
1	-26.236	-5.3837	-10.928	-2.4092	10.648	0.094063	0.11393	-0.1238	0.00011456	0.023909	0
2	-18.952	-7.6213	-19.515	-3.2738	15.176	-1.8872	7.3604	4.6194	-0.0022986	-4.8141	-0.0075117
3	-37.969	-9.268	-33.874	0.60571	23.957	-1.9813	7.2439	4.7432	-0.0024131	-4.8374	0.0075117
4	-2.1504	1.2694	-1.6165	-12.229	2.018	-6.118	-6.0758	6.0586	6.3091	-0.0040246	-0.0075117
5	5062.2	5062.2	5062.2	5062.2	-10515	-395.45	-395.45	-395.45	-395.45	822.34	0
6	1.8521	3.883	11.135	-2.5544	-7.2443	-22.076	-0.30834	-1.1617	-0.002933	-133.07	1.5777e-25
7	-2.2391	37.909	81.891	0.22721	-58.392	-4.0559	-155.12	-24.834	-0.00494	21.229	7.8886e-26
8	-0.63853	42.439	94.882	-2.7529	-66.775	29.329	-99.624	-82.052	0.035722	-108.72	0
9	-17.248	-18.147	15.644	23.759	0.32963	45.02	44.673	-44.425	-100.56	-3.8237	-1.6156e-22
10	-5402.6	-5402.6	-5402.6	-5402.6	11222	5781.6	5781.6	5781.6	5781.6	-12211	1.034e-20
11	0	0	0	0	0	0	0	0	0	0	-31.273

Eigenvalues: -0.0023471421; -0.0043851487; -0.01163729; -0.033333333; -0.034089148; -0.059754965; -0.10762392; -0.13674655; -0.2; -8.7062483; -15.428727.

C4a, expression 0.0

State: 26.27248, 175.9622, 80.16363, 26.07319, 44.0093, 1.537802, 12.6342, 3.575436, 15.90681, 0.4022997, 0.3028266, 0.1078454, 1.

Solver flag: True; evaluations: 14; residual: 1.89753e-12; physiological: True.

Row/column	1	2	3	4	5	6	7	8	9	10	11
1	-26.098	-4.396	-11.523	-2.8703	10.491	0.095614	0.11561	-0.12549	0.00016683	0.023966	0
2	-18.924	-7.1443	-19.277	-2.9448	14.838	-1.9106	7.1506	4.6393	-0.0033337	-4.8104	9.8608e-27
3	-38.942	-9.003	-32.966	0.40619	23.539	-2.0062	7.0327	4.7648	-0.0035005	-4.8338	0
4	-2.2542	1.2664	-1.638	-10.941	1.9222	-6.1083	-6.066	6.0485	6.3029	0.0072486	3.9443e-26
5	4937.9	4937.9	4937.9	4937.9	-10257	-395.15	-395.15	-395.15	-395.15	821.73	0
6	2.3531	3.819	10.698	-2.4291	-7.1041	-22.259	-0.38918	-1.123	-0.0042541	-132.38	1.5777e-25
7	-2.49	36.305	79.485	0.21607	-56.935	-4.107	-152.66	-24.725	-0.0071659	20.472	0
8	-0.36073	40.761	91.966	-2.6179	-65.15	29.698	-96.645	-82.506	0.051818	-108.84	0
9	-15.246	-16.106	13.687	20.211	0.49211	44.458	44.119	-43.864	-100.63	-3.6256	1.5777e-25
10	-5270	-5270	-5270	-5270	10947	5777.2	5777.2	5777.2	5777.2	-12202	0
11	0	0	0	0	0	0	0	0	0	0	-31.273

Eigenvalues: -0.0021743099; -0.0033552801; -0.010624601; -0.033333333; -0.034309271; -0.060398218; -0.10759125; -0.13377908; -0.2; -8.5679955; -15.281905.

C4b, expression 1.0

State: 22.66805, 160.1109, 83.63703, 17.07847, 21.08015, 1.454865, 12.43563, 3.771254, 15.9278, 0.5505061, 0.279085, 0.1083882, 1.

Solver flag: True; evaluations: 144; residual: 6.53858e-10; physiological: False.

Row/column	1	2	3	4	5	6	7	8	9	10	11
1	-39.416	-13.923	-0.47993	-4.1728	11.104	0.072571	0.086243	-0.093393	3.7508e-05	0.016532	-0.081856
2	-20.47	-8.6104	-16.814	-2.4762	14.882	-2.1266	6.4364	4.8656	-0.0010991	-4.7928	3.1554e-25
3	-38.845	-5.5328	-34.984	5.3947	23.374	-2.1992	6.3482	4.959	-0.0011366	-4.8089	0.081856
4	-4.5846	-1.5583	1.7589	-20.505	2.3046	-6.1265	-6.0972	6.0848	6.2744	-0.0758	-0.081856
5	5219.4	5219.4	5219.4	5219.4	-10842	-393.17	-393.17	-393.17	-393.17	817.69	1.2925e-21
6	3.4429	3.4443	11.139	-2.0425	-7.8285	-23.491	-0.79286	-0.8418	-0.0013958	-128.84	-1.5777e-25
7	-3.4486	39.966	76.105	0.18169	-60.098	-4.5066	-146.97	-24.276	-0.0023292	18.411	-6.3109e-25
8	-0.28069	43.985	89.101	-2.2013	-69.14	32.578	-86.872	-86.284	0.016838	-107.65	0
9	-30.67	-31.519	29.089	45.731	-0.57958	40.433	40.117	-39.843	-101.07	-4.2729	1.6156e-22
10	-5570.5	-5570.5	-5570.5	-5570.5	11571	5748.3	5748.3	5748.3	5748.3	-12142	1.034e-20
11	0	0	0	0	0	0	0	0	0	0	-31.273

Eigenvalues: -0.0031884057; -0.010473454; -0.02962534; -0.033213585; -0.033333333; -0.065089593; -0.10766528; -0.12936487; -0.2; -8.8500786; -15.561552.

C4b, expression 0.05

State: 25.598, 173.6653, 80.7659, 25.0776, 40.43568, 1.526629, 12.60206, 3.60516, 15.90774, 0.4267956, 0.2994836, 0.1079204, 1.

Solver flag: True; evaluations: 67; residual: 2.89099e-12; physiological: True.

Row/column	1	2	3	4	5	6	7	8	9	10	11
1	-27.716	-5.3029	-10.464	-2.454	10.565	0.096493	0.11644	-0.12629	0.00011366	0.023849	-0.0075593
2	-19.342	-7.3572	-18.941	-2.9587	14.894	-1.9362	7.0468	4.6657	-0.0022806	-4.8071	0
3	-39.748	-9.0217	-32.745	0.57532	23.627	-2.0327	6.9281	4.792	-0.0023943	-4.8304	0.0075593
4	-2.2674	1.3424	-1.6802	-11.871	1.9836	-6.1041	-6.062	6.0445	6.3	-0.0041261	-0.0075593
5	4974.1	4974.1	4974.1	4974.1	-10332	-394.88	-394.88	-394.88	-394.88	821.17	-8.0779e-23
6	2.4911	3.7627	10.738	-2.4406	-7.1784	-22.428	-0.44347	-1.0891	-0.0029101	-131.84	3.9443e-26
7	-2.6505	36.795	78.882	0.2171	-57.339	-4.1612	-151.73	-24.668	-0.0049014	20.155	0
8	-0.40031	41.185	91.409	-2.6303	-65.638	30.091	-95.158	-83.03	0.035443	-108.65	-3.9443e-26
9	-17.324	-18.192	15.776	23.573	0.32177	43.929	43.594	-43.336	-100.69	-3.738	0
10	-5308.6	-5308.6	-5308.6	-5308.6	11027	5773.2	5773.2	5773.2	5773.2	-12193	0
11	0	0	0	0	0	0	0	0	0	0	-31.273

Eigenvalues: -0.0023790986; -0.0043642238; -0.011479333; -0.033333333; -0.034609562; -0.061023623; -0.10761517; -0.13298557; -0.2; -8.6060273; -15.315437.

C4b, expression 0.0

State: 26.27248, 175.9622, 80.16363, 26.07319, 44.0093, 1.537802, 12.6342, 3.575436, 15.90681, 0.4022997, 0.3028266, 0.1078454, 1.

Solver flag: True; evaluations: 14; residual: 1.89753e-12; physiological: True.

Row/column	1	2	3	4	5	6	7	8	9	10	11
1	-26.098	-4.396	-11.523	-2.8703	10.491	0.095614	0.11561	-0.12549	0.00016683	0.023966	0
2	-18.924	-7.1443	-19.277	-2.9448	14.838	-1.9106	7.1506	4.6393	-0.0033337	-4.8104	9.8608e-27
3	-38.942	-9.003	-32.966	0.40619	23.539	-2.0062	7.0327	4.7648	-0.0035005	-4.8338	0
4	-2.2542	1.2664	-1.638	-10.941	1.9222	-6.1083	-6.066	6.0485	6.3029	0.0072486	3.9443e-26
5	4937.9	4937.9	4937.9	4937.9	-10257	-395.15	-395.15	-395.15	-395.15	821.73	0
6	2.3531	3.819	10.698	-2.4291	-7.1041	-22.259	-0.38918	-1.123	-0.0042541	-132.38	1.5777e-25
7	-2.49	36.305	79.485	0.21607	-56.935	-4.107	-152.66	-24.725	-0.0071659	20.472	0
8	-0.36073	40.761	91.966	-2.6179	-65.15	29.698	-96.645	-82.506	0.051818	-108.84	0
9	-15.246	-16.106	13.687	20.211	0.49211	44.458	44.119	-43.864	-100.63	-3.6256	1.5777e-25
10	-5270	-5270	-5270	-5270	10947	5777.2	5777.2	5777.2	5777.2	-12202	0
11	0	0	0	0	0	0	0	0	0	0	-31.273

Eigenvalues: -0.0021743099; -0.0033552801; -0.010624601; -0.033333333; -0.034309271; -0.060398218; -0.10759125; -0.13377908; -0.2; -8.5679955; -15.281905.

Source record: evidence/task44/output/equilibria.json.

32 Complete expression sensitivities and modal accounting

Table 11: Implicit expression derivatives at all four verified points. Flux derivatives are per unit expression.

e_4	Q_e	N_e	$(J_4)_e$	P_e	H_e	$(pH)_e$	C_N
1	1.28527e-06	-0.0117368	0.0252674	0.000673673	0.00309526	-0.0724609	0.929011
0.5	9.59704e-06	-0.0283978	0.0620609	0.00147416	0.00529833	-0.175657	0.915159
0.05	4.41072e-05	-0.071996	0.1603	0.00306229	0.00560557	-0.507229	0.898266
0	5.06256e-05	-0.0795571	0.177417	0.0032697	0.00516036	-0.580009	0.896838

Table 12: Independent nearby root checks for each expression derivative and difference step.

e_4	Step	Rule	Max component error	State error	Root residual
1	0.0001	centred	1.0247e-07	6.01819e-09	2.2979e-12
1	5e-05	centred	1.03313e-07	1.67435e-09	5.68295e-12
0.5	0.0001	centred	3.22151e-08	1.22956e-08	3.70048e-12
0.5	5e-05	centred	1.68938e-08	3.93657e-09	2.95344e-12
0.05	0.0001	centred	7.33576e-08	3.24719e-08	2.94958e-12
0.05	5e-05	centred	6.19127e-08	8.34278e-09	1.52112e-12
0	0.0001	second_order_forward	3.95616e-08	8.06978e-08	1.43811e-12
0	5e-05	second_order_forward	6.2377e-08	2.02032e-08	4.95907e-12

Table 13: All modal projections at expression 1.0. Projections use the declared scaled coordinates.

Mode	λ (1/s)	τ (s)	Input	Output	Residue	Stationary contribution
1	-0.00289169	345.819	-0.000148937	0.00108248	-1.61222e-07	-5.57536e-05
2	-0.00447476	223.476	5.83148e-05	0.00122877	7.16556e-08	1.60133e-05
3	-0.0326698	30.6093	0.000702471	-0.00018535	-1.30203e-07	-3.98541e-06
4	-0.0333333	30	0	5.97958e-05	0	0
5	-0.0413017	24.2121	0.00113294	0.000393944	4.46316e-07	1.08062e-05
6	-0.0617906	16.1837	0.000114472	-0.00134877	-1.54397e-07	-2.49871e-06
7	-0.108063	9.25389	-0.000294285	0.00851219	-2.50501e-06	-2.31811e-05
8	-0.136466	7.32784	-0.00188052	-0.0110079	2.07006e-05	0.000151691
9	-0.2	5	-0.00145479	0.0127143	-1.84965e-05	-9.24826e-05
10	-9.10146	0.109872	0.000695322	0.020258	1.40858e-05	1.54764e-06
11	-15.905	0.0628732	0.00067473	-0.0205372	-1.3857e-05	-8.71236e-07

Table 14: All modal projections at expression 0.0. Projections use the declared scaled coordinates.

Mode	λ (1/s)	τ (s)	Input	Output	Residue	Stationary contribution
1	-0.00217431	459.916	0.00112772	-0.000690139	-7.78285e-07	-0.000357946
2	-0.00335528	298.038	0.000156605	0.000943594	1.47771e-07	4.40415e-05
3	-0.0106246	94.1212	-0.00217902	-0.0014848	3.23539e-06	0.000304519
4	-0.0333333	30	0	4.33681e-14	0	0
5	-0.0343093	29.1466	0.000667533	0.000348579	2.32688e-07	6.78207e-06
6	-0.0603982	16.5568	-9.9719e-05	0.00128035	-1.27676e-07	-2.1139e-06
7	-0.107591	9.29444	0.000198567	-0.00855199	-1.69814e-06	-1.57833e-05
8	-0.133779	7.47501	0.00284998	0.0109128	3.11014e-05	0.000232483
9	-0.2	5	-0.00255957	0.0127325	-3.25896e-05	-0.000162948
10	-8.568	0.116713	0.00153443	0.019815	3.04048e-05	3.54864e-06
11	-15.2819	0.0654369	0.0014486	-0.0206601	-2.99283e-05	-1.95841e-06

Table 15: Recorded coordinate chain rule attribution. This is not a unique causal allocation to individual transporters.

e_4	Coordinate	dQ/de_4 contribution
1	na_i_fmol	-0
1	k_i_fmol	-0
1	cl_i_fmol	-0
1	tic_i_fmol	-0
1	volume_i_pL	-0
1	na_l_fmol	0
1	k_l_fmol	-0
1	cl_l_fmol	0
1	tic_l_fmol	-0
1	volume_l_pL	1.28527e-06
1	ae4_effective_activation_fraction	-0
0.5	na_i_fmol	-0
0.5	k_i_fmol	-0
0.5	cl_i_fmol	-0
0.5	tic_i_fmol	-0
0.5	volume_i_pL	-0
0.5	na_l_fmol	0
0.5	k_l_fmol	-0
0.5	cl_l_fmol	0
0.5	tic_l_fmol	-0
0.5	volume_l_pL	9.59704e-06
0.5	ae4_effective_activation_fraction	0
0.05	na_i_fmol	-0
0.05	k_i_fmol	-0
0.05	cl_i_fmol	-0
0.05	tic_i_fmol	-0
0.05	volume_i_pL	-0
0.05	na_l_fmol	0
0.05	k_l_fmol	-0
0.05	cl_l_fmol	0
0.05	tic_l_fmol	-0
0.05	volume_l_pL	4.41072e-05
0.05	ae4_effective_activation_fraction	-0

Source record: evidence/task44/output/equilibrium_review.json.

Source record: evidence/task44/output/coordinate_attribution.csv.

33 All extended parameter sensitivity outputs

For each of the sixteen parameters, each recorded metric is listed as an unnormalised derivative with respect to $\log p$, a log elasticity, and the recorded difference step disagreement. For pH the unnormalised derivative is the physically useful quantity; the source also stores a formal log elasticity of the numerical pH value. It is retained here as metadata, not reinterpreted as a hydrogen concentration elasticity. Near zero metrics can make a log elasticity undefined; n/a preserves that fact.

NBC capacity

Active setting: `nbc.capacity_fmol_s` = 0.115709132. No justified physiological uncertainty interval; local analysis only.

Table 16: Complete metric derivatives for NBC capacity.

Metric	$dY/d\log p$	$d\log Y/d\log p$	Step disagreement
WT_Q	3.15588e-05	0.0199226	1.93608e-09
WT_pH	0.064832	0.00919553	4.95274e-10
null_stationary_deficit	0.0197256	2.08477	2.07061e-07
NBC_alkalinity_share	0.0517877	0.0542385	5.62998e-09
noNBC_observed_support_share	-0.0517877	-1.14613	1.18945e-07
AE4_alkalinity_demand	0.0906026	0.603435	5.78306e-08
NHE_physiological_upper	0	0	0
noNBC_AE4_capacity_upper	0	0	0
noNBC_capacity_to_current_demand	-0.349759	-0.603435	6.07007e-08
noNBC_capacity_gap	0.0453013	1.43543	1.37565e-07
chloride_compensation_gain	-0.0598054	-0.0643754	3.71371e-09
WT_slowest_decay	37.0061	0.10701	7.16627e-08
null_slowest_decay	-61.7063	-0.134169	7.42118e-08

Hydrogen concentration elasticity: -0.149281; deficit percentage points per log parameter: 1.97256; maximum IFT relative error: 1.94488e-06.

AE4 carrier

Active setting: `ae4.carrier_amount_fmole = 0.1501264265`. No justified physiological uncertainty interval; local analysis only.

Table 17: Complete metric derivatives for AE4 carrier.

Metric	$dY/d\log p$	$d\log Y/d\log p$	Step disagreement
WT_Q	1.28527e-06	0.000811371	2.84796e-10
WT_pH	-0.0724609	-0.0102776	3.95816e-10
null_stationary_deficit	0.000803694	0.0849416	3.00151e-08
NBC_alkalinity_share	-0.011048	-0.0115708	9.00734e-10
noNBC_observed_support_share	0.011048	0.244507	1.89537e-08
AE4_alkalinity_demand	0.0505347	0.336573	1.46268e-08
NHE_physiological_upper	0	0	0
noNBC_AE4_capacity_upper	0	0	0
noNBC_capacity_to_current_demand	-0.195082	-0.336573	6.14006e-08
noNBC_capacity_gap	0.0252674	0.800627	3.47935e-08
chloride_compensation_gain	0.021727	0.0233873	6.44511e-07
WT_slowest_decay	-22.1213	-0.063968	6.58077e-08
null_slowest_decay	0	0	0

Hydrogen concentration elasticity: 0.166847; deficit percentage points per log parameter: 0.0803694; maximum IFT relative error: 4.4693e-08.

NKCC scale

Active setting: `nkcc1.alpha_eff_fmole = 0.01733474689`. No justified physiological uncertainty interval; local analysis only.

Table 18: Complete metric derivatives for NKCC scale.

Metric	$dY/d\log p$	$d\log Y/d\log p$	Step disagreement
WT_Q	8.76953e-05	0.0553609	2.20164e-09
WT_pH	-0.00155802	-0.000220984	4.1509e-11
null_stationary_deficit	-0.0141328	-1.49368	6.60673e-08
NBC_alkalinity_share	-0.00173929	-0.0018216	5.73242e-11
noNBC_observed_support_share	0.00173929	0.0384928	1.03305e-09
AE4_alkalinity_demand	-0.00761267	-0.0507022	3.61768e-10
NHE_physiological_upper	0	0	0
noNBC_AE4_capacity_upper	0	0	0

Metric	$dY/d\log p$	$d\log Y/d\log p$	Step disagreement
noNBC_capacity_to_current_demand	0.0293876	0.0507022	6.45221e-10
noNBC_capacity_gap	-0.00380634	-0.120608	8.6056e-10
chloride_compensation_gain	0.0594955	0.0640418	2.41813e-07
WT_slowest_decay	65.7564	0.190147	1.11422e-07
null_slowest_decay	67.5683	0.146915	1.31417e-07

Hydrogen concentration elasticity: 0.00358748; deficit percentage points per log parameter: -1.41328; maximum IFT relative error: 7.24353e-08.

NHE carrier

Active setting: `parameters.homeostasis.nhe1_cha_carrier_amount_fmole = 2.339370006e-05`. No justified physiological uncertainty interval; local analysis only.

Table 19: Complete metric derivatives for NHE carrier.

Metric	$dY/d\log p$	$d\log Y/d\log p$	Step disagreement
WT_Q	-5.16575e-06	-0.00326106	9.55481e-11
WT_pH	0.0708381	0.0100474	1.43927e-10
null_stationary_deficit	-0.0021694	-0.229282	1.42273e-08
NBC_alkalinity_share	-0.0186507	-0.0195333	5.03534e-10
noNBC_observed_support_share	0.0186507	0.412764	1.30975e-08
AE4_alkalinity_demand	0.00171704	0.0114359	2.82002e-10
NHE_physiological_upper	0.0820259	1	1.25e-07
noNBC_AE4_capacity_upper	0.041013	0.942546	1.17818e-07
noNBC_capacity_to_current_demand	0.539683	0.93111	1.13769e-07
noNBC_capacity_gap	-0.0401544	-1.27234	1.63114e-07
chloride_compensation_gain	0.00713379	0.00767891	2.5761e-07
WT_slowest_decay	-5.09309	-0.0147276	1.49055e-07
null_slowest_decay	-12.6158	-0.0274307	5.74744e-08

Hydrogen concentration elasticity: -0.163111; deficit percentage points per log parameter: -0.21694; maximum IFT relative error: 6.30036e-08.

pump capacity

Active setting: `parameters.membranes.nak_capacity_fmole_s = 0.08`. No justified physiological uncertainty interval; local analysis only.

Table 20: Complete metric derivatives for pump capacity.

Metric	$dY/d\log p$	$d\log Y/d\log p$	Step disagreement
WT_Q	0.000833109	0.52593	1.55267e-07
WT_pH	0.120613	0.0171072	5.97384e-09
null_stationary_deficit	0.0438432	4.63374	3.22697e-06
NBC_alkalinity_share	0.0149391	0.0156461	4.95122e-09
noNBC_observed_support_share	-0.0149391	-0.330623	1.04462e-07
AE4_alkalinity_demand	0.0299113	0.199216	1.87764e-07
NHE_physiological_upper	0	0	0
noNBC_AE4_capacity_upper	0	0	0
noNBC_capacity_to_current_demand	-0.115468	-0.199217	2.65905e-07
noNBC_capacity_gap	0.0149557	0.473888	4.46646e-07
chloride_compensation_gain	-0.253957	-0.273362	2.6333e-07
WT_slowest_decay	-526.404	-1.5222	1.85117e-06
null_slowest_decay	-339.522	-0.738226	2.16652e-06

Hydrogen concentration elasticity: -0.277721; deficit percentage points per log parameter: 4.38432; maximum IFT relative error: 5.22959e-07.

CaCC conductance

Active setting: `parameters.membranes.g_cl_apical_S = 3.14e-08`. No justified physiological uncertainty interval; local analysis only.

Table 21: Complete metric derivatives for CaCC conductance.

Metric	$dY/d\log p$	$d\log Y/d\log p$	Step disagreement
WT_Q	5.14341e-05	0.0324697	3.8224e-09
WT_pH	-0.0055239	-0.000783489	6.05944e-11
null_stationary_deficit	-0.00209253	-0.221157	1.005e-08
NBC_alkalinity_share	-0.000296827	-0.000310874	8.5463e-12
noNBC_observed_support_share	0.000296827	0.00656919	6.38455e-10
AE4_alkalinity_demand	-0.000789119	-0.00525571	1.85413e-10
NHE_physiological_upper	0	0	0
noNBC_AE4_capacity_upper	0	0	0
noNBC_capacity_to_current_demand	0.00304628	0.00525571	1.7038e-10
noNBC_capacity_gap	-0.000394559	-0.0125021	4.41053e-10
chloride_compensation_gain	0.0102626	0.0110468	5.26855e-07
WT_slowest_decay	2.86118	0.00827365	3.13325e-08
null_slowest_decay	-5.37412	-0.011685	2.37091e-08

Hydrogen concentration elasticity: 0.0127192; deficit percentage points per log parameter: -0.209253; maximum IFT relative error: 2.18379e-07.

buffer pool

Active setting: `parameters.geometry.cell_buffer_total_fmol = 30`. No justified physiological uncertainty interval; local analysis only.

Table 22: Complete metric derivatives for buffer pool.

Metric	$dY/d\log p$	$d\log Y/d\log p$	Step disagreement
WT_Q	-2.4947e-06	-0.00157487	1.04035e-11
WT_pH	0.00139566	0.000197955	1.13378e-12
null_stationary_deficit	-0.00195204	-0.206309	4.10684e-10
NBC_alkalinity_share	0.000302635	0.000316956	4.18594e-12
noNBC_observed_support_share	-0.000302635	-0.00669771	2.49547e-11
AE4_alkalinity_demand	-0.000118028	-0.000786094	8.31864e-12
NHE_physiological_upper	0	0	0
noNBC_AE4_capacity_upper	0	0	0
noNBC_capacity_to_current_demand	0.00045563	0.000786094	8.52378e-12
noNBC_capacity_gap	-5.9014e-05	-0.00186993	1.9788e-11
chloride_compensation_gain	0.00882912	0.00950379	1.07406e-07
WT_slowest_decay	106.704	0.308555	2.03105e-07
null_slowest_decay	184.733	0.401666	2.00556e-07

Hydrogen concentration elasticity: -0.00321362; deficit percentage points per log parameter: -0.195204; maximum IFT relative error: 6.10481e-06.

water apical

Active setting: `parameters.water.apical_hydraulic_pL_s_mOsm = 0.00432`. No justified physiological uncertainty interval; local analysis only.

Table 23: Complete metric derivatives for water apical.

Metric	$dY/d\log p$	$d\log Y/d\log p$	Step disagreement
WT_Q	3.37326e-06	0.00212949	1.7686e-10

Metric	$dY/d\log p$	$d\log Y/d\log p$	Step disagreement
WT_pH	-8.31675e-05	-1.17962e-05	5.66891e-13
null_stationary_deficit	-3.36644e-05	-0.00355795	3.81349e-10
NBC_alkalinity_share	-1.51011e-06	-1.58158e-06	1.02904e-11
noNBC_observed_support_share	1.51012e-06	3.3421e-05	2.03476e-11
AE4_alkalinity_demand	-1.61264e-05	-0.000107406	1.75616e-12
NHE_physiological_upper	0	0	0
noNBC_AE4_capacity_upper	0	0	0
noNBC_capacity_to_current_demand	6.22536e-05	0.000107406	1.72391e-12
noNBC_capacity_gap	-8.0632e-06	-0.000255492	4.17747e-12
chloride_compensation_gain	0.000207998	0.000223892	3.81471e-07
WT_slowest_decay	-0.0129329	-3.73979e-05	5.44652e-09
null_slowest_decay	-0.16216	-0.000352587	1.68272e-08

Hydrogen concentration elasticity: 0.0001915; deficit percentage points per log parameter: -0.00336644; maximum IFT relative error: 1.30553e-07.

AE4 tau

Active setting: `regulation.ae4_regulatory_model.tau_activation_s = 30`. No justified physiological uncertainty interval; local analysis only.

Table 24: Complete metric derivatives for AE4 tau.

Metric	$dY/d\log p$	$d\log Y/d\log p$	Step disagreement
WT_Q	0	0	0
WT_pH	0	0	0
null_stationary_deficit	0	0	0
NBC_alkalinity_share	0	0	0
noNBC_observed_support_share	0	0	0
AE4_alkalinity_demand	0	0	0
NHE_physiological_upper	0	0	0
noNBC_AE4_capacity_upper	0	0	0
noNBC_capacity_to_current_demand	0	0	0
noNBC_capacity_gap	0	0	0
chloride_compensation_gain	0	0	0
WT_slowest_decay	0	0	0
null_slowest_decay	0	0	0

Hydrogen concentration elasticity: -0; deficit percentage points per log parameter: 0; maximum IFT relative error: 0.

AE4 gain

Active setting: `regulation.ae4_regulatory_model.gain.fully_activated_increment = 0.25`. No justified physiological uncertainty interval; local analysis only.

Table 25: Complete metric derivatives for AE4 gain.

Metric	$dY/d\log p$	$d\log Y/d\log p$	Step disagreement
WT_Q	2.57053e-07	0.000162274	2.62141e-11
WT_pH	-0.0144922	-0.00205552	5.82638e-11
null_stationary_deficit	0.000160739	0.0169883	2.75158e-09
NBC_alkalinity_share	-0.0022096	-0.00231416	1.01974e-10
noNBC_observed_support_share	0.0022096	0.0489014	2.17919e-09
AE4_alkalinity_demand	0.0101069	0.0673146	1.6665e-09
NHE_physiological_upper	0	0	0
noNBC_AE4_capacity_upper	0	0	0
noNBC_capacity_to_current_demand	-0.0390164	-0.0673146	4.44003e-10
noNBC_capacity_gap	0.00505347	0.160125	3.9642e-09
chloride_compensation_gain	0.00434561	0.00467767	1.9282e-07

Metric	$dY/d\log p$	$d\log Y/d\log p$	Step disagreement
WT_slowest_decay	-4.42423	-0.0127935	8.89569e-08
null_slowest_decay	0	0	0

Hydrogen concentration elasticity: 0.0333695; deficit percentage points per log parameter: 0.0160739; maximum IFT relative error: 4.64675e-08.

apical pump fraction

Active setting: `parameters.membranes.apical_pump_fraction = 0.075075`. No justified physiological uncertainty interval; local analysis only.

Table 26: Complete metric derivatives for apical pump fraction.

Metric	$dY/d\log p$	$d\log Y/d\log p$	Step disagreement
WT_Q	7.85135e-05	0.0495645	6.45148e-09
WT_pH	-0.00357128	-0.000506537	7.77271e-11
null_stationary_deficit	-0.000566061	-0.0598263	1.4679e-08
NBC_alkalinity_share	-0.000208311	-0.000218169	4.79058e-11
noNBC_observed_support_share	0.000208311	0.00461021	5.66893e-10
AE4_alkalinity_demand	-0.000673156	-0.00448338	7.88607e-10
NHE_physiological_upper	0	0	0
noNBC_AE4_capacity_upper	0	0	0
noNBC_capacity_to_current_demand	0.00259862	0.00448338	8.05832e-10
noNBC_capacity_gap	-0.000336578	-0.0106649	1.87591e-09
chloride_compensation_gain	0.00292689	0.00315054	7.22836e-07
WT_slowest_decay	15.6203	0.045169	1.76137e-08
null_slowest_decay	9.32054	0.0202657	1.83547e-08

Hydrogen concentration elasticity: 0.00822317; deficit percentage points per log parameter: -0.056606; maximum IFT relative error: 2.53979e-07.

NHE stimulated multiplier

Active setting: `nbc.nhe1_stimulated_multiplier = 2.3`. No justified physiological uncertainty interval; local analysis only.

Table 27: Complete metric derivatives for NHE stimulated multiplier.

Metric	$dY/d\log p$	$d\log Y/d\log p$	Step disagreement
WT_Q	-5.16575e-06	-0.00326106	9.28788e-11
WT_pH	0.0708381	0.0100474	1.44179e-10
null_stationary_deficit	-0.0021694	-0.229282	1.39457e-08
NBC_alkalinity_share	-0.0186507	-0.0195333	5.16906e-10
noNBC_observed_support_share	0.0186507	0.412764	1.30769e-08
AE4_alkalinity_demand	0.00171704	0.0114359	2.91707e-10
NHE_physiological_upper	0.0820259	1	1.25e-07
noNBC_AE4_capacity_upper	0.041013	0.942546	1.17818e-07
noNBC_capacity_to_current_demand	0.539683	0.93111	1.13778e-07
noNBC_capacity_gap	-0.0401544	-1.27234	1.63137e-07
chloride_compensation_gain	0.0071334	0.00767849	1.63473e-07
WT_slowest_decay	-5.09309	-0.0147276	1.50992e-07
null_slowest_decay	-12.6158	-0.0274307	5.84941e-08

Hydrogen concentration elasticity: -0.163111; deficit percentage points per log parameter: -0.21694; maximum IFT relative error: 6.30036e-08.

CO2 basolateral permeability

Active setting: `parameters.homeostasis.co2_basolateral_permeability_fmols_mM = 0.02`.
No justified physiological uncertainty interval; local analysis only.

Table 28: Complete metric derivatives for CO2 basolateral permeability.

Metric	$dY/d\log p$	$d\log Y/d\log p$	Step disagreement
WT_Q	-7.04598e-06	-0.00444803	1.98899e-10
WT_pH	-0.0831861	-0.0117988	9.07026e-11
null_stationary_deficit	-0.00410429	-0.433779	1.97128e-08
NBC_alkalinity_share	-0.0232637	-0.0243646	1.37398e-09
noNBC_observed_support_share	0.0232637	0.514857	2.31545e-08
AE4_alkalinity_demand	0.00200029	0.0133224	7.64668e-10
NHE_physiological_upper	0	0	0
noNBC_AE4_capacity_upper	0	0	0
noNBC_capacity_to_current_demand	-0.00772182	-0.0133224	7.90221e-10
noNBC_capacity_gap	0.00100014	0.0316907	1.81896e-09
chloride_compensation_gain	0.047022	0.0506151	1.90077e-07
WT_slowest_decay	-23.5774	-0.0681786	1.41451e-07
null_slowest_decay	-69.9883	-0.152176	1.7382e-07

Hydrogen concentration elasticity: 0.191543; deficit percentage points per log parameter: -0.410429; maximum IFT relative error: 6.49468e-08.

AE2 capacity

Active setting: `parameters.homeostasis.ae2_capacity_fmols = 0.005`. No justified physiological uncertainty interval; local analysis only.

Table 29: Complete metric derivatives for AE2 capacity.

Metric	$dY/d\log p$	$d\log Y/d\log p$	Step disagreement
WT_Q	-1.16185e-07	-7.33458e-05	7.80264e-12
WT_pH	0.000362969	5.14821e-05	6.2358e-12
null_stationary_deficit	-0.000289197	-0.030565	3.58468e-09
NBC_alkalinity_share	-0.00469991	-0.00492232	5.78649e-10
noNBC_observed_support_share	0.00469991	0.104015	1.22851e-08
AE4_alkalinity_demand	0.000505812	0.00336883	4.03731e-10
NHE_physiological_upper	0	0	0
noNBC_AE4_capacity_upper	0.0025	0.0574542	7.18169e-09
noNBC_capacity_to_current_demand	0.0313486	0.0540853	6.64251e-09
noNBC_capacity_gap	-0.00224709	-0.0712018	8.94145e-09
chloride_compensation_gain	-0.0314286	-0.0338301	3.54413e-09
WT_slowest_decay	0.0259648	7.50822e-05	8.17993e-08
null_slowest_decay	-1.70281	-0.00370243	7.08077e-09

Hydrogen concentration elasticity: -0.000835766; deficit percentage points per log parameter: -0.0289197; maximum IFT relative error: 1.65073e-07.

NBC log width

Active setting: `nbc.log_width = 2`. No justified physiological uncertainty interval; local analysis only.

Table 30: Complete metric derivatives for NBC log width.

Metric	$dY/d\log p$	$d\log Y/d\log p$	Step disagreement
WT_Q	-2.2734e-05	-0.0143517	9.58356e-10

Metric	$dY/d\log p$	$d\log Y/d\log p$	Step disagreement
WT_pH	-0.0467032	-0.0066242	3.9695e-10
null_stationary_deficit	-0.0142074	-1.50156	9.9409e-08
NBC_alkalinity_share	-0.0373064	-0.0390719	4.01519e-09
noNBC_observed_support_share	0.0373064	0.825641	8.49456e-08
AE4_alkalinity_demand	-0.0652676	-0.434697	2.87085e-08
NHE_physiological_upper	0	0	0
noNBC_AE4_capacity_upper	0	0	0
noNBC_capacity_to_current_demand	0.251956	0.434697	4.32029e-08
noNBC_capacity_gap	-0.0326338	-1.03404	6.82907e-08
chloride_compensation_gain	0.0165089	0.0177704	2.63759e-07
WT_slowest_decay	-25.118	-0.0726336	9.34528e-08
null_slowest_decay	61.7062	0.134168	1.74507e-08

Hydrogen concentration elasticity: 0.107538; deficit percentage points per log parameter: -1.42074; maximum IFT relative error: 3.34413e-06.

NKCC stimulated multiplier

Active setting: `regulation.nkcc1_regulatory_model.fully_activated_multiplier = 1.75`.
No justified physiological uncertainty interval; local analysis only.

Table 31: Complete metric derivatives for NKCC stimulated multiplier.

Metric	$dY/d\log p$	$d\log Y/d\log p$	Step disagreement
WT_Q	8.76953e-05	0.0553609	2.20164e-09
WT_pH	-0.00155802	-0.000220984	4.27058e-11
null_stationary_deficit	-0.0141328	-1.49368	6.63607e-08
NBC_alkalinity_share	-0.00173929	-0.0018216	3.86618e-11
noNBC_observed_support_share	0.00173929	0.0384928	1.10238e-09
AE4_alkalinity_demand	-0.00761267	-0.0507022	3.48736e-10
NHE_physiological_upper	0	0	0
noNBC_AE4_capacity_upper	0	0	0
noNBC_capacity_to_current_demand	0.0293876	0.0507022	6.58247e-10
noNBC_capacity_gap	-0.00380634	-0.120608	8.29559e-10
chloride_compensation_gain	0.0594962	0.0640426	6.93126e-07
WT_slowest_decay	65.7564	0.190147	1.39192e-09
null_slowest_decay	67.5683	0.146914	6.06859e-08

Hydrogen concentration elasticity: 0.00358748; deficit percentage points per log parameter: -1.41328; maximum IFT relative error: 7.28046e-08.

Source record: `evidence/task44/output/sensitivity_extended.csv`.

Source record: `evidence/task44/output/sensitivity_extended.json`.

34 Complete inverse scan and both crossing sensitivity observables

Table 32: All seventeen ordered scan points in the prescribed inverse family. Deficits are percentages; output is pL.

ρ	b	I_A	I_S	$100D_A$	$100D_S$	Gate pass
0	1	0.95425792	0.95423805	3.857411	3.857486	Yes
0.1	0.9	0.94933258	0.94931332	4.353645	4.353669	Yes
0.2	0.8	0.94325604	0.94323753	4.965863	4.965824	Yes
0.3	0.7	0.9355714	0.9355538	5.7401	5.739984	Yes
0.4	0.6	0.92554309	0.92552664	6.750464	6.750252	Yes
0.5	0.5	0.91190632	0.91189136	8.124384	8.124049	Yes

ρ	b	I_A	I_S	$100D_A$	$100D_S$	Gate pass
0.6	0.4	0.89228667	0.89227373	10.10109	10.10059	Yes
0.7	0.3	0.8616407	0.86163063	13.1887	13.18798	Yes
0.75	0.25	0.83880188	0.83879376	15.48974	15.48886	Yes
0.8	0.2	0.80702962	0.80702397	18.69083	18.68977	Yes
0.85	0.15	0.75978276	0.75978035	23.45101	23.44972	Yes
0.875	0.125	0.72630434	0.72630396	26.824	26.82257	Yes
0.89	0.11	0.70132171	0.70132271	29.34103	29.33951	Yes
0.894881272	0.10511873	0.69217371	0.6921752	30.2627	30.26115	Yes
0.9	0.1	0.68195476	0.68195677	31.29227	31.29069	Yes
0.925	0.075	0.62025661	0.62026151	37.50843	37.50668	Yes
0.95	0.05	0.52794581	0.52795425	46.80885	46.80693	Yes

Table 33: All sixteen parameter and observable pairs for the inverse threshold.

Parameter	Observable	$d\rho/d\log p$	E_ρ	E_b	Validation error
NBC capacity	adaptive	-0.00533011	-0.00595495	0.0507982	1.61947e-06
NBC capacity	sampled	-0.00532963	-0.00595436	0.0507974	1.63388e-06
AE4 carrier	adaptive	-0.00452626	-0.00505686	0.0431371	1.61496e-06
AE4 carrier	sampled	-0.00452603	-0.00505657	0.0431382	1.62948e-06
NKCC scale	adaptive	-0.00196257	-0.00219264	0.0187041	1.6351e-06
NKCC scale	sampled	-0.00196294	-0.00219303	0.018709	1.64811e-06
NHE carrier	adaptive	0.000717662	0.000801792	-0.00683961	1.62144e-06
NHE carrier	sampled	0.00071767	0.000801794	-0.0068402	1.63572e-06
pump capacity	adaptive	-0.0318489	-0.0355825	0.303533	1.61727e-06
pump capacity	sampled	-0.0318503	-0.0355837	0.303569	1.63166e-06
CaCC conductance	adaptive	0.0880143	0.098332	-0.838812	1.70519e-06
CaCC conductance	sampled	0.0880092	0.0983254	-0.838827	1.67871e-06
buffer pool	adaptive	0.00640488	0.00715571	-0.0610411	1.61876e-06
buffer pool	sampled	0.00642313	0.00717603	-0.0612196	1.63561e-06
water apical	adaptive	-0.000266308	-0.000297527	0.00253802	1.62359e-06
water apical	sampled	-0.000263734	-0.000294648	0.00251368	1.63281e-06

Source record: `evidence/task44/inverse_detail/ordered_family_scan.json`.

Source record: `evidence/task44/inverse_detail/threshold_sensitivities.json`.

35 Complete scaling and capacity metadata

The saved group file also contains loose kinetic turnover bounds and the sharper pH conditioned bounds. They are distinct calculations, not contradictory versions of one bound. The fields named NHE global bound and AE4 without NBC global bound use the broad printed turnover bound; the physiological fields use the specified pH floor. Flux bound fields retain fmol/s even though they share the dimensionless group file.

Table 34: All saved scale and group scalars.

Field	Recorded value
AE4_without_NBC_global_bound	0.2849789282
CO_mM	145
JO_fmols	0.2246180791
LO_pL	0.1053884074
NBC_lower_for_AE4_demand_0075_fmols	0.03148704525
NHE_global_bound	0.5649578564
NKCC_eta2	56.40285308
NKCC_eta4	594.1476089
VO_pL	1.453334393
ae2_capacity_ratio	0.02226000694
epsilon_lumen	0.07251490634
g_cl_apical_S_dimensionless	38.72293949
g_k_total_S_dimensionless	17.26500487
g_para_cl_S_dimensionless	0.3083036584

Field	Recorded value
g_para_hco3_S_dimensionless	0.06166073167
g_para_k_S_dimensionless	0.06166073167
g_para_na_S_dimensionless	0.36996439
gamma_AE4_reference	0.6683630592
n0_fm1	210.7334869
nbc_capacity_ratio	0.5151372162
nkcc_forward_bound	4.636027181
nkcc_reversal_product_mM4	7837380.573
nkcc_reverse_bound	0.4401013419
n10_fm1	15.28131907
outflow_rate_dimensionless	187.6371553
physiological_AE4_without_NBC_upper_fm1_s	0.04351295475
physiological_NHE_upper_fm1_s	0.08202590949
pump_capacity_ratio	0.3561601111
t0_s	938.1857764
thermal_voltage_V	0.02672665911
theta_ae4	0.03197660928
water_apical	404.3663821
water_basolateral	4820.571453
water_paracellular	24.33686559

Table 35: Additional scale interpretation fields.

Field	Recorded value
AE4_to_NKCC_chloride_initial	0.01098293198
AE4_to_NKCC_cycle_initial	0.02196586395
Ka1_over_CO	5.478125757e-06
Ka2_over_CO	3.45646368e-10
Kw_over_CO_squared	4.756242568e-13
carrier_QSS_time_ratio_initial	0.0001392109809
cell_buffer_K_over_CO	6.896551724e-07
cell_buffer_scaled	0.1423599089
cell_other_impermeant_scaled	0.5917396954
co2_apical	6.455402013
co2_basolateral	12.91080403
electrical_relaxation_group	n/a
electrical_relaxation_reason	No capacitance or voltage differential equation in the frozen model; conductance groups measure currents and do not establish an electrical relaxation time.
fixed_cell_charge_scaled	0.3704288452
lumen_buffer_K_over_CO	6.896551724e-07
lumen_buffer_scaled	6.543937704e-11

Source record: evidence/task44/output/dimensionless_groups.json.

Source record: evidence/task44/output/exact_verification.json.

36 Detailed provenance for every inventory record

All 137 records are included, not only the active or sensitive subset. The full original JSON and CSV additionally preserve all raw metadata and complete code location lists. Repeated generic caveats are not evidence of an independently measured constraint. Values, units, status and the following explanatory fields are transcribed from the audit without turning logical domains into physiological uncertainty intervals.

Record 1: temperature K

constants.temperature_K

Value: 310.15; units: K; active: Yes; classification: effective_assumption.

Default: 310.15; differs from default: False.

Equation role. Thermal voltage RT/F , Nernst potentials, and conversion between current and ionic amount flux.

Constraint. 37 C reference temperature; protocol-specific temperature must replace it

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. `genuinely_unknown_for_physiological_transfer`

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. `src/modern_full_model/membranes.py:50`. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 2: gas constant J mol K

`constants.gas_constant_J_mol_K`

Value: 8.31446261815324; units: $\text{J mol}^{-1} \text{K}^{-1}$; active: Yes; classification: `physical_constant`.

Default: 8.31446261815324; differs from default: False.

Equation role. Thermal voltage RT/F , Nernst potentials, and conversion between current and ionic amount flux.

Constraint. CODATA exact/standard physical constant

Limit of justification. No physiological fitting or salivary uncertainty interval applies to a standard physical constant.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. `fixed_standard_constant_at_stored_precision`

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. No physiological experiment required; retain standard value and unit convention.

Source. `src/modern_full_model/membranes.py:50`. Standard physical constant, rounded to the stored precision; not a physiological estimate.

Record 3: faraday C mol

`constants.faraday_C_mol`

Value: 96485.33212000001; units: C mol^{-1} ; active: Yes; classification: `physical_constant`.

Default: 96485.33212; differs from default: False.

Equation role. Thermal voltage RT/F , Nernst potentials, and conversion between current and ionic amount flux.

Constraint. Faraday conversion between charge current and molar flux

Limit of justification. No physiological fitting or salivary uncertainty interval applies to a standard physical constant.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. `fixed_standard_constant_at_stored_precision`

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. No physiological experiment required; retain standard value and unit convention.

Source. `src/modern_full_model/membranes.py:50`. Standard physical constant, rounded to the stored precision; not a physiological estimate.

Record 4: carbon pka1

`acid_base.carbon_pka1`

Value: 6.1; units: dimensionless (p-scale); active: Yes; classification: `inherited_model_setting`.

Default: 6.1; differs from default: False.

Equation role. Algebraic $TA = \text{carbonate alkalinity} + \text{finite buffer base} + OH - H$; pH is solved from TIC, TA and volume.

Constraint. effective CO_2/HCO_3 pKa; temperature/context refinement required Source metadata identifies a model lineage; independent experimental measurement is not established here.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Finite p scale dissociation constant; environmental applicability must be measured.

Uncertainty status. `genuinely_unknown_for_physiological_transfer`

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. `src/modern_full_model/acid_base.py:93`. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 5: carbon pka2

`acid_base.carbon_pka2`

Value: 10.3; units: dimensionless (p-scale); active: Yes; classification: `inherited_model_setting`.

Default: 10.3; differs from default: False.

Equation role. Algebraic $TA = \text{carbonate alkalinity} + \text{finite buffer base} + OH - H$; pH is solved from TIC, TA and volume.

Constraint. effective HCO_3/CO_3 pKa; carbonate retained for charge closure Source metadata identifies a model lineage; independent experimental measurement is not established here.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Finite p scale dissociation constant; environmental applicability must be measured.

Uncertainty status. `genuinely_unknown_for_physiological_transfer`

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. `src/modern_full_model/acid_base.py:93`. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 6: water pkw

`acid_base.water_pkw`

Value: 14; units: dimensionless (p-scale); active: Yes; classification: inherited_model_setting.

Default: 14.0; differs from default: False.

Equation role. Algebraic $TA = \text{carbonate alkalinity} + \text{finite buffer base} + OH - H$; pH is solved from TIC, TA and volume.

Constraint. effective water ionization value; temperature refinement required Source metadata identifies a model lineage; independent experimental measurement is not established here.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Finite p scale dissociation constant; environmental applicability must be measured.

Uncertainty status. genuinely_unknown_for_physiological_transfer

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. src/modern_full_model/acid_base.py:93. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 7: cell buffer pka

acid_base.cell_buffer_pka

Value: 7; units: dimensionless (p-scale); active: Yes; classification: effective_assumption.

Default: 7.0; differs from default: False.

Equation role. Algebraic $TA = \text{carbonate alkalinity} + \text{finite buffer base} + OH - H$; pH is solved from TIC, TA and volume.

Constraint. single effective non-carbonate buffer site; not identified

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Finite p scale dissociation constant; environmental applicability must be measured.

Uncertainty status. genuinely_unknown_for_physiological_transfer

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. src/modern_full_model/acid_base.py:93. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 8: lumen buffer pka

acid_base.lumen_buffer_pka

Value: 7; units: dimensionless (p-scale); active: Yes; classification: effective_assumption.

Default: 7.0; differs from default: False.

Equation role. Algebraic $TA = \text{carbonate alkalinity} + \text{finite buffer base} + OH - H$; pH is solved from TIC, TA and volume.

Constraint. irrelevant when luminal buffer amount is zero

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Finite p scale dissociation constant; environmental applicability must be measured.

Uncertainty status. genuinely_unknown_for_physiological_transfer

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. src/modern_full_model/acid_base.py:93. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 9: ph lower

acid_base.ph_lower

Value: 3; units: dimensionless (p-scale); active: Yes; classification: numerical_bracket.

Default: 3.0; differs from default: False.

Equation role. Algebraic $TA = \text{carbonate alkalinity} + \text{finite buffer base} + OH - H$; pH is solved from TIC, TA and volume.

Constraint. Numerical pH inversion bracket, not an experimental physiological limit.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Ordered numerical inversion endpoints [3,11]; not the inherited physiological pH gate [6.6,7.3].

Uncertainty status. not_a_measured_uncertainty_parameter

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Verify numerical convergence and successful closure on the stated domain; this is not an abundance measurement.

Source. src/modern_full_model/acid_base.py:93. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 10: ph upper

acid_base.ph_upper

Value: 11; units: dimensionless (p-scale); active: Yes; classification: numerical_bracket.

Default: 11.0; differs from default: False.

Equation role. Algebraic $TA = \text{carbonate alkalinity} + \text{finite buffer base} + OH - H$; pH is solved from TIC, TA and volume.

Constraint. Numerical pH inversion bracket, not an experimental physiological limit.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Ordered numerical inversion endpoints [3,11]; not the inherited physiological pH gate [6.6,7.3].

Uncertainty status. not_a_measured_uncertainty_parameter

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Verify numerical convergence and successful closure on the stated domain; this is not an abundance measurement.

Source. src/modern_full_model/acid_base.py:93. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 11: na mM

bath.na_mM

Value: 145; units: mM; active: Yes; classification: effective_assumption.

Default: 145.0; differs from default: False.

Equation role. Fixed reservoir composition determines bath pH species, osmolarity, transporter affinities and reversal potentials.

Constraint. representative open-bath value; protocol ledger must replace it

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. genuinely_unknown_for_physiological_transfer

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. src/modern_full_model/model.py:374. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 12: k mM

bath.k_mM

Value: 5; units: mM; active: Yes; classification: effective_assumption.

Default: 5.0; differs from default: False.

Equation role. Fixed reservoir composition determines bath pH species, osmolarity, transporter affinities and reversal potentials.

Constraint. representative open-bath value; protocol ledger must replace it

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. genuinely_unknown_for_physiological_transfer

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. src/modern_full_model/model.py:374. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 13: cl mM

bath.cl_mM

Value: 126.1615928036609; units: mM; active: Yes; classification: reference_constraint_derived.

Default: 126.16159280366088; differs from default: False.

Equation role. Fixed reservoir composition determines bath pH species, osmolarity, transporter affinities and reversal potentials.

Constraint. Bath electroneutrality using the adopted bath sodium, potassium and carbonate speciation.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical

uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. conditional_derivation_no_experimental_interval

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. src/modern_full_model/model.py:374. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 14: tic mM

bath.tic_mM

Value: 25; units: mM; active: Yes; classification: effective_assumption.

Default: 25.0; differs from default: False.

Equation role. Fixed reservoir composition determines bath pH species, osmolarity, transporter affinities and reversal potentials.

Constraint. open-bath total inorganic carbon starting value

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. genuinely_unknown_for_physiological_transfer

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. src/modern_full_model/model.py:374. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 15: ph

bath.ph

Value: 7.4; units: dimensionless (p-scale); active: Yes; classification: effective_assumption.

Default: 7.4; differs from default: False.

Equation role. Fixed reservoir composition determines bath pH species, osmolarity, transporter affinities and reversal potentials.

Constraint. representative bicarbonate-buffered bath value

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. genuinely_unknown_for_physiological_transfer

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. src/modern_full_model/model.py:374. No independently verified primary measurement

attached to this numerical setting; model lineage is the source.

Record 16: buffer total mM

`bath.buffer_total_mM`

Value: 0; units: mM sites; active: Yes; classification: effective_assumption.

Default: 0.0; differs from default: False.

Equation role. Fixed reservoir composition determines bath pH species, osmolarity, transporter affinities and reversal potentials.

Constraint. bath non-carbonate buffer omitted from exchanged species An exact zero is an explicit structural omission, not a small positive fitted value.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Exact zero records a declared omitted contribution or protocol normalisation; it is not a confidence bound.

Uncertainty status. genuinely_unknown_for_physiological_transfer

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. `src/modern_full_model/model.py:374`. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 17: untracked osmolyte mM

`bath.untracked_osmolyte_mM`

Value: 0; units: mOsm L⁻¹; active: Yes; classification: effective_assumption.

Default: 0.0; differs from default: False.

Equation role. Fixed reservoir composition determines bath pH species, osmolarity, transporter affinities and reversal potentials.

Constraint. explicit correction only if protocol osmolarity requires it An exact zero is an explicit structural omission, not a small positive fitted value.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Exact zero records a declared omitted contribution or protocol normalisation; it is not a confidence bound.

Uncertainty status. genuinely_unknown_for_physiological_transfer

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. `src/modern_full_model/model.py:374`. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 18: cell buffer total fMol

`geometry.cell_buffer_total_fMol`

Value: 30; units: fMol sites; active: Yes; classification: effective_assumption.

Default: 30.0; differs from default: False.

Equation role. Finite buffer and impermeant pools enter algebraic pH, cell osmolarity, or the neutral charge constraint.

Constraint. effective finite buffer pool; must be WT-calibrated independently

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. genuinely_unknown_for_physiological_transfer

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value. Local numerical dependence is quantified in the linked crosswalk; it establishes neither a measured uncertainty range nor global robustness.

Discriminating measurement. Measure intracellular titration capacity together with TIC and cell volume across the experimental pH interval.

Source. src/modern_full_model/model.py:374. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 19: lumen buffer total fmol

geometry.lumen_buffer_total_fmol

Value: 1e-09; units: fmol sites; active: Yes; classification: numerical_regularisation.

Default: 1e-09; differs from default: False.

Equation role. Finite buffer and impermeant pools enter algebraic pH, cell osmolarity, or the neutral charge constraint.

Constraint. Positive approximation to no fixed luminal buffer. Its size is numerical, not measured.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. not_a_measured_uncertainty_parameter

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Verify numerical convergence and successful closure on the stated domain; this is not an abundance measurement.

Source. src/modern_full_model/model.py:374. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 20: fixed cell anion equivalents fmol

geometry.fixed_cell_anion_equivalents_fmol

Value: 78.06176220594485; units: fmol charge equivalents; active: Yes; classification: reference_constraint_derived.

Default: 78.06176220594485; differs from default: False.

Equation role. Finite buffer and impermeant pools enter algebraic pH, cell osmolarity, or the neutral charge constraint.

Constraint. Derived from the reference initial cations, chloride and total alkalinity. Conditional on that reference tuple, not independently measured.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. conditional_derivation_no_experimental_interval

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. `src/modern_full_model/model.py:374`. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 21: cell impermeant osmoles fmol

`geometry.cell_impermeant_osmoles_fmol`

Value: 124.6993693712438; units: fmol osmoles; active: Yes; classification: `inherited_effective_setting`.

Default: 56.16159280366088; differs from default: True.

Equation role. Finite buffer and impermeant pools enter algebraic pH, cell osmolarity, or the neutral charge constraint.

Constraint. Active R09 background contains 124.69936937124379 fmol, not the dataclass default. An effective osmotic closure setting, not a directly measured molecular pool.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. `genuinely_unknown_for_physiological_transfer`

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure resting and stimulated cell volume and osmolarity to separate fixed buffer sites from other impermeant osmotic particles.

Source. `src/modern_full_model/model.py:374`. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 22: nkcc1 capacity fmol s

`homeostasis.nkcc1_capacity_fmol_s`

Value: 0.32; units: fmol s⁻¹; active: No; classification: `inactive_legacy`.

Default: 0.08; differs from default: True.

Equation role. NHE1 supplies Na and TA; AE2 imports Cl and exports carbon/TA; CO2 transfers carbon only.

Constraint. Not used by selected Palk NKCC1 or Cha NHE1 flux. Retained for older models and payload compatibility.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. `not_a_measured_uncertainty_parameter`

Dependence of claims. No independent parent RHS sensitivity. Used only for the explicitly stated metadata, initial reference, design lineage or inverse family role.

Discriminating measurement. No new production parameter experiment is implied for this inactive field; measure the actual corresponding state or active pathway instead.

Source. `src/modern_full_model/membranes.py:206`. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 23: nhe1 capacity fmol s

homeostasis.nhe1_capacity_fmол_s

Value: 0.007440980590414482; units: fmol s⁻¹; active: No; classification: inactive_legacy.

Default: 0.02; differs from default: True.

Equation role. NHE1 supplies Na and TA; AE2 imports Cl and exports carbon/TA; CO₂ transfers carbon only.

Constraint. Not used by selected Palk NKCC1 or Cha NHE1 flux. Retained for older models and payload compatibility.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. not_a_measured_uncertainty_parameter

Dependence of claims. No independent parent RHS sensitivity. Used only for the explicitly stated metadata, initial reference, design lineage or inverse family role.

Discriminating measurement. No new production parameter experiment is implied for this inactive field; measure the actual corresponding state or active pathway instead.

Source. src/modern_full_model/membranes.py:206. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 24: nhe1 model

homeostasis.nhe1_model

Value: cha_2009_eight_state_mod2; units: model identifier; active: Yes; classification: law_selection.

Default: legacy_tanh; differs from default: True.

Equation role. NHE1 supplies Na and TA; AE2 imports Cl and exports carbon/TA; CO₂ transfers carbon only.

Constraint. Production explicitly selects Cha 2009 eight-state Mod2; the default field note refers to an older selection.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Selected discrete setting or stated numerical reference.

Uncertainty status. not_a_measured_uncertainty_parameter

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. src/modern_full_model/membranes.py:206. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 25: nhe1 published g fmol s

homeostasis.nhe1_published_g_fmол_s

Value: 0.0305; units: fmol s⁻¹; active: No; classification: inactive_legacy.

Default: 0.0305; differs from default: False.

Equation role. NHE1 supplies Na and TA; AE2 imports Cl and exports carbon/TA; CO₂ transfers carbon only.

Constraint. Not used by selected Palk NKCC1 or Cha NHE1 flux. Retained for older models and payload compatibility.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. not_a_measured_uncertainty_parameter

Dependence of claims. No independent parent RHS sensitivity. Used only for the explicitly stated metadata, initial reference, design lineage or inverse family role.

Discriminating measurement. No new production parameter experiment is implied for this inactive field; measure the actual corresponding state or active pathway instead.

Source. src/modern_full_model/membranes.py:206. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 26: nhe1 published k h mM

homeostasis.nhe1_published_k_h_mM

Value: 0.00045; units: mM; active: No; classification: inactive_legacy.

Default: 0.00045; differs from default: False.

Equation role. NHE1 supplies Na and TA; AE2 imports Cl and exports carbon/TA; CO2 transfers carbon only.

Constraint. Not used by selected Palk NKCC1 or Cha NHE1 flux. Retained for older models and payload compatibility.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. not_a_measured_uncertainty_parameter

Dependence of claims. No independent parent RHS sensitivity. Used only for the explicitly stated metadata, initial reference, design lineage or inverse family role.

Discriminating measurement. No new production parameter experiment is implied for this inactive field; measure the actual corresponding state or active pathway instead.

Source. src/modern_full_model/membranes.py:206. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 27: nhe1 published k na mM

homeostasis.nhe1_published_k_na_mM

Value: 15; units: mM; active: No; classification: inactive_legacy.

Default: 15.0; differs from default: False.

Equation role. NHE1 supplies Na and TA; AE2 imports Cl and exports carbon/TA; CO2 transfers carbon only.

Constraint. Not used by selected Palk NKCC1 or Cha NHE1 flux. Retained for older models and payload compatibility.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. not_a_measured_uncertainty_parameter

Dependence of claims. No independent parent RHS sensitivity. Used only for the explicitly stated metadata, initial reference, design lineage or inverse family role.

Discriminating measurement. No new production parameter experiment is implied for this inactive field; measure the actual corresponding state or active pathway instead.

Source. src/modern_full_model/membranes.py:206. No independently verified primary measurement

attached to this numerical setting; model lineage is the source.

Record 28: nhe1 cha carrier amount fmol

homeostasis.nhe1_cha_carrier_amount_fmol

Value: 2.339370005697548e-05; units: fmol of NHE1 carriers; active: Yes; classification: WT_constraint_derived.

Default: None; differs from default: True.

Equation role. NHE1 supplies Na and TA; AE2 imports Cl and exports carbon/TA; CO2 transfers carbon only.

Constraint. One salivary carrier amount calibrated against a resting pH target 6.91 on the selected R09 background. It is not the cardiac carrier count or a direct salivary abundance measurement.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. conditional_derivation_no_experimental_interval

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value. Local numerical dependence is quantified in the linked crosswalk; it establishes neither a measured uncertainty range nor global robustness.

Discriminating measurement. Measure resting and stimulated Na/H exchange turnover during controlled acid loads, with pH and cell volume; distinguish carrier abundance from kinetic activity.

Source. src/modern_full_model/membranes.py:206. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 29: ae2 capacity fmol s

homeostasis.ae2_capacity_fmol_s

Value: 0.005; units: fmol s⁻¹; active: Yes; classification: effective_assumption.

Default: 0.005; differs from default: False.

Equation role. NHE1 supplies Na and TA; AE2 imports Cl and exports carbon/TA; CO2 transfers carbon only.

Constraint. small placeholder capacity; AE2-null is a validation gate

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. genuinely_unknown_for_physiological_transfer

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value. Local numerical dependence is quantified in the linked crosswalk; it establishes neither a measured uncertainty range nor global robustness.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. src/modern_full_model/membranes.py:206. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 30: thermodynamic saturation log width

homeostasis.thermodynamic_saturation_log_width

Value: 2; units: dimensionless log-affinity; active: Yes; classification: effective_assumption.

Default: 2.0; differs from default: False.

Equation role. NHE1 supplies Na and TA; AE2 imports Cl and exports carbon/TA; CO2 transfers carbon only.

Constraint. sets tanh approach to capacity without moving reversal

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. genuinely_unknown_for_physiological_transfer

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. src/modern_full_model/membranes.py:206. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 31: co2 basolateral permeability fmol s mM

homeostasis.co2_basolateral_permeability_fmole_s_mM

Value: 0.02; units: fmole s⁻¹ mM⁻¹; active: Yes; classification: effective_assumption.

Default: 0.02; differs from default: False.

Equation role. $J_{CO_2,b} = P_b([CO_2]_{bath} - [CO_2]_{cell})$; $J_{CO_2,a} = P_a([CO_2]_{lumen} - [CO_2]_{cell})$; modifies carbon amounts, not alkalinity.

Constraint. linear neutral CO2 exchange coefficient

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. genuinely_unknown_for_physiological_transfer

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value. Local numerical dependence is quantified in the linked crosswalk; it establishes neither a measured uncertainty range nor global robustness.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. src/modern_full_model/nbc_minimal.py:589. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 32: co2 apical permeability fmol s mM

homeostasis.co2_apical_permeability_fmole_s_mM

Value: 0.01; units: fmole s⁻¹ mM⁻¹; active: Yes; classification: effective_assumption.

Default: 0.01; differs from default: False.

Equation role. $J_{CO_2,b} = P_b([CO_2]_{bath} - [CO_2]_{cell})$; $J_{CO_2,a} = P_a([CO_2]_{lumen} - [CO_2]_{cell})$; modifies carbon amounts, not alkalinity.

Constraint. linear neutral CO2 exchange coefficient

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. genuinely_unknown_for_physiological_transfer

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. `src/modern_full_model/nbc_minimal.py:589`. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 33: nak capacity fmol s

`membranes.nak_capacity_fmols`

Value: 0.08; units: fmol cycles s^{-1} ; active: Yes; classification: `effective_assumption`.

Default: 0.08; differs from default: False.

Equation role. Pump and channel currents determine two algebraic voltages and conserved cell/lumen source vectors.

Constraint. total pump capacity shared between membranes

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. `genuinely_unknown_for_physiological_transfer`

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value. Local numerical dependence is quantified in the linked crosswalk; it establishes neither a measured uncertainty range nor global robustness.

Discriminating measurement. Measure pump current or turnover versus intracellular Na during the secretion protocol and its apical distribution.

Source. `src/modern_full_model/nbc_minimal.py:248`. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 34: nak na half mM

`membranes.nak_na_half_mM`

Value: 10; units: mM; active: Yes; classification: `inherited_model_setting`.

Default: 10.0; differs from default: False.

Equation role. Pump and channel currents determine two algebraic voltages and conserved cell/lumen source vectors.

Constraint. effective intracellular Na half-saturation starting value Source metadata identifies a model lineage; independent experimental measurement is not established here.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. `genuinely_unknown_for_physiological_transfer`

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. `src/modern_full_model/nbc_minimal.py:248`. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 35: nak k half mM

`membranes.nak_k_half_mM`

Value: 1.5; units: mM; active: Yes; classification: `inherited_model_setting`.

Default: 1.5; differs from default: False.

Equation role. Pump and channel currents determine two algebraic voltages and conserved cell/lumen source vectors.

Constraint. effective extracellular K half-saturation starting value Source metadata identifies a model lineage; independent experimental measurement is not established here.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. `genuinely_unknown_for_physiological_transfer`

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. `src/modern_full_model/nbc_minimal.py:248`. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 36: apical pump fraction

`membranes.apical_pump_fraction`

Value: 0.075075; units: fraction; active: Yes; classification: `effective_assumption`.

Default: 0.1; differs from default: True.

Equation role. Pump and channel currents determine two algebraic voltages and conserved cell/lumen source vectors.

Constraint. apical localization is supported; quantitative fraction is not measured

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. [0,1] by fractional definition only; not a measured uncertainty interval.

Uncertainty status. `genuinely_unknown_for_physiological_transfer`

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value. Local numerical dependence is quantified in the linked crosswalk; it establishes neither a measured uncertainty range nor global robustness.

Discriminating measurement. Quantify apical versus basolateral functional pump turnover, rather than inferring the fraction from localisation alone.

Source. `src/modern_full_model/nbc_minimal.py:248`. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 37: apical k fraction

`membranes.apical_k_fraction`

Value: 0.3; units: fraction; active: Yes; classification: `effective_assumption`.

Default: 0.1; differs from default: True.

Equation role. Pump and channel currents determine two algebraic voltages and conserved cell/lumen source vectors.

Constraint. apical K current is supported; quantitative fraction is not measured

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. [0,1] by fractional definition only; not a measured uncertainty interval.

Uncertainty status. genuinely_unknown_for_physiological_transfer

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. src/modern_full_model/nbc_minimal.py:248. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 38: g k total S

membranes.g_k_total_S

Value: 1.4e-08; units: S; active: Yes; classification: effective_assumption.

Default: 4e-10; differs from default: True.

Equation role. Pump and channel currents determine two algebraic voltages and conserved cell/lumen source vectors.

Constraint. total Ca-activated K conductance shared between membranes

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. genuinely_unknown_for_physiological_transfer

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. src/modern_full_model/nbc_minimal.py:248. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 39: g cl apical S

membranes.g_cl_apical_S

Value: 3.14e-08; units: S; active: Yes; classification: effective_assumption.

Default: 6e-10; differs from default: True.

Equation role. Pump and channel currents determine two algebraic voltages and conserved cell/lumen source vectors.

Constraint. apical Ca-activated Cl conductance

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. genuinely_unknown_for_physiological_transfer

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value. Local numerical dependence is quantified in the linked crosswalk; it establishes neither a measured uncertainty range nor global robustness.

Discriminating measurement. Measure CaCC current versus calcium and voltage in matched WT and AE4 deficient cells, including onset and sustained recruitment.

Source. `src/modern_full_model/nbc_minimal.py:248`. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 40: calcium_half uM

`membranes.calcium_half_uM`

Value: 0.26; units: uM; active: Yes; classification: effective_assumption.

Default: 0.3; differs from default: True.

Equation role. Pump and channel currents determine two algebraic voltages and conserved cell/lumen source vectors.

Constraint. common test gating scale, not a frozen WT estimate

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. genuinely_unknown_for_physiological_transfer

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. `src/modern_full_model/nbc_minimal.py:248`. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 41: calcium_hill

`membranes.calcium_hill`

Value: 1.46; units: dimensionless; active: Yes; classification: effective_assumption.

Default: 2.0; differs from default: True.

Equation role. Pump and channel currents determine two algebraic voltages and conserved cell/lumen source vectors.

Constraint. effective Ca gating exponent

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. genuinely_unknown_for_physiological_transfer

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. `src/modern_full_model/nbc_minimal.py:248`. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 42: g para na S

`membranes.g_para_na_S`

Value: 3e-10; units: S; active: Yes; classification: inherited_model_setting.

Default: 3e-10; differs from default: False.

Equation role. Pump and channel currents determine two algebraic voltages and conserved cell/lumen source vectors.

Constraint. historical-lineage paracellular Na pathway; value not frozen Source metadata identifies a model lineage; independent experimental measurement is not established here.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. genuinely_unknown_for_physiological_transfer

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. src/modern_full_model/nbc_minimal.py:248. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 43: g para k S

membranes.g_para_k_S

Value: 5e-11; units: S; active: Yes; classification: inherited_model_setting.

Default: 5e-11; differs from default: False.

Equation role. Pump and channel currents determine two algebraic voltages and conserved cell/lumen source vectors.

Constraint. historical-lineage paracellular K pathway; value not frozen Source metadata identifies a model lineage; independent experimental measurement is not established here.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. genuinely_unknown_for_physiological_transfer

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. src/modern_full_model/nbc_minimal.py:248. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 44: g para cl S

membranes.g_para_cl_S

Value: 2.5e-10; units: S; active: Yes; classification: effective_assumption.

Default: 2.5e-10; differs from default: False.

Equation role. Pump and channel currents determine two algebraic voltages and conserved cell/lumen source vectors.

Constraint. charge-complete paracellular Cl pathway

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. genuinely_unknown_for_physiological_transfer

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. `src/modern_full_model/nbc_minimal.py:248`. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 45: g para hco3 S

`membranes.g_para_hco3_S`

Value: 5e-11; units: S; active: Yes; classification: effective_assumption.

Default: 5e-11; differs from default: False.

Equation role. Pump and channel currents determine two algebraic voltages and conserved cell/lumen source vectors.

Constraint. charge-complete paracellular bicarbonate pathway

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. `genuinely_unknown_for_physiological_transfer`

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. `src/modern_full_model/nbc_minimal.py:248`. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 46: g apical background S

`membranes.g_apical_background_S`

Value: 0; units: S; active: Yes; classification: effective_assumption.

Default: 0.0; differs from default: False.

Equation role. Pump and channel currents determine two algebraic voltages and conserved cell/lumen source vectors.

Constraint. zero preserves the exact no-apical-K nesting limit An exact zero is an explicit structural omission, not a small positive fitted value.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Exact zero records a declared omitted contribution or protocol normalisation; it is not a confidence bound.

Uncertainty status. `genuinely_unknown_for_physiological_transfer`

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. `src/modern_full_model/nbc_minimal.py:248`. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 47: g basolateral background S

`membranes.g_basolateral_background_S`

Value: 0; units: S; active: Yes; classification: effective_assumption.

Default: 5e-12; differs from default: True.

Equation role. Pump and channel currents determine two algebraic voltages and conserved cell/lumen source vectors.

Constraint. small K-selective regularizer; set zero only if closure remains nonsingular An exact zero is an explicit structural omission, not a small positive fitted value.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Exact zero records a declared omitted contribution or protocol normalisation; it is not a confidence bound.

Uncertainty status. genuinely_unknown_for_physiological_transfer

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. `src/modern_full_model/nbc_minimal.py:248`. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 48: apical hydraulic pL s mOsm

`water.apical_hydraulic_pL_s_mOsm`

Value: 0.00432; units: $\text{pL s}^{-1} (\text{mOsm L}^{-1})^{-1}$; active: Yes; classification: published_model_conversion.

Default: 0.00432; differs from default: False.

Equation role. $Q_b = L_b(O_i - O_b)$, $Q_a = L_a(O_l - O_i)$, $Q_p = L_p(O_l - O_b)$, $Q_{out} = k_{out} \max(V_l - V_{dead}, 0)$.

Constraint. Palk 2010/2018 $\text{Pa} = 4.32 \times 10^{-12} \text{ L}^2 \text{ mol}^{-1} \text{ s}^{-1}$ converted by $1 \text{ mM} = 10^{-3} \text{ mol/L}$ and $1 \text{ L} = 10^{12} \text{ pL}$ Inherited model coefficient, not a new direct measurement.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. genuinely_unknown_for_physiological_transfer

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value. Local numerical dependence is quantified in the linked crosswalk; it establishes neither a measured uncertainty range nor global robustness.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. `src/modern_full_model/water.py:93`. Palk 2010/2018 inherited whole cell water coefficients; dimensional conversion is recorded in `parameters.py`.

Record 49: basolateral hydraulic pL s mOsm

`water.basolateral_hydraulic_pL_s_mOsm`

Value: 0.0515; units: $\text{pL s}^{-1} (\text{mOsm L}^{-1})^{-1}$; active: Yes; classification: published_model_conversion.

Default: 0.0515; differs from default: False.

Equation role. $Q_b = L_b(O_i - O_b)$, $Q_a = L_a(O_l - O_i)$, $Q_p = L_p(O_l - O_b)$, $Q_{out} = k_{out}$

$\max(V_l - V_{\text{dead}}, 0)$.

Constraint. Palk 2010/2018 $P_b = 5.15 \times 10^{-11} \text{ L}^2 \text{ mol}^{-1} \text{ s}^{-1}$ converted by $1 \text{ mM} = 10^{-3} \text{ mol/L}$ and $1 \text{ L} = 10^{12} \text{ pL}$ Inherited model coefficient, not a new direct measurement.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. `genuinely_unknown_for_physiological_transfer`

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. `src/modern_full_model/water.py:93`. Palk 2010/2018 inherited whole cell water coefficients; dimensional conversion is recorded in `parameters.py`.

Record 50: paracellular hydraulic pL s mOsm

`water.paracellular_hydraulic_pL_s_mOsm`

Value: 0.00026; units: $\text{pL s}^{-1} (\text{mOsm L}^{-1})^{-1}$; active: Yes; classification: `published_model_conversion`.
Default: 0.00026; differs from default: False.

Equation role. $Q_b = L_b(O_i - O_b)$, $Q_a = L_a(O_l - O_i)$, $Q_p = L_p(O_l - O_b)$, $Q_{\text{out}} = k_{\text{out}} \max(V_l - V_{\text{dead}}, 0)$.

Constraint. Palk 2010/2018 $P_t = 2.6 \times 10^{-13} \text{ L}^2 \text{ mol}^{-1} \text{ s}^{-1}$ converted by $1 \text{ mM} = 10^{-3} \text{ mol/L}$ and $1 \text{ L} = 10^{12} \text{ pL}$ Inherited model coefficient, not a new direct measurement.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. `genuinely_unknown_for_physiological_transfer`

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. `src/modern_full_model/water.py:93`. Palk 2010/2018 inherited whole cell water coefficients; dimensional conversion is recorded in `parameters.py`.

Record 51: lumen dead volume pL

`water.lumen_dead_volume_pL`

Value: 0.1; units: pL; active: Yes; classification: `effective_assumption`.
Default: 0.1; differs from default: False.

Equation role. $Q_b = L_b(O_i - O_b)$, $Q_a = L_a(O_l - O_i)$, $Q_p = L_p(O_l - O_b)$, $Q_{\text{out}} = k_{\text{out}} \max(V_l - V_{\text{dead}}, 0)$.

Constraint. non-draining local lumen volume

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. `genuinely_unknown_for_physiological_transfer`

Dependence of claims. Numerical state, current and secretion claims are conditional on this active

setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. `src/modern_full_model/water.py:93`. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 52: outflow rate s

`water.outflow_rate_s`

Value: 0.2; units: s^{-1} ; active: Yes; classification: `effective_assumption`.

Default: 0.2; differs from default: False.

Equation role. $Q_b = L_b(O_i - O_b)$, $Q_a = L_a(O_l - O_i)$, $Q_p = L_p(O_l - O_b)$, $Q_{out} = k_{out} \max(V_l - V_{dead}, 0)$.

Constraint. local compliance/outflow closure, not secretion calibration

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. `genuinely_unknown_for_physiological_transfer`

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. `src/modern_full_model/water.py:93`. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 53: cell na mM

`initial.cell_na_mM`

Value: 15; units: mM; active: No; classification: `constructor_seed`.

Default: 15.0; differs from default: False.

Equation role. Constructor reference concentrations and volumes define a seed and charge consistency check, not the frozen production initial condition.

Constraint. Constructor reference tuple, not the actual production initial state. Task 40 uses the independently frozen WT state; this tuple also defines charge consistency checks.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. `not_a_measured_uncertainty_parameter`

Dependence of claims. No independent parent RHS sensitivity. Used only for the explicitly stated metadata, initial reference, design lineage or inverse family role.

Discriminating measurement. No new production parameter experiment is implied for this inactive field; measure the actual corresponding state or active pathway instead.

Source. `src/modern_full_model/model.py:319`. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 54: cell k mM

`initial.cell_k_mM`

Value: 140; units: mM; active: No; classification: constructor_seed.

Default: 140.0; differs from default: False.

Equation role. Constructor reference concentrations and volumes define a seed and charge consistency check, not the frozen production initial condition.

Constraint. Constructor reference tuple, not the actual production initial state. Task 40 uses the independently frozen WT state; this tuple also defines charge consistency checks.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. not_a_measured_uncertainty_parameter

Dependence of claims. No independent parent RHS sensitivity. Used only for the explicitly stated metadata, initial reference, design lineage or inverse family role.

Discriminating measurement. No new production parameter experiment is implied for this inactive field; measure the actual corresponding state or active pathway instead.

Source. src/modern_full_model/model.py:319. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 55: cell cl mM

initial.cell_cl_mM

Value: 40; units: mM; active: No; classification: constructor_seed.

Default: 40.0; differs from default: False.

Equation role. Constructor reference concentrations and volumes define a seed and charge consistency check, not the frozen production initial condition.

Constraint. Constructor reference tuple, not the actual production initial state. Task 40 uses the independently frozen WT state; this tuple also defines charge consistency checks.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. not_a_measured_uncertainty_parameter

Dependence of claims. No independent parent RHS sensitivity. Used only for the explicitly stated metadata, initial reference, design lineage or inverse family role.

Discriminating measurement. No new production parameter experiment is implied for this inactive field; measure the actual corresponding state or active pathway instead.

Source. src/modern_full_model/model.py:319. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 56: cell tic mM

initial.cell_tic_mM

Value: 20; units: mM; active: No; classification: constructor_seed.

Default: 20.0; differs from default: False.

Equation role. Constructor reference concentrations and volumes define a seed and charge consistency check, not the frozen production initial condition.

Constraint. Constructor reference tuple, not the actual production initial state. Task 40 uses the independently frozen WT state; this tuple also defines charge consistency checks.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. not_a_measured_uncertainty_parameter

Dependence of claims. No independent parent RHS sensitivity. Used only for the explicitly stated metadata, initial reference, design lineage or inverse family role.

Discriminating measurement. No new production parameter experiment is implied for this inactive field; measure the actual corresponding state or active pathway instead.

Source. src/modern_full_model/model.py:319. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 57: cell ph

initial.cell_ph

Value: 7.2; units: dimensionless (p-scale); active: No; classification: constructor_seed.

Default: 7.2; differs from default: False.

Equation role. Constructor reference concentrations and volumes define a seed and charge consistency check, not the frozen production initial condition.

Constraint. Constructor reference tuple, not the actual production initial state. Task 40 uses the independently frozen WT state; this tuple also defines charge consistency checks.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. not_a_measured_uncertainty_parameter

Dependence of claims. No independent parent RHS sensitivity. Used only for the explicitly stated metadata, initial reference, design lineage or inverse family role.

Discriminating measurement. No new production parameter experiment is implied for this inactive field; measure the actual corresponding state or active pathway instead.

Source. src/modern_full_model/model.py:319. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 58: cell volume pL

initial.cell_volume_pL

Value: 1; units: pL; active: No; classification: constructor_seed.

Default: 1.0; differs from default: False.

Equation role. Constructor reference concentrations and volumes define a seed and charge consistency check, not the frozen production initial condition.

Constraint. Constructor reference tuple, not the actual production initial state. Task 40 uses the independently frozen WT state; this tuple also defines charge consistency checks.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. not_a_measured_uncertainty_parameter

Dependence of claims. No independent parent RHS sensitivity. Used only for the explicitly stated metadata, initial reference, design lineage or inverse family role.

Discriminating measurement. No new production parameter experiment is implied for this inactive field; measure the actual corresponding state or active pathway instead.

Source. src/modern_full_model/model.py:319. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 59: lumen na mM

`initial.lumen_na_mM`

Value: 145; units: mM; active: No; classification: constructor_seed.

Default: 145.0; differs from default: False.

Equation role. Constructor reference concentrations and volumes define a seed and charge consistency check, not the frozen production initial condition.

Constraint. Constructor reference tuple, not the actual production initial state. Task 40 uses the independently frozen WT state; this tuple also defines charge consistency checks.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. not_a_measured_uncertainty_parameter

Dependence of claims. No independent parent RHS sensitivity. Used only for the explicitly stated metadata, initial reference, design lineage or inverse family role.

Discriminating measurement. No new production parameter experiment is implied for this inactive field; measure the actual corresponding state or active pathway instead.

Source. `src/modern_full_model/model.py:319`. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 60: lumen k mM

`initial.lumen_k_mM`

Value: 5; units: mM; active: No; classification: constructor_seed.

Default: 5.0; differs from default: False.

Equation role. Constructor reference concentrations and volumes define a seed and charge consistency check, not the frozen production initial condition.

Constraint. Constructor reference tuple, not the actual production initial state. Task 40 uses the independently frozen WT state; this tuple also defines charge consistency checks.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. not_a_measured_uncertainty_parameter

Dependence of claims. No independent parent RHS sensitivity. Used only for the explicitly stated metadata, initial reference, design lineage or inverse family role.

Discriminating measurement. No new production parameter experiment is implied for this inactive field; measure the actual corresponding state or active pathway instead.

Source. `src/modern_full_model/model.py:319`. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 61: lumen cl mM

`initial.lumen_cl_mM`

Value: 126.1615927965083; units: mM; active: No; classification: constructor_seed.

Default: 126.16159279650834; differs from default: False.

Equation role. Constructor reference concentrations and volumes define a seed and charge consistency check, not the frozen production initial condition.

Constraint. Constructor reference tuple, not the actual production initial state. Task 40 uses the independently frozen WT state; this tuple also defines charge consistency checks.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. not_a_measured_uncertainty_parameter

Dependence of claims. No independent parent RHS sensitivity. Used only for the explicitly stated metadata, initial reference, design lineage or inverse family role.

Discriminating measurement. No new production parameter experiment is implied for this inactive field; measure the actual corresponding state or active pathway instead.

Source. src/modern_full_model/model.py:319. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 62: lumen tic mM

initial.lumen_tic_mM

Value: 25; units: mM; active: No; classification: constructor_seed.

Default: 25.0; differs from default: False.

Equation role. Constructor reference concentrations and volumes define a seed and charge consistency check, not the frozen production initial condition.

Constraint. Constructor reference tuple, not the actual production initial state. Task 40 uses the independently frozen WT state; this tuple also defines charge consistency checks.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. not_a_measured_uncertainty_parameter

Dependence of claims. No independent parent RHS sensitivity. Used only for the explicitly stated metadata, initial reference, design lineage or inverse family role.

Discriminating measurement. No new production parameter experiment is implied for this inactive field; measure the actual corresponding state or active pathway instead.

Source. src/modern_full_model/model.py:319. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 63: lumen ph

initial.lumen_ph

Value: 7.4; units: dimensionless (p-scale); active: No; classification: constructor_seed.

Default: 7.4; differs from default: False.

Equation role. Constructor reference concentrations and volumes define a seed and charge consistency check, not the frozen production initial condition.

Constraint. Constructor reference tuple, not the actual production initial state. Task 40 uses the independently frozen WT state; this tuple also defines charge consistency checks.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. not_a_measured_uncertainty_parameter

Dependence of claims. No independent parent RHS sensitivity. Used only for the explicitly stated metadata, initial reference, design lineage or inverse family role.

Discriminating measurement. No new production parameter experiment is implied for this inactive field; measure the actual corresponding state or active pathway instead.

Source. src/modern_full_model/model.py:319. No independently verified primary measurement

attached to this numerical setting; model lineage is the source.

Record 64: lumen volume pL

`initial.lumen_volume_pL`

Value: 0.1; units: pL; active: No; classification: `constructor_seed`.

Default: 0.1; differs from default: False.

Equation role. Constructor reference concentrations and volumes define a seed and charge consistency check, not the frozen production initial condition.

Constraint. Constructor reference tuple, not the actual production initial state. Task 40 uses the independently frozen WT state; this tuple also defines charge consistency checks.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. `not_a_measured_uncertainty_parameter`

Dependence of claims. No independent parent RHS sensitivity. Used only for the explicitly stated metadata, initial reference, design lineage or inverse family role.

Discriminating measurement. No new production parameter experiment is implied for this inactive field; measure the actual corresponding state or active pathway instead.

Source. `src/modern_full_model/model.py:319`. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 65: carrier amount fmol

`ae4.carrier_amount_fmol`

Value: 0.1501264265070657; units: fmol; active: Yes; classification: `inherited_effective_setting`.

Default: ; differs from default: .

Equation role. Shared carrier QSS determines scalar J4; equal routing imposes (-J4/2,-J4/2,J4,-2J4,-2J4) in cell Na,K,Cl,TIC,TA.

Constraint. Shared-carrier model setting. Thermodynamic construction fixes forward/reverse ratios, not this absolute abundance or attempt rate.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. `genuinely_unknown_for_physiological_transfer`

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value. Local numerical dependence is quantified in the linked crosswalk; it establishes neither a measured uncertainty range nor global robustness.

Discriminating measurement. Measure absolute AE4 abundance and exchange turnover in the same preparation; expression alone does not fix per carrier turnover.

Source. `src/modern_full_model/transporters.py:458`. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 66: common cl attempt rate s

`ae4.common_cl_attempt_rate_s`

Value: 1; units: s^{-1} ; active: Yes; classification: `inherited_effective_setting`.

Default: ; differs from default: .

Equation role. Shared carrier QSS determines scalar J_4 ; equal routing imposes $(-J_4/2, -J_4/2, J_4, -2J_4, -2J_4)$ in cell Na,K,Cl,TIC,TA.

Constraint. Shared-carrier model setting. Thermodynamic construction fixes forward/reverse ratios, not this absolute abundance or attempt rate.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. genuinely_unknown_for_physiological_transfer

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. src/modern_full_model/transporters.py:458. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 67: na loaded attempt rate s

ae4.na_loaded_attempt_rate_s

Value: 0.1; units: s^{-1} ; active: Yes; classification: inherited_effective_setting.

Default: ; differs from default: .

Equation role. Shared carrier QSS determines scalar J_4 ; equal routing imposes $(-J_4/2, -J_4/2, J_4, -2J_4, -2J_4)$ in cell Na,K,Cl,TIC,TA.

Constraint. Shared-carrier model setting. Thermodynamic construction fixes forward/reverse ratios, not this absolute abundance or attempt rate.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. genuinely_unknown_for_physiological_transfer

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. src/modern_full_model/transporters.py:458. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 68: k loaded attempt rate s

ae4.k_loaded_attempt_rate_s

Value: 1.9; units: s^{-1} ; active: Yes; classification: inherited_effective_setting.

Default: ; differs from default: .

Equation role. Shared carrier QSS determines scalar J_4 ; equal routing imposes $(-J_4/2, -J_4/2, J_4, -2J_4, -2J_4)$ in cell Na,K,Cl,TIC,TA.

Constraint. Shared-carrier model setting. Thermodynamic construction fixes forward/reverse ratios, not this absolute abundance or attempt rate.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. genuinely_unknown_for_physiological_transfer

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. src/modern_full_model/transporters.py:458. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 69: cooperative gate

ae4.cooperative_gate

Value: None; units: configuration; active: No; classification: inherited_effective_setting.

Default: ; differs from default: .

Equation role. Shared carrier QSS determines scalar J4; equal routing imposes $(-J4/2, -J4/2, J4, -2J4, -2J4)$ in cell Na,K,Cl,TIC,TA.

Constraint. Shared-carrier model setting. Thermodynamic construction fixes forward/reverse ratios, not this absolute abundance or attempt rate.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Selected discrete setting or stated numerical reference.

Uncertainty status. genuinely_unknown_for_physiological_transfer

Dependence of claims. No independent parent RHS sensitivity. Used only for the explicitly stated metadata, initial reference, design lineage or inverse family role.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. src/modern_full_model/transporters.py:458. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 70: capacity fmol s

nbc.capacity_fmол_s

Value: 0.115709131974644; units: fmol/s; active: Yes; classification: WT_architecture_derived.

Default: ; differs from default: .

Equation role. $JB = G_B u(Ca) \tanh(\text{affinity}/\log_width)$, with electrogenic 1 Na and 2 bicarbonate inward per cycle.

Constraint. Derived from an assumed 70/30 positive chloride loading architecture at the Task 31 reference state, a 2.3 NHE1 multiplier and coupled electrical closure. Derived conditional on those assumptions, not measured or identified by knockout data.

Limit of justification. The 70/30 partition is assumed, not measured. Solving the coupled current balance does not establish the physiological truth or unique identification of that partition.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. conditional_derivation_no_experimental_interval

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value. Local numerical dependence is quantified in the linked crosswalk; it establishes neither a measured uncertainty range nor global robustness.

Discriminating measurement. Measure Na dependent bicarbonate loading or alkalinity recovery during stimulation with calibrated pH, TIC and volume, separating NHE1 and AE2 contributions; establish molecular identity independently.

Source. `src/modern_full_model/nbc_minimal.py:190`. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 71: log width

`nbc.log_width`

Value: 2; units: dimensionless; active: Yes; classification: `effective_assumption`.

Default: ; differs from default: .

Equation role. $JB = G_B u(Ca) \tanh(\text{affinity}/\text{log_width})$, with electrogenic 1 Na and 2 bicarbonate inward per cycle.

Constraint. Inherited NBC recruitment parameter; not an isoform-specific measurement.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. `genuinely_unknown_for_physiological_transfer`

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value. Local numerical dependence is quantified in the linked crosswalk; it establishes neither a measured uncertainty range nor global robustness.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. `src/modern_full_model/nbc_minimal.py:190`. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 72: resting calcium uM

`nbc.resting_calcium_uM`

Value: 0.058; units: uM; active: Yes; classification: `effective_assumption`.

Default: ; differs from default: .

Equation role. $JB = G_B u(Ca) \tanh(\text{affinity}/\text{log_width})$, with electrogenic 1 Na and 2 bicarbonate inward per cycle.

Constraint. Inherited NBC recruitment parameter; not an isoform-specific measurement.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. `genuinely_unknown_for_physiological_transfer`

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. `src/modern_full_model/nbc_minimal.py:190`. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 73: fully recruited calcium uM

`nbc.fully_recruited_calcium_uM`

Value: 0.25; units: uM; active: Yes; classification: `effective_assumption`.

Default: ; differs from default: .

Equation role. $J_B = G_B u(\text{Ca}) \tanh(\text{affinity}/\log_width)$, with electrogenic 1 Na and 2 bicarbonate inward per cycle.

Constraint. Inherited NBC recruitment parameter; not an isoform-specific measurement.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. genuinely_unknown_for_physiological_transfer

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. src/modern_full_model/nbc_minimal.py:190. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 74: nhe1 stimulated multiplier

nbc.nhe1_stimulated_multiplier

Value: 2.3; units: dimensionless; active: Yes; classification: effective_assumption.

Default: ; differs from default: .

Equation role. $J_H_stim = [1 + (\text{multiplier} - 1) * u(\text{Ca})] * J_H_Cha$; shares the NBC recruitment endpoints.

Constraint. Inherited source-fixed 2.3 multiplier. Fixing it before a test prevents retuning but does not make its transfer to this tissue/protocol uncertainty-free.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. At least 1 by the selected model constructor; this is a structural choice, not an experimental lower bound.

Uncertainty status. genuinely_unknown_for_physiological_transfer

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value. Local numerical dependence is quantified in the linked crosswalk; it establishes neither a measured uncertainty range nor global robustness.

Discriminating measurement. Measure the ratio of stimulated to resting NHE1 flux in the same salivary preparation and calcium protocol.

Source. src/modern_full_model/nbc_minimal.py:149. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 75: alpha eff fmo1 s

nkcc.alpha_eff_fmo1_s

Value: 0.01733474689409605; units: fmo1/s; active: Yes; classification: WT_constraint_derived.

Default: ; differs from default: .

Equation role. $J_N = \alpha * m_N * (A1 - A2[Na]i[K]i[Cl]i^2) / (A3 + A4[Na]i[K]i[Cl]i^2)$; inward 1 Na, 1 K, 2 Cl per cycle.

Constraint. J_N ,rest divided by the published shape at the accepted resting concentrations. The resting flux is itself an inherited model value, not a direct new measurement.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. conditional_derivation_no_experimental_interval

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value. Local numerical dependence is quantified in the linked crosswalk; it establishes neither a measured uncertainty range nor global robustness.

Discriminating measurement. Measure acute intact cell NKCC chloride uptake during AE4 loss with simultaneous intracellular Na, K and Cl; compare separately with the isolated 2015 uptake assay.

Source. `src/modern_full_model/nkcc1_palk2010.py:31`. Palk et al. 2010/Benjamin and Johnson 1997 law, reproduced as Vera Siguenza et al. 2018 Eq. 17; coefficients in `src/modern_full_model/nkcc1_palk2010.py`.

Record 76: A1

`nkcc.A1`

Value: 157.5; units: dimensionless; active: Yes; classification: `published_kinetic_coefficient`.

Default: ; differs from default: .

Equation role. $JN = \alpha \cdot m_N \cdot (A1 - A2[Na]i[K]i[Cl]i^2) / (A3 + A4[Na]i[K]i[Cl]i^2)$; inward 1 Na, 1 K, 2 Cl per cycle.

Constraint. Palk/Benjamin effective two-state law in the fixed bath, with fourth-order concentration factors converted to mM. Do not vary bath composition without revisiting these fixed-bath coefficients.

Limit of justification. Publication of the kinetic law does not measure salivary carrier abundance or establish an uncertainty interval for transfer to this preparation.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. `genuinely_unknown_for_physiological_transfer`

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. `src/modern_full_model/nkcc1_palk2010.py:31`. Palk et al. 2010/Benjamin and Johnson 1997 law, reproduced as Vera Siguenza et al. 2018 Eq. 17; coefficients in `src/modern_full_model/nkcc1_palk2010.py`.

Record 77: A2

`nkcc.A2`

Value: 2.0096e-05; units: mM^{-4} ; active: Yes; classification: `published_kinetic_coefficient`.

Default: ; differs from default: .

Equation role. $JN = \alpha \cdot m_N \cdot (A1 - A2[Na]i[K]i[Cl]i^2) / (A3 + A4[Na]i[K]i[Cl]i^2)$; inward 1 Na, 1 K, 2 Cl per cycle.

Constraint. Palk/Benjamin effective two-state law in the fixed bath, with fourth-order concentration factors converted to mM. Do not vary bath composition without revisiting these fixed-bath coefficients.

Limit of justification. Publication of the kinetic law does not measure salivary carrier abundance or establish an uncertainty interval for transfer to this preparation.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. `genuinely_unknown_for_physiological_transfer`

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. `src/modern_full_model/nkcc1_palk2010.py:31`. Palk et al. 2010/Benjamin

and Johnson 1997 law, reproduced as Vera Siguenza et al. 2018 Eq. 17; coefficients in `src/modern_full_model/nkcc1_palk2010.py`.

Record 78: A3

`nkcc.A3`

Value: 1.0306; units: dimensionless; active: Yes; classification: `published_kinetic_coefficient`.

Default: ; differs from default: .

Equation role. $JN = \alpha \cdot m_N \cdot (A1 - A2[Na]i[K]i[Cl]i^2) / (A3 + A4[Na]i[K]i[Cl]i^2)$; inward 1 Na, 1 K, 2 Cl per cycle.

Constraint. Palk/Benjamin effective two-state law in the fixed bath, with fourth-order concentration factors converted to mM. Do not vary bath composition without revisiting these fixed-bath coefficients.

Limit of justification. Publication of the kinetic law does not measure salivary carrier abundance or establish an uncertainty interval for transfer to this preparation.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. `genuinely_unknown_for_physiological_transfer`

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. `src/modern_full_model/nkcc1_palk2010.py:31`. Palk et al. 2010/Benjamin and Johnson 1997 law, reproduced as Vera Siguenza et al. 2018 Eq. 17; coefficients in `src/modern_full_model/nkcc1_palk2010.py`.

Record 79: A4

`nkcc.A4`

Value: 1.3852e-06; units: mM^{-4} ; active: Yes; classification: `published_kinetic_coefficient`.

Default: ; differs from default: .

Equation role. $JN = \alpha \cdot m_N \cdot (A1 - A2[Na]i[K]i[Cl]i^2) / (A3 + A4[Na]i[K]i[Cl]i^2)$; inward 1 Na, 1 K, 2 Cl per cycle.

Constraint. Palk/Benjamin effective two-state law in the fixed bath, with fourth-order concentration factors converted to mM. Do not vary bath composition without revisiting these fixed-bath coefficients.

Limit of justification. Publication of the kinetic law does not measure salivary carrier abundance or establish an uncertainty interval for transfer to this preparation.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. `genuinely_unknown_for_physiological_transfer`

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. `src/modern_full_model/nkcc1_palk2010.py:31`. Palk et al. 2010/Benjamin and Johnson 1997 law, reproduced as Vera Siguenza et al. 2018 Eq. 17; coefficients in `src/modern_full_model/nkcc1_palk2010.py`.

Record 80: k1 plus per ms

`cha.k1_plus_per_ms`

Value: 10.5; units: ms^{-1} ; active: Yes; classification: published_kinetic_coefficient.

Default: ; differs from default: .

Equation role. $JH=N_H*m_H*Mod2*(a*b-c*d)/(a+b+c+d)$, with rate conversion from ms to s and printed fourth reverse rate retained.

Constraint. Printed Cha 2009 Table S1 coefficient, transferred as a kinetic law. The printed fourth reverse rate is retained despite its documented rounding discrepancy from exact microscopic reversibility.

Limit of justification. Publication of the kinetic law does not measure salivary carrier abundance or establish an uncertainty interval for transfer to this preparation.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. genuinely_unknown_for_physiological_transfer

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. src/modern_full_model/nhe1_cha2009.py:113. Cha et al. 2009, Eqs. 3 and 5, Table S1; DOI 10.1016/j.bpj.2009.08.053; repository results/31_nhe1_mechanistic_repair/source_table_s1.json.

Record 81: k1 minus per ms

cha.k1_minus_per_ms

Value: 0.201; units: ms^{-1} ; active: Yes; classification: published_kinetic_coefficient.

Default: ; differs from default: .

Equation role. $JH=N_H*m_H*Mod2*(a*b-c*d)/(a+b+c+d)$, with rate conversion from ms to s and printed fourth reverse rate retained.

Constraint. Printed Cha 2009 Table S1 coefficient, transferred as a kinetic law. The printed fourth reverse rate is retained despite its documented rounding discrepancy from exact microscopic reversibility.

Limit of justification. Publication of the kinetic law does not measure salivary carrier abundance or establish an uncertainty interval for transfer to this preparation.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. genuinely_unknown_for_physiological_transfer

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. src/modern_full_model/nhe1_cha2009.py:113. Cha et al. 2009, Eqs. 3 and 5, Table S1; DOI 10.1016/j.bpj.2009.08.053; repository results/31_nhe1_mechanistic_repair/source_table_s1.json.

Record 82: k2 plus per ms

cha.k2_plus_per_ms

Value: 15.8; units: ms^{-1} ; active: Yes; classification: published_kinetic_coefficient.

Default: ; differs from default: .

Equation role. $JH=N_H*m_H*Mod2*(a*b-c*d)/(a+b+c+d)$, with rate conversion from ms to s and printed fourth reverse rate retained.

Constraint. Printed Cha 2009 Table S1 coefficient, transferred as a kinetic law. The printed fourth reverse rate is retained despite its documented rounding discrepancy from exact microscopic reversibility.

Limit of justification. Publication of the kinetic law does not measure salivary carrier abundance or

establish an uncertainty interval for transfer to this preparation.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. genuinely_unknown_for_physiological_transfer

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. src/modern_full_model/nhe1_cha2009.py:113. Cha et al. 2009, Eqs. 3 and 5, Table S1; DOI 10.1016/j.bpj.2009.08.053; repository results/31_nhe1_mechanistic_repair/source_table_s1.json.

Record 83: k na o mM

cha.k_na_o_mM

Value: 195; units: mM; active: Yes; classification: published_kinetic_coefficient.

Default: ; differs from default: .

Equation role. $JH = N_H \cdot m_H \cdot \text{Mod}2(a \cdot b \cdot c \cdot d) / (a + b + c + d)$, with rate conversion from ms to s and printed fourth reverse rate retained.

Constraint. Printed Cha 2009 Table S1 coefficient, transferred as a kinetic law. The printed fourth reverse rate is retained despite its documented rounding discrepancy from exact microscopic reversibility.

Limit of justification. Publication of the kinetic law does not measure salivary carrier abundance or establish an uncertainty interval for transfer to this preparation.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. genuinely_unknown_for_physiological_transfer

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. src/modern_full_model/nhe1_cha2009.py:113. Cha et al. 2009, Eqs. 3 and 5, Table S1; DOI 10.1016/j.bpj.2009.08.053; repository results/31_nhe1_mechanistic_repair/source_table_s1.json.

Record 84: k na i mM

cha.k_na_i_mM

Value: 16.2; units: mM; active: Yes; classification: published_kinetic_coefficient.

Default: ; differs from default: .

Equation role. $JH = N_H \cdot m_H \cdot \text{Mod}2(a \cdot b \cdot c \cdot d) / (a + b + c + d)$, with rate conversion from ms to s and printed fourth reverse rate retained.

Constraint. Printed Cha 2009 Table S1 coefficient, transferred as a kinetic law. The printed fourth reverse rate is retained despite its documented rounding discrepancy from exact microscopic reversibility.

Limit of justification. Publication of the kinetic law does not measure salivary carrier abundance or establish an uncertainty interval for transfer to this preparation.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. genuinely_unknown_for_physiological_transfer

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before

making global robustness claims.

Source. `src/modern_full_model/nhe1_cha2009.py:113`. Cha et al. 2009, Eqs. 3 and 5, Table S1; DOI 10.1016/j.bpj.2009.08.053; repository results/31_nhe1_mechanistic_repair/source_table_s1.json.

Record 85: k h o mM

`cha.k_h_o_mM`

Value: 0.00162; units: mM; active: Yes; classification: published_kinetic_coefficient.

Default: ; differs from default: .

Equation role. $JH = N_H m_H \text{Mod}2(a*b-c*d)/(a+b+c+d)$, with rate conversion from ms to s and printed fourth reverse rate retained.

Constraint. Printed Cha 2009 Table S1 coefficient, transferred as a kinetic law. The printed fourth reverse rate is retained despite its documented rounding discrepancy from exact microscopic reversibility.

Limit of justification. Publication of the kinetic law does not measure salivary carrier abundance or establish an uncertainty interval for transfer to this preparation.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. genuinely_unknown_for_physiological_transfer

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. `src/modern_full_model/nhe1_cha2009.py:113`. Cha et al. 2009, Eqs. 3 and 5, Table S1; DOI 10.1016/j.bpj.2009.08.053; repository results/31_nhe1_mechanistic_repair/source_table_s1.json.

Record 86: k h i mM

`cha.k_h_i_mM`

Value: 0.000605; units: mM; active: Yes; classification: published_kinetic_coefficient.

Default: ; differs from default: .

Equation role. $JH = N_H m_H \text{Mod}2(a*b-c*d)/(a+b+c+d)$, with rate conversion from ms to s and printed fourth reverse rate retained.

Constraint. Printed Cha 2009 Table S1 coefficient, transferred as a kinetic law. The printed fourth reverse rate is retained despite its documented rounding discrepancy from exact microscopic reversibility.

Limit of justification. Publication of the kinetic law does not measure salivary carrier abundance or establish an uncertainty interval for transfer to this preparation.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. genuinely_unknown_for_physiological_transfer

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. `src/modern_full_model/nhe1_cha2009.py:113`. Cha et al. 2009, Eqs. 3 and 5, Table S1; DOI 10.1016/j.bpj.2009.08.053; repository results/31_nhe1_mechanistic_repair/source_table_s1.json.

Record 87: modifier k i mM

`cha.modifier_k_i_mM`

Value: 3.07e-05; units: mM; active: Yes; classification: published_kinetic_coefficient.

Default: ; differs from default: .

Equation role. $JH=N_H*m_H*Mod2*(a*b-c*d)/(a+b+c+d)$, with rate conversion from ms to s and printed fourth reverse rate retained.

Constraint. Printed Cha 2009 Table S1 coefficient, transferred as a kinetic law. The printed fourth reverse rate is retained despite its documented rounding discrepancy from exact microscopic reversibility.

Limit of justification. Publication of the kinetic law does not measure salivary carrier abundance or establish an uncertainty interval for transfer to this preparation.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. genuinely_unknown_for_physiological_transfer

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. src/modern_full_model/nhe1_cha2009.py:113. Cha et al. 2009, Eqs. 3 and 5, Table S1; DOI 10.1016/j.bpj.2009.08.053; repository results/31_nhe1_mechanistic_repair/source_table_s1.json.

Record 88: modifier k o mM

cha.modifier_k_o_mM

Value: 4.8e-07; units: mM; active: Yes; classification: published_kinetic_coefficient.

Default: ; differs from default: .

Equation role. $JH=N_H*m_H*Mod2*(a*b-c*d)/(a+b+c+d)$, with rate conversion from ms to s and printed fourth reverse rate retained.

Constraint. Printed Cha 2009 Table S1 coefficient, transferred as a kinetic law. The printed fourth reverse rate is retained despite its documented rounding discrepancy from exact microscopic reversibility.

Limit of justification. Publication of the kinetic law does not measure salivary carrier abundance or establish an uncertainty interval for transfer to this preparation.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. genuinely_unknown_for_physiological_transfer

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. src/modern_full_model/nhe1_cha2009.py:113. Cha et al. 2009, Eqs. 3 and 5, Table S1; DOI 10.1016/j.bpj.2009.08.053; repository results/31_nhe1_mechanistic_repair/source_table_s1.json.

Record 89: k2 minus table per ms

cha.k2_minus_table_per_ms

Value: 183; units: ms⁻¹; active: Yes; classification: published_kinetic_coefficient.

Default: ; differs from default: .

Equation role. $JH=N_H*m_H*Mod2*(a*b-c*d)/(a+b+c+d)$, with rate conversion from ms to s and printed fourth reverse rate retained.

Constraint. Printed Cha 2009 Table S1 coefficient, transferred as a kinetic law. The printed fourth reverse rate is retained despite its documented rounding discrepancy from exact microscopic reversibility.

Limit of justification. Publication of the kinetic law does not measure salivary carrier abundance or

establish an uncertainty interval for transfer to this preparation.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. genuinely_unknown_for_physiological_transfer

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. src/modern_full_model/nhe1_cha2009.py:113. Cha et al. 2009, Eqs. 3 and 5, Table S1; DOI 10.1016/j.bpj.2009.08.053; repository results/31_nhe1_mechanistic_repair/source_table_s1.json.

Record 90: tau ae4 s

regulation.tau_ae4_s

Value: 30; units: s; active: Yes; classification: effective_assumption.

Default: ; differs from default: .

Equation role. $da/dt = (\beta - a)/\tau$; AE4 multiplier = basal + increment * coupling * construct_scale * a.

Constraint. A 30 s effective response time, not a resolved measurement of the intermediate signalling kinetics.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. genuinely_unknown_for_physiological_transfer

Dependence of claims. Regulatory transient claims depend on the selected family/timescale and gain; constant full stimulus equilibrium is independent of tau within R1. Local numerical dependence is quantified in the linked crosswalk; it establishes neither a measured uncertainty range nor global robustness.

Discriminating measurement. Acquire acute beta input, cAMP/PKA activity and AE4 transport time series to distinguish regulation from slow ionic and volume modes.

Source. src/modern_full_model/camp_pka.py:90. Pena Munzenmayer et al. 2021 gives pathway and endpoint activation evidence; acute kinetic times and exact common capacity gain remain effective settings.

Record 91: basal capacity multiplier

regulation.basal_capacity_multiplier

Value: 1; units: dimensionless; active: Yes; classification: effective_assumption.

Default: ; differs from default: .

Equation role. $da/dt = (\beta - a)/\tau$; AE4 multiplier = basal + increment * coupling * construct_scale * a.

Constraint. Effective normalisation or gain; pathway activation evidence does not identify each coefficient independently.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. genuinely_unknown_for_physiological_transfer

Dependence of claims. Regulatory transient claims depend on the selected family/timescale and gain; constant full stimulus equilibrium is independent of tau within R1.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. src/modern_full_model/camp_pka.py:90. Pena Munzenmayer et al. 2021 gives pathway and

endpoint activation evidence; acute kinetic times and exact common capacity gain remain effective settings.

Record 92: fully activated increment

`regulation.fully_activated_increment`

Value: 0.25; units: dimensionless; active: Yes; classification: effective_assumption.

Default: ; differs from default: .

Equation role. $da/dt = (\beta - a)/\tau$; AE4 multiplier = basal + increment * coupling * construct_scale * a.

Constraint. Effective normalisation or gain; pathway activation evidence does not identify each coefficient independently.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. genuinely_unknown_for_physiological_transfer

Dependence of claims. Regulatory transient claims depend on the selected family/timescale and gain; constant full stimulus equilibrium is independent of tau within R1. Local numerical dependence is quantified in the linked crosswalk; it establishes neither a measured uncertainty range nor global robustness.

Discriminating measurement. Measure the AE4 transport increment under matched stimulation and substrate gradients; resolve the discrepant approximate 2021 gain readings.

Source. `src/modern_full_model/camp_pka.py:90`. Pena Munzenmayer et al. 2021 gives pathway and endpoint activation evidence; acute kinetic times and exact common capacity gain remain effective settings.

Record 93: coupling scale

`regulation.coupling_scale`

Value: 1; units: dimensionless; active: Yes; classification: effective_assumption.

Default: ; differs from default: .

Equation role. $da/dt = (\beta - a)/\tau$; AE4 multiplier = basal + increment * coupling * construct_scale * a.

Constraint. Effective normalisation or gain; pathway activation evidence does not identify each coefficient independently.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. genuinely_unknown_for_physiological_transfer

Dependence of claims. Regulatory transient claims depend on the selected family/timescale and gain; constant full stimulus equilibrium is independent of tau within R1.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. `src/modern_full_model/camp_pka.py:90`. Pena Munzenmayer et al. 2021 gives pathway and endpoint activation evidence; acute kinetic times and exact common capacity gain remain effective settings.

Record 94: fully activated multiplier

`nkcc_activation.fully_activated_multiplier`

Value: 1.75; units: dimensionless; active: Yes; classification: inherited_effective_setting.

Default: ; differs from default: .

Equation role. $m_N = 1 + (m_{\max} - 1) * \text{clip}((Ca - Ca_{\text{rest}})/(Ca_{\text{sat}} - Ca_{\text{rest}}), 0, 1)$.

Constraint. Inherited algebraic capacity modulation. Its 0.10 uM saturation endpoint is distinct from the NBC 0.25 uM endpoint; at the production input both are fully recruited.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. genuinely_unknown_for_physiological_transfer

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value. Local numerical dependence is quantified in the linked crosswalk; it establishes neither a measured uncertainty range nor global robustness.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. src/modern_full_model/nkcc_stimulation.py:78. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 95: resting calcium uM

nkcc_activation.resting_calcium_uM

Value: 0.058; units: uM; active: Yes; classification: inherited_effective_setting.

Default: ; differs from default: .

Equation role. $m_N=1+(m_max-1)*clip((Ca-Ca_rest)/(Ca_sat-Ca_rest),0,1)$.

Constraint. Inherited algebraic capacity modulation. Its 0.10 uM saturation endpoint is distinct from the NBC 0.25 uM endpoint; at the production input both are fully recruited.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. genuinely_unknown_for_physiological_transfer

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. src/modern_full_model/nkcc_stimulation.py:78. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 96: stimulated calcium uM

nkcc_activation.stimulated_calcium_uM

Value: 0.1; units: uM; active: Yes; classification: inherited_effective_setting.

Default: ; differs from default: .

Equation role. $m_N=1+(m_max-1)*clip((Ca-Ca_rest)/(Ca_sat-Ca_rest),0,1)$.

Constraint. Inherited algebraic capacity modulation. Its 0.10 uM saturation endpoint is distinct from the NBC 0.25 uM endpoint; at the production input both are fully recruited.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. genuinely_unknown_for_physiological_transfer

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. `src/modern_full_model/nkcc_stimulation.py:78`. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 97: arm

`protocol.arm`

Value: CCH_IPR; units: configuration; active: Yes; classification: protocol_setting.

Default: ; differs from default: .

Equation role. Declared input arm and time window select calcium and beta activation steps; no agonist dose response map is present.

Constraint. Declared experimental-input representation, including a step waveform; this is not evidence for measured intracellular calcium kinetics.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Selected discrete setting or stated numerical reference.

Uncertainty status. `genuinely_unknown_for_physiological_transfer`

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. `src/modern_full_model/validation.py:135`. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 98: onset s

`protocol.onset_s`

Value: 0; units: s; active: Yes; classification: protocol_setting.

Default: ; differs from default: .

Equation role. Declared input arm and time window select calcium and beta activation steps; no agonist dose response map is present.

Constraint. Declared experimental-input representation, including a step waveform; this is not evidence for measured intracellular calcium kinetics.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Exact zero records a declared omitted contribution or protocol normalisation; it is not a confidence bound.

Uncertainty status. `genuinely_unknown_for_physiological_transfer`

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. `src/modern_full_model/validation.py:135`. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 99: duration s

`protocol.duration_s`

Value: 600; units: s; active: Yes; classification: protocol_setting.

Default: ; differs from default: .

Equation role. Declared input arm and time window select calcium and beta activation steps; no agonist dose response map is present.

Constraint. Declared experimental-input representation, including a step waveform; this is not evidence for measured intracellular calcium kinetics.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. genuinely_unknown_for_physiological_transfer

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. `src/modern_full_model/validation.py:135`. 2015 protocol lineage recorded by WTProtocol.provenance in `validation.py`; implementation transfer retains dose labels and duration.

Record 100: cch dose uM

`protocol.cch_dose_uM`

Value: 0.3; units: uM; active: No; classification: protocol_setting.

Default: ; differs from default: .

Equation role. Declared input arm and time window select calcium and beta activation steps; no agonist dose response map is present.

Constraint. Nominal experiment dose labels retained for provenance. The selected model has no dose response map from these fields to calcium or beta activation.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. genuinely_unknown_for_physiological_transfer

Dependence of claims. No independent parent RHS sensitivity. Used only for the explicitly stated metadata, initial reference, design lineage or inverse family role.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. `src/modern_full_model/validation.py:135`. 2015 protocol lineage recorded by WTProtocol.provenance in `validation.py`; implementation transfer retains dose labels and duration.

Record 101: ipr dose uM

`protocol.ipr_dose_uM`

Value: 5; units: uM; active: No; classification: protocol_setting.

Default: ; differs from default: .

Equation role. Declared input arm and time window select calcium and beta activation steps; no agonist dose response map is present.

Constraint. Nominal experiment dose labels retained for provenance. The selected model has no dose response map from these fields to calcium or beta activation.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. genuinely_unknown_for_physiological_transfer

Dependence of claims. No independent parent RHS sensitivity. Used only for the explicitly stated metadata, initial reference, design lineage or inverse family role.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. src/modern_full_model/validation.py:135. 2015 protocol lineage recorded by WTProtocol.provenance in validation.py; implementation transfer retains dose labels and duration.

Record 102: resting calcium uM

protocol.resting_calcium_uM

Value: 0.058; units: uM; active: Yes; classification: protocol_setting.

Default: ; differs from default: .

Equation role. Declared input arm and time window select calcium and beta activation steps; no agonist dose response map is present.

Constraint. Declared experimental-input representation, including a step waveform; this is not evidence for measured intracellular calcium kinetics.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. genuinely_unknown_for_physiological_transfer

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. src/modern_full_model/validation.py:135. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 103: stimulated calcium uM

protocol.stimulated_calcium_uM

Value: 0.25; units: uM; active: Yes; classification: protocol_setting.

Default: ; differs from default: .

Equation role. Declared input arm and time window select calcium and beta activation steps; no agonist dose response map is present.

Constraint. Declared experimental-input representation, including a step waveform; this is not evidence for measured intracellular calcium kinetics.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. genuinely_unknown_for_physiological_transfer

Dependence of claims. Numerical state, current and secretion claims are conditional on this active

setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. `src/modern_full_model/validation.py:135`. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 104: beta occupancy on

`protocol.beta_occupancy_on`

Value: 1; units: dimensionless fraction; active: Yes; classification: `protocol_setting`.

Default: ; differs from default: .

Equation role. Declared input arm and time window select calcium and beta activation steps; no agonist dose response map is present.

Constraint. Declared experimental-input representation, including a step waveform; this is not evidence for measured intracellular calcium kinetics.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. [0,1] by fractional definition only; not a measured uncertainty interval.

Uncertainty status. `genuinely_unknown_for_physiological_transfer`

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. `src/modern_full_model/validation.py:135`. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 105: b

`Task41.b`

Value: 0.1051187284328845; units: dimensionless; active: No; classification: `target_selected`.

Default: ; differs from default: .

Equation role. In the prescribed inverse family, residual stimulated CaCC recruitment is b and the AE4 dependent fraction is $\rho=1-b$.

Constraint. Residual CaCC recruitment in the deliberately inverse Task 41 construction. Not part of the parent Task 40 parameter set.

Limit of justification. Matching the target secretion comparison is inverse selection, not independent validation or experimental estimation of CaCC recruitment.

Logical domain. $0 < b \leq 1$, equivalently $0 \leq \rho < 1$, by the prescribed family constructor. $b=0$ is excluded; this is not a measured uncertainty interval.

Uncertainty status. `genuinely_unknown_for_physiological_transfer`

Dependence of claims. Directly controls the target selected Task 41 CaCC recruitment mechanism and its 600 s inverse crossing; outside the parent Task 40 model.

Discriminating measurement. Measure the fraction of stimulated CaCC recruitment lost with acute AE4 depletion while controlling calcium, pH, chloride gradient, voltage and expression; test the prescribed family rather than use the secretion target again.

Source. `src/modern_full_model/ae4_cacc_recruitment.py:24`. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 106: intracellular modifier hill

`cha.intracellular_modifier_hill`

Value: 3; units: dimensionless; active: Yes; classification: published_kinetic_coefficient.

Default: ; differs from default: .

Equation role. $JH = N_H \cdot m_H \cdot \text{Mod2}(a \cdot b - c \cdot d) / (a + b + c + d)$, with rate conversion from ms to s and printed fourth reverse rate retained.

Constraint. Cha 2009 Eq. 5 Mod2 exponent; a fixed published exponent, not an identified salivary abundance.

Limit of justification. Publication of the kinetic law does not measure salivary carrier abundance or establish an uncertainty interval for transfer to this preparation.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. genuinely_unknown_for_physiological_transfer

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. `src/modern_full_model/nhe1_cha2009.py:113`. Cha et al. 2009, Eqs. 3 and 5, Table S1; DOI 10.1016/j.bpj.2009.08.053; repository results/31_nhe1_mechanistic_repair/source_table_s1.json.

Record 107: extracellular modifier hill

`cha.extracellular_modifier_hill`

Value: 1; units: dimensionless; active: Yes; classification: published_kinetic_coefficient.

Default: ; differs from default: .

Equation role. $JH = N_H \cdot m_H \cdot \text{Mod2}(a \cdot b - c \cdot d) / (a + b + c + d)$, with rate conversion from ms to s and printed fourth reverse rate retained.

Constraint. Cha 2009 Eq. 5 Mod2 exponent; a fixed published exponent, not an identified salivary abundance.

Limit of justification. Publication of the kinetic law does not measure salivary carrier abundance or establish an uncertainty interval for transfer to this preparation.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. genuinely_unknown_for_physiological_transfer

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. `src/modern_full_model/nhe1_cha2009.py:113`. Cha et al. 2009, Eqs. 3 and 5, Table S1; DOI 10.1016/j.bpj.2009.08.053; repository results/31_nhe1_mechanistic_repair/source_table_s1.json.

Record 108: law

`nkcc.law`

Value: `palk_benjamin_2010_eq17`; units: model identifier; active: Yes; classification: law_selection.

Default: ; differs from default: .

Equation role. $JN = \alpha \cdot m_N \cdot (A1 - A2[Na]i[K]i[Cl]i^2) / (A3 + A4[Na]i[K]i[Cl]i^2)$; inward 1 Na, 1 K, 2 Cl per cycle.

Constraint. Explicit Palk/Benjamin law selected by the Task 39 builder and inherited by Task 40.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Selected discrete setting or stated numerical reference.

Uncertainty status. not_a_measured_uncertainty_parameter

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. `src/modern_full_model/nkcc1_palk2010.py:31`. Palk et al. 2010/Benjamin and Johnson 1997 law, reproduced as Vera Siguenza et al. 2018 Eq. 17; coefficients in `src/modern_full_model/nkcc1_palk2010.py`.

Record 109: law

`ae4.law`

Value: `equal_routing_adapter`; units: model identifier; active: Yes; classification: `law_selection`.

Default: ; differs from default: .

Equation role. Shared carrier QSS determines scalar J4; equal routing imposes $(-J4/2, -J4/2, J4, -2J4, -2J4)$ in cell Na,K,Cl,TIC,TA.

Constraint. Equal cation routing fixes source shares at one half each while retaining the inherited total scalar cycle flux.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Selected discrete setting or stated numerical reference.

Uncertainty status. not_a_measured_uncertainty_parameter

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. `src/modern_full_model/transporters.py:458`. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 110: family

`regulation.family`

Value: R1; units: model identifier; active: Yes; classification: `law_selection`.

Default: ; differs from default: .

Equation role. $da/dt = (\beta a - a)/\tau$; AE4 multiplier = `basal + increment * coupling * construct_scale * a`.

Constraint. One retained effective AE4 activation state; no measured separation of cAMP and PKA kinetics.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Selected discrete setting or stated numerical reference.

Uncertainty status. not_a_measured_uncertainty_parameter

Dependence of claims. Regulatory transient claims depend on the selected family/timescale and gain; constant full stimulus equilibrium is independent of τ within R1.

Discriminating measurement. Measure this quantity or its identifiable combination in the same

preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. `src/modern_full_model/camp_pka.py:90`. Pena Munzenmayer et al. 2021 gives pathway and endpoint activation evidence; acute kinetic times and exact common capacity gain remain effective settings.

Record 111: construct

`regulation.construct`

Value: WT; units: construct identifier; active: Yes; classification: `protocol_setting`.

Default: ; differs from default: .

Equation role. $da/dt = (\beta - a)/\tau$; AE4 multiplier=basal+increment*coupling*construct_scale*a.

Constraint. WT construct selects unit response scale in the gain map.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Selected discrete setting or stated numerical reference.

Uncertainty status. `genuinely_unknown_for_physiological_transfer`

Dependence of claims. Regulatory transient claims depend on the selected family/timescale and gain; constant full stimulus equilibrium is independent of tau within R1.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. `src/modern_full_model/camp_pka.py:90`. Pena Munzenmayer et al. 2021 gives pathway and endpoint activation evidence; acute kinetic times and exact common capacity gain remain effective settings.

Record 112: family

`nkcc_activation.family`

Value: N1_CCH_ALGEBRAIC; units: model identifier; active: Yes; classification: `law_selection`.

Default: ; differs from default: .

Equation role. $m_N = 1 + (m_{\max} - 1) * \text{clip}((Ca - Ca_{\text{rest}})/(Ca_{\text{sat}} - Ca_{\text{rest}}), 0, 1)$.

Constraint. Selected instantaneous calcium recruitment family; no added NKCC dynamic state.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Selected discrete setting or stated numerical reference.

Uncertainty status. `not_a_measured_uncertainty_parameter`

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. `src/modern_full_model/nkcc_stimulation.py:78`. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 113: TARGET NKCC1 POSITIVE CL SHARE

`nbc_design.TARGET_NKCC1_POSITIVE_CL_SHARE`

Value: 0.7; units: dimensionless; active: No; classification: `effective_assumption`.

Default: ; differs from default: .

Equation role. Upstream construction: $J4_req=((1-s)/s)*L_N$; $JB_req=J4_req-H_stim/2$; capacity includes the self consistent voltage.

Constraint. Design input or conditional derived quantity upstream of the frozen NBC capacity; not independently evaluated in the production RHS. The 70/30 share is assumed, and the reference fluxes are modelled.

Limit of justification. The 70/30 partition is assumed, not measured. Solving the coupled current balance does not establish the physiological truth or unique identification of that partition.

Logical domain. [0,1] by fractional definition only; not a measured uncertainty interval.

Uncertainty status. conditional_design_value_no_experimental_interval

Dependence of claims. No independent parent RHS sensitivity. Used only for the explicitly stated metadata, initial reference, design lineage or inverse family role.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. src/modern_full_model/nbc_minimal.py:77. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 114: TASK31 R09 NKCC1 CL FMOL S

nbc_design.TASK31_R09_NKCC1_CL_FMOL_S

Value: 0.2562444047732706; units: fmol/s; active: No; classification: WT_architecture_derived.

Default: ; differs from default: .

Equation role. Upstream construction: $J4_req=((1-s)/s)*L_N$; $JB_req=J4_req-H_stim/2$; capacity includes the self consistent voltage.

Constraint. Design input or conditional derived quantity upstream of the frozen NBC capacity; not independently evaluated in the production RHS. The 70/30 share is assumed, and the reference fluxes are modelled.

Limit of justification. The 70/30 partition is assumed, not measured. Solving the coupled current balance does not establish the physiological truth or unique identification of that partition.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. conditional_design_value_no_experimental_interval

Dependence of claims. No independent parent RHS sensitivity. Used only for the explicitly stated metadata, initial reference, design lineage or inverse family role.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. src/modern_full_model/nbc_minimal.py:77. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 115: TASK31 R09 NHE1 FMOL S

nbc_design.TASK31_R09_NHE1_FMOL_S

Value: 0.007890580489555009; units: fmol/s; active: No; classification: WT_architecture_derived.

Default: ; differs from default: .

Equation role. Upstream construction: $J4_req=((1-s)/s)*L_N$; $JB_req=J4_req-H_stim/2$; capacity includes the self consistent voltage.

Constraint. Design input or conditional derived quantity upstream of the frozen NBC capacity; not independently evaluated in the production RHS. The 70/30 share is assumed, and the reference fluxes are modelled.

Limit of justification. The 70/30 partition is assumed, not measured. Solving the coupled current

balance does not establish the physiological truth or unique identification of that partition.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. conditional_design_value_no_experimental_interval

Dependence of claims. No independent parent RHS sensitivity. Used only for the explicitly stated metadata, initial reference, design lineage or inverse family role.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. src/modern_full_model/nbc_minimal.py:77. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 116: DERIVED AE4 CL FMOL S

nbc_design.DERIVED_AE4_CL_FMOL_S

Value: 0.109819030617116; units: fmol/s; active: No; classification: WT_architecture_derived.

Default: ; differs from default: .

Equation role. Upstream construction: $J4_req = ((1-s)/s) * L_N$; $JB_req = J4_req - H_stim/2$; capacity includes the self consistent voltage.

Constraint. Design input or conditional derived quantity upstream of the frozen NBC capacity; not independently evaluated in the production RHS. The 70/30 share is assumed, and the reference fluxes are modelled.

Limit of justification. The 70/30 partition is assumed, not measured. Solving the coupled current balance does not establish the physiological truth or unique identification of that partition.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. conditional_design_value_no_experimental_interval

Dependence of claims. No independent parent RHS sensitivity. Used only for the explicitly stated metadata, initial reference, design lineage or inverse family role.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. src/modern_full_model/nbc_minimal.py:77. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 117: DERIVED STIMULATED NHE1 FMOL S

nbc_design.DERIVED_STIMULATED_NHE1_FMOL_S

Value: 0.01814833512597652; units: fmol/s; active: No; classification: WT_architecture_derived.

Default: ; differs from default: .

Equation role. Upstream construction: $J4_req = ((1-s)/s) * L_N$; $JB_req = J4_req - H_stim/2$; capacity includes the self consistent voltage.

Constraint. Design input or conditional derived quantity upstream of the frozen NBC capacity; not independently evaluated in the production RHS. The 70/30 share is assumed, and the reference fluxes are modelled.

Limit of justification. The 70/30 partition is assumed, not measured. Solving the coupled current balance does not establish the physiological truth or unique identification of that partition.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. conditional_design_value_no_experimental_interval

Dependence of claims. No independent parent RHS sensitivity. Used only for the explicitly stated metadata, initial reference, design lineage or inverse family role.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. `src/modern_full_model/nbc_minimal.py:77`. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 118: DERIVED REQUIRED NBC CYCLE FMOL S

`nbc_design.DERIVED_REQUIRED_NBC_CYCLE_FMOI_S`

Value: 0.1007448630541277; units: fmol/s; active: No; classification: WT_architecture_derived.

Default: ; differs from default: .

Equation role. Upstream construction: $J4_{req} = ((1-s)/s) * L_N$; $JB_{req} = J4_{req} - H_{stim}/2$; capacity includes the self consistent voltage.

Constraint. Design input or conditional derived quantity upstream of the frozen NBC capacity; not independently evaluated in the production RHS. The 70/30 share is assumed, and the reference fluxes are modelled.

Limit of justification. The 70/30 partition is assumed, not measured. Solving the coupled current balance does not establish the physiological truth or unique identification of that partition.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. `conditional_design_value_no_experimental_interval`

Dependence of claims. No independent parent RHS sensitivity. Used only for the explicitly stated metadata, initial reference, design lineage or inverse family role.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. `src/modern_full_model/nbc_minimal.py:77`. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 119: na i fmol

`frozen_initial.na_i_fmol`

Value: 16.72035873274549; units: fmol; active: Yes; classification: WT_constraint_derived.

Default: ; differs from default: .

Equation role. Full conserved state and regulatory initial value at $t=0$ for the finite duration protocol; all amounts and both volumes evolve subsequently.

Constraint. Solved, accepted WT resting state used as the shared initial condition for production genotype trajectories; not the constructor seed and not an independent measurement.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. `conditional_root_value_no_experimental_interval`

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. `src/modern_full_model/model.py:559`. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 120: k i fmol

`frozen_initial.k_i_fmol`

Value: 170.0020085218768; units: fmol; active: Yes; classification: WT_constraint_derived.

Default: ; differs from default: .

Equation role. Full conserved state and regulatory initial value at t=0 for the finite duration protocol; all amounts and both volumes evolve subsequently.

Constraint. Solved, accepted WT resting state used as the shared initial condition for production genotype trajectories; not the constructor seed and not an independent measurement.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. conditional_root_value_no_experimental_interval

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. `src/modern_full_model/model.py:559`. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 121: cl i fmol

`frozen_initial.cl_i_fmol`

Value: 88.03180196356951; units: fmol; active: Yes; classification: WT_constraint_derived.

Default: ; differs from default: .

Equation role. Full conserved state and regulatory initial value at t=0 for the finite duration protocol; all amounts and both volumes evolve subsequently.

Constraint. Solved, accepted WT resting state used as the shared initial condition for production genotype trajectories; not the constructor seed and not an independent measurement.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. conditional_root_value_no_experimental_interval

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. `src/modern_full_model/model.py:559`. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 122: tic i fmol

`frozen_initial.tic_i_fmol`

Value: 8.263511808402752; units: fmol; active: Yes; classification: WT_constraint_derived.

Default: ; differs from default: .

Equation role. Full conserved state and regulatory initial value at t=0 for the finite duration protocol; all amounts and both volumes evolve subsequently.

Constraint. Solved, accepted WT resting state used as the shared initial condition for production genotype trajectories; not the constructor seed and not an independent measurement.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. conditional_root_value_no_experimental_interval

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. src/modern_full_model/model.py:559. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 123: alk i fmo1

frozen_initial.alk_i_fmo1

Value: 20.62880308510795; units: fmo1; active: Yes; classification: WT_constraint_derived.

Default: ; differs from default: .

Equation role. Full conserved state and regulatory initial value at t=0 for the finite duration protocol; all amounts and both volumes evolve subsequently.

Constraint. Solved, accepted WT resting state used as the shared initial condition for production genotype trajectories; not the constructor seed and not an independent measurement.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. conditional_root_value_no_experimental_interval

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. src/modern_full_model/model.py:559. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 124: volume i pL

frozen_initial.volume_i_pL

Value: 1.45333439258746; units: pL; active: Yes; classification: WT_constraint_derived.

Default: ; differs from default: .

Equation role. Full conserved state and regulatory initial value at t=0 for the finite duration protocol; all amounts and both volumes evolve subsequently.

Constraint. Solved, accepted WT resting state used as the shared initial condition for production genotype trajectories; not the constructor seed and not an independent measurement.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. conditional_root_value_no_experimental_interval

Dependence of claims. Numerical state, current and secretion claims are conditional on this active

setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. `src/modern_full_model/model.py:559`. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 125: na_l fmol

`frozen_initial.na_l_fmol`

Value: 13.12776866822541; units: fmol; active: Yes; classification: WT_constraint_derived.

Default: ; differs from default: .

Equation role. Full conserved state and regulatory initial value at $t=0$ for the finite duration protocol; all amounts and both volumes evolve subsequently.

Constraint. Solved, accepted WT resting state used as the shared initial condition for production genotype trajectories; not the constructor seed and not an independent measurement.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. `conditional_root_value_no_experimental_interval`

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. `src/modern_full_model/model.py:559`. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 126: k_l fmol

`frozen_initial.k_l_fmol`

Value: 2.718972608990927; units: fmol; active: Yes; classification: WT_constraint_derived.

Default: ; differs from default: .

Equation role. Full conserved state and regulatory initial value at $t=0$ for the finite duration protocol; all amounts and both volumes evolve subsequently.

Constraint. Solved, accepted WT resting state used as the shared initial condition for production genotype trajectories; not the constructor seed and not an independent measurement.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. `conditional_root_value_no_experimental_interval`

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. `src/modern_full_model/model.py:559`. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 127: cl l fmol

`frozen_initial.cl_l_fmol`

Value: 15.37868785453798; units: fmol; active: Yes; classification: WT_constraint_derived.

Default: ; differs from default: .

Equation role. Full conserved state and regulatory initial value at t=0 for the finite duration protocol; all amounts and both volumes evolve subsequently.

Constraint. Solved, accepted WT resting state used as the shared initial condition for production genotype trajectories; not the constructor seed and not an independent measurement.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. conditional_root_value_no_experimental_interval

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. `src/modern_full_model/model.py:559`. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 128: tic l fmol

`frozen_initial.tic_l_fmol`

Value: 0.5402623164056106; units: fmol; active: Yes; classification: WT_constraint_derived.

Default: ; differs from default: .

Equation role. Full conserved state and regulatory initial value at t=0 for the finite duration protocol; all amounts and both volumes evolve subsequently.

Constraint. Solved, accepted WT resting state used as the shared initial condition for production genotype trajectories; not the constructor seed and not an independent measurement.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. conditional_root_value_no_experimental_interval

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. `src/modern_full_model/model.py:559`. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 129: alk l fmol

`frozen_initial.alk_l_fmol`

Value: 0.4680534226783657; units: fmol; active: Yes; classification: WT_constraint_derived.

Default: ; differs from default: .

Equation role. Full conserved state and regulatory initial value at t=0 for the finite duration protocol; all amounts and both volumes evolve subsequently.

Constraint. Solved, accepted WT resting state used as the shared initial condition for production genotype trajectories; not the constructor seed and not an independent measurement.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. conditional_root_value_no_experimental_interval

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. src/modern_full_model/model.py:559. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 130: volume 1 pL

frozen_initial.volume_1_pL

Value: 0.1053884073536874; units: pL; active: Yes; classification: WT_constraint_derived.

Default: ; differs from default: .

Equation role. Full conserved state and regulatory initial value at t=0 for the finite duration protocol; all amounts and both volumes evolve subsequently.

Constraint. Solved, accepted WT resting state used as the shared initial condition for production genotype trajectories; not the constructor seed and not an independent measurement.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Positive for the retained law; no finite upper physiological bound established.

Uncertainty status. conditional_root_value_no_experimental_interval

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. src/modern_full_model/model.py:559. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 131: ae4 effective activation fraction

frozen_initial.ae4_effective_activation_fraction

Value: 0; units: dimensionless; active: Yes; classification: WT_constraint_derived.

Default: ; differs from default: .

Equation role. Full conserved state and regulatory initial value at t=0 for the finite duration protocol; all amounts and both volumes evolve subsequently.

Constraint. Solved, accepted WT resting state used as the shared initial condition for production genotype trajectories; not the constructor seed and not an independent measurement.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. [0,1] by fractional definition only; not a measured uncertainty interval.

Uncertainty status. conditional_root_value_no_experimental_interval

Dependence of claims. Numerical state, current and secretion claims are conditional on this active

setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. `src/modern_full_model/model.py:559`. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 132: name

`genotype.name`

Value: WT; units: genotype label; active: No; classification: `protocol_setting`.

Default: ; differs from default: .

Equation role. Externally passed expression multipliers scale the corresponding transport flux; zero AE4 expression is an exact deletion of its source.

Constraint. Externally passed WT expression scale, equal to one for all retained transporters; not a measured abundance. Task 40 alters only the AE4 expression scale.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Selected discrete setting or stated numerical reference.

Uncertainty status. `genuinely_unknown_for_physiological_transfer`

Dependence of claims. No independent parent RHS sensitivity. Used only for the explicitly stated metadata, initial reference, design lineage or inverse family role.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. `src/modern_full_model/model.py:93`. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 133: ae4 expression

`genotype.ae4_expression`

Value: 1; units: dimensionless expression scale; active: Yes; classification: `protocol_setting`.

Default: ; differs from default: .

Equation role. Externally passed expression multipliers scale the corresponding transport flux; zero AE4 expression is an exact deletion of its source.

Constraint. Externally passed WT expression scale, equal to one for all retained transporters; not a measured abundance. Task 40 alters only the AE4 expression scale.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Nonnegative expression scale; values 1, 0.05 and 0 are specified interventions, not a fitted interval.

Uncertainty status. `genuinely_unknown_for_physiological_transfer`

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. `src/modern_full_model/model.py:93`. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 134: ae2 expression

`genotype.ae2_expression`

Value: 1; units: dimensionless expression scale; active: Yes; classification: protocol_setting.

Default: ; differs from default: .

Equation role. Externally passed expression multipliers scale the corresponding transport flux; zero AE4 expression is an exact deletion of its source.

Constraint. Externally passed WT expression scale, equal to one for all retained transporters; not a measured abundance. Task 40 alters only the AE4 expression scale.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Nonnegative expression scale; values 1, 0.05 and 0 are specified interventions, not a fitted interval.

Uncertainty status. `genuinely_unknown_for_physiological_transfer`

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. `src/modern_full_model/model.py:93`. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 135: nkcc1 expression

`genotype.nkcc1_expression`

Value: 1; units: dimensionless expression scale; active: Yes; classification: protocol_setting.

Default: ; differs from default: .

Equation role. Externally passed expression multipliers scale the corresponding transport flux; zero AE4 expression is an exact deletion of its source.

Constraint. Externally passed WT expression scale, equal to one for all retained transporters; not a measured abundance. Task 40 alters only the AE4 expression scale.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Nonnegative expression scale; values 1, 0.05 and 0 are specified interventions, not a fitted interval.

Uncertainty status. `genuinely_unknown_for_physiological_transfer`

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. `src/modern_full_model/model.py:93`. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 136: nhe1 expression

`genotype.nhe1_expression`

Value: 1; units: dimensionless expression scale; active: Yes; classification: protocol_setting.

Default: ; differs from default: .

Equation role. Externally passed expression multipliers scale the corresponding transport flux; zero AE4

expression is an exact deletion of its source.

Constraint. Externally passed WT expression scale, equal to one for all retained transporters; not a measured abundance. Task 40 alters only the AE4 expression scale.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Nonnegative expression scale; values 1, 0.05 and 0 are specified interventions, not a fitted interval.

Uncertainty status. genuinely_unknown_for_physiological_transfer

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. src/modern_full_model/model.py:93. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Record 137: ae4 expression cases

protocol.ae4_expression_cases

Value: [1.0, 0.05, 0.0]; units: dimensionless expression scales; active: Yes; classification: protocol_setting.

Default: ; differs from default: .

Equation role. Task40 calls the same model with e4=1,0.05,0; eNKCC=eNHE=eAE2=1 and all other parameters fixed.

Constraint. Predeclared WT, 5 percent and null interventions; only the AE4 expression field differs. These are protocol conditions, not estimated abundance uncertainty.

Limit of justification. An accepted physiological point does not identify this setting or supply a numerical uncertainty interval. No knockout fit or new measurement is performed in this audit.

Logical domain. Selected discrete setting or stated numerical reference.

Uncertainty status. genuinely_unknown_for_physiological_transfer

Dependence of claims. Numerical state, current and secretion claims are conditional on this active setting; exact stoichiometric identities are independent of its numerical value.

Discriminating measurement. Measure this quantity or its identifiable combination in the same preparation and stimulus protocol, with independent WT data; determine physiological ranges before making global robustness claims.

Source. analysis/40_ae4_equal_cation_routing/run_case.py:78. No independently verified primary measurement attached to this numerical setting; model lineage is the source.

Part V

Verification, coverage and use in an article

37 Scientific conclusions suitable for an article

The completed analysis establishes a reproducible mathematical diagnosis of the frozen reconstruction. It does not establish a new validated biological model. The most defensible central conclusion is that explicit conservation and acceptable reference physiology do not suffice to reproduce the AE4 secretion phenotype: the adopted network largely replaces the missing AE4 chloride input, while the prescribed apical recruitment construction recovers the selected magnitude only under an additional strong and unvalidated coupling.

This conclusion is supported by several complementary results. The conserved amount formulation identifies the alkalinity cost of productive AE4 transport. The exact chloride identity identifies which explicit source contributions survive in a stationary balance. The full equilibrium and transient calculations quantify the state mediated response that the identity alone cannot determine. The parameter audit exposes which effective settings are independently supported and which remain assumptions. Finally, the inverse calculation separates mathematical sufficiency of a chosen coupling from biological evidence for that coupling. None of those steps can be replaced by a single comparison with the experimental secretion percentage.

37.1 What the model explains internally

Within the frozen configuration, a fall in the AE4 chloride contribution is substantially offset by the rest of the transport network, particularly NKCC1. The local equilibrium gain, the integrated replacement fraction and the relative increase in NKCC1 loading all support this interpretation, provided their distinct definitions are retained. The small stationary effect and the larger, but still small, 600 s cumulative effect are compatible with the presence of changing ionic stores and a finite lumen. They are not competing estimates of one observable.

The model also explains why alkalinity must be examined together with carbon and volume. A sustained AE4 demand must be supplied by the admissible alkalinity sources or by depletion of a finite store. An NBC absent root with much smaller AE4 loading is consistent with that requirement. It does not invalidate the budget. Conversely, observing NBC support in one WT trajectory does not establish a universal necessity theorem or identify the molecular pathway.

The local mode analysis shows that the system already possesses dynamics on a scale of several minutes. Its modal contributions to secretion are not all zero, and their signs can cancel in the stationary sensitivity. Accordingly, a failed temporal phenotype should not be attributed to absence of a slow state without analysing how the perturbation excites the existing modes and how secretion observes them. The documented correction of the regulatory clamp derivative is a numerical differentiation correction, not a change to the production mechanism.

37.2 What fails as an account of the biological phenotype

The reference finite protocol does not reproduce the large sustained secretion loss described in the source experimental comparison. The compensation responsible for much of this attenuation has not been independently established in the corresponding physiological preparation. In addition, the substitution of source vectors into inherited scalar AE4 kinetics leaves explicit thermodynamic qualifications. These are substantive limitations, not typesetting or numerical convergence problems.

The Task 41 family illustrates a distinct limitation. It can approach the selected finite duration deficit while retaining the declared 600 s gates, but it requires most fully stimulated CaCC recruitment to depend on AE4 in that particular formula. The original temporal and ionic discrepancies remain, and the separate extended null trajectories later cross the upper pH gate. Thus a finite duration fit to one chosen magnitude does not establish a sustained, independently validated secretion mechanism.

The source analysis does not isolate NKCC1 as the uniquely incorrect component. The existing rational law already has finite forward and reverse limits. Its strong compensatory response could depend on its effective capacity or stimulation, on ionic state dependence, on the pump and bicarbonate pathways, on the common initial condition, or on missing coupling elsewhere. Establishing the responsible defect requires a new comparison of mechanistically distinct candidates and matched experimental protocols. The results here justify that comparison and identify quantities to measure; they do not predetermine its winner.

Source record: `evidence/task44/integrated_context.tex`;`evidence/task44/literature_comparison.tex`;`evidence/task44/inverse_results.tex`;`evidence/task43/report.tex`.

38 Claims that must remain qualified

Claim to avoid	Supported formulation
The full model is biologically correct because all checks pass.	Conservation, numerical agreement, physiological gates and thermodynamic compatibility are separate tests. Some physiological mathematical roots retain an opposed source cycle. Numerical replay does not validate biology.
NKCC1 is proved to be the sole defective mechanism.	The frozen network compensates strongly, and several poorly constrained settings affect that response. Attribution to a unique defective component remains unresolved.
NKCC1 must never increase after AE4 loss.	The source reports no detectable increase in the particular isolated initial uptake assay. That observation must be compared with an appropriate assay model; it is not an identity across all states and protocols.
The chloride cancellation proves no AE4 effect.	It removes an explicit source term from one stationary balance. The state mediated derivative remains and is quantified locally.
The rank result proves whole cell observations cannot distinguish source classes.	The five specified joint source projections have rank five. Whether experimental observables identify a source vector also depends on kinetics, excitation, measurement and uncertainty.
NBC is necessary for all physiological secretion.	The obstruction is conditional on a specified AE4 loading demand and admissible support bounds. The source includes an NBC absent physiological root at substantially smaller AE4 loading.
A missing slow regulator is proved necessary.	Slow local modes already occur and project onto secretion. Their existence alone does not reproduce the experimental temporal signature or exclude missing regulation.
The inverse crossing is a global minimum or a confidence bound.	It is a first numerical crossing in one ordered and locally refined family. The 30.3% comparison equals a reported mean minus one standard error.

Claim to avoid	Supported formulation
A local sensitivity proves robustness over physiological variation.	Derivatives are verified near a reference point. The audit supplies no defensible uncertainty intervals for many effective parameters, so finite range robustness is not established.
The 5% model case independently validates an expression experiment.	It is a prescribed expression perturbation in this source record. Independent experimental status requires an appropriate source, not the presence of a simulated curve.
All failed historical numerical attempts have been reconstructed.	Four early unsuccessful continuation attempts survive only as an aggregate statement. Individual seeds and messages were unavailable and have not been invented.

39 What must be constrained in a subsequent reconstruction

The source sensitivities prioritise functional pump capacity and apical distribution, effective NKCC1 capacity and stimulation, NBC capacity and saturation, NHE1 abundance and recruitment, the finite buffer pool and CaCC conductance or recruitment. These priorities are local to the analysed configuration. Their importance does not supply their physiological ranges, and a small sensitivity does not prove that a parameter is unimportant in all regimes.

The most informative measurements would separate transporter abundance from its state dependence and measure relevant quantities under comparable stimulation and ionic conditions. The source report identifies combined sodium and bicarbonate flux, current and pH measurements for alkalinity support; ion conditioned NKCC, NHE and pump measurements; simultaneous total inorganic carbon, buffering, chloride and volume time courses; and matched stimulated CaCC current or surface recruitment during acute AE4 perturbation. Combined cation, chloride, carbon and reversal measurements are needed to discriminate proposed AE4 source cycles and compatible kinetic implementations. These are proposed discriminating experiments, not data already available in this analysis.

The comparison of acute perturbations with established knockout preparations also requires an explicit initial state model. Reusing the inherited WT resting state across expression cases is a controlled computational intervention, but it does not reproduce every resting adaptation present in a knockout gland. Likewise, a single acinar cell with a finite lumen does not include ductal modification or an entire gland. Those differences should be modelled or bounded when they matter to an experimental comparison, not hidden by equating all secretion observables.

A subsequent model should therefore explain both the observed output and the mechanism that changes it, while preserving the conserved balances. It should distinguish data used for construction and selection from observations left for independent testing. If an observation has already been used to guide candidate selection, it is no longer an untouched validation test. This is a methodological consequence of the diagnosis, rather than an additional result proved by Tasks 43 and 44.

40 Completeness, source limits and reproducible use

The report contains the complete final Task 44 technical exposition and the complete Task 43 provenance report, with explicit source namespaces. It expands the numerical tables beyond those printed in the original 26 and 11 page reports. All 26 state vectors, all 26 dynamic Jacobian matrices and all eigenvalues are printed. Every inventory record is given its own detailed entry, and all extended parameter metrics and both inverse threshold sensitivity observables are tabulated. The electronic evidence is part of the report, not an optional source of convenient extra facts.

It preserves full precision numbers, complete modal vectors, every stored nearby trial, source locations, diagnostic residuals, failure records, verification exclusions and original scripts. The PDF rounds table values for reading, and the source manifest supplies hashes of the unchanged underlying files. A reader can follow the report to a particular source record and then to the exact pinned commit. The build does not depend on live branch heads or regenerate scientific simulations.

Some detail is unavailable in the source record, most notably the individual seeds and errors of the four historical continuation failures and experimentally justified ranges for many effective parameters. No report can legitimately fill those gaps by inventing plausible values. They are therefore stated explicitly. The source literature review also distinguishes full article inspection from abstract or caption level access. This compilation does not silently upgrade that review to a systematic search of every later experiment.

The final technical package is a basis for rigorous article development, not a declaration that all scientific questions are closed. It preserves the results that survive, the conditional assumptions that make them true, the calculations that contradict earlier interpretations, and the evidence needed to test a better model. That is the contribution of the completed mathematical analysis.

Source record: `evidence/task44/FINAL_STATUS.md;evidence/task43/FINAL_STATUS.md;SOURCE_MANIFEST.json;COVERAGE.md.`

41 Recorded verification results and their scope

The tables below expose scalar receipts from the original verification. Nested trial arrays and per file comparisons are retained in full in the original JSON. This report has not rerun the numerical replay and does not claim that reproducibility alone supplies experimental validation.

exact verification

Table 37: Scalar receipt fields.

Field	Recorded value
dimensionless_additional_groups.AE4_to_NKCC_chlo ride_initial	0.01098293198
dimensionless_additional_groups.AE4_to_NKCC_cycl e_initial	0.02196586395
dimensionless_additional_groups.Ka1_over_CO	5.478125757e-06
dimensionless_additional_groups.Ka2_over_CO	3.45646368e-10
dimensionless_additional_groups.Kw_over_CO_squar ed	4.756242568e-13
dimensionless_additional_groups.carrier_QSS_time _ratio_initial	0.0001392109809
dimensionless_additional_groups.cell_buffer_K_ov er_CO	6.896551724e-07
dimensionless_additional_groups.cell_buffer_scal ed	0.1423599089
dimensionless_additional_groups.cell_other_imper meant_scaled	0.5917396954
dimensionless_additional_groups.co2_apical	6.455402013
dimensionless_additional_groups.co2_basolateral	12.91080403
dimensionless_additional_groups.electrical_relax ation_group	<i>n/a</i>
dimensionless_additional_groups.electrical_relax ation_reason	No capacitance or voltage differential equation in the frozen model; conductance groups measure currents and do not establish an electrical relaxation time.
dimensionless_additional_groups.fixed_cell_charg e_scaled	0.3704288452
dimensionless_additional_groups.lumen_buffer_K_o ver_CO	6.896551724e-07

Field	Recorded value
dimensionless_additional_groups.lumen_buffer_sca	6.543937704e-11
led	
joint_projection_kernel	zero
joint_projection_rank	5
maxima.carbon_alkalinity_identity	0
maxima.carbonate_derivative_relative	2.113318143e-08
maxima.carbonate_ph	4.112266083e-13
maxima.charge_rate	1.607047828e-14
maxima.electrical_slope_relative	3.105515844e-12
maxima.electrical_voltage	8.326672685e-17
maxima.rhs_scaled	4.5092998e-12
maxima.structural_identity	2.220446049e-16
minimum_electrical_slope_S	5.430912552e-09
proof_scope	Exact identities and conditional closure proofs; random checks establish implementation agreement, not global physiology or global stability.
random_seed	45044
regulatory_jacobian_caveat	At r=1 the executable trial clip is not differentiable. A centred difference halves its column. Use the inward derivative or the stated smooth analytical extension $dr/dt=(1-r)/30$ for full spectrum and modal calculations.
states	84
states_per_class	12
status	passed

Source record: evidence/task44/output/exact_verification.json.

rescaling verification

Table 38: Scalar receipt fields.

Field	Recorded value
adaptive_integral_pL	0.9925444427
canonical_flow_max_error	1.361907982e-12
dimensionless_max_scaled_error	7.523696111e-10
original_sampled_cumulative_pL	0.9925245441
reproduced_sampled_cumulative_pL	0.9925245447

Source record: evidence/task44/output/rescaling_verification.json.

sensitivity verification

Table 39: Scalar receipt fields.

Field	Recorded value
NHE_bound_random_checks	128
bounds_scope	Random checks supplement the algebraic inequality; they are not its proof.
independent_nearby_WT_null_roots	128
largest_sampled_NHE_to_bound_ratio	0.9761054535
max_IFT_relative_error	6.104805193e-06
max_eigenvalue_method_relative_difference	6.680615237e-09
max_elasticity_step_disagreement	3.226967386e-06
max_equilibrium_residual	5.235111754e-12
max_expression_gain_FD_relative_error	1.403296157e-08
parameter_count	16
status	passed
uncertainty_interval_inferred	No

Source record: evidence/task44/output/sensitivity_verification.json.

completion verification

Table 40: Scalar receipt fields.

Field	Recorded value
analysed_commit	f783785a863440df459e9c5530beba4c7eb59110
dimensional_recovered_cumulative_water_error_pL	1.443289932e-15
equilibrium_roots	26
independently_solved_nearby_roots	128
inverse_parameter_observable_pairs	16
physiological_roots	22
random_valid_exact_states	84
scope	Cross-file and fresh dimensional recovery checks. Full clean-checkout numerical replay is recorded separately.
status	passed
uncertain_equilibrium_parameters	16

Source record: evidence/task44/output/completion_verification.json.

clean recomputation verification

Table 41: Scalar receipt fields.

Field	Recorded value
authoritative_files	47
bit_identical_numeric_values	331157
comparison_helper_sha256	6d4f9f19101b4aa1d506c822e521a3966961a4465e64b95c9ec06c2cc009debd
files_compared	47
independent_clean_source.acquisition_detail	Independent –no-hardlinks clone of the root repository object history immediately after its exact remote checkpoint fetch; detached at f783785a863440df459e9c5530beba4c7eb59110 with clean status. Direct GitHub checkout is unavailable in this runtime. No working-tree files from the root checkout were reused.
independent_clean_source.checkout	/workspace/scratch/32c3c5dd39a9/clean_task44_f783785
independent_clean_source.clean_git_status	Yes
independent_clean_source.initial_inverse_trial_c	0
independent_clean_source.runtime_python	3.12.14 (main, Aug 25 2026, 14:00:49) [Clang 22.1.3]
independent_clean_source.source_acquisition	Fresh checkout of the remotely published numerical checkpoint; exact HEAD and clean status checked before removing generated outputs.
independent_clean_source.source_python_sha256.analysis/44_full_system_mathematics/core.py	9658e08688680e840ae2f032f7d99e7e94c112bed316dc41bdb0800334ebf555
independent_clean_source.source_python_sha256.analysis/44_full_system_mathematics/equilibrium_review.py	e5063719355c0492373ca4d7361735545e3671632bd9ea90e35898a18d9868c8
independent_clean_source.source_python_sha256.analysis/44_full_system_mathematics/inverse_analysis.py	30b8d80ce445606d24da76e5a700761204e6df797df04dda216d1ef868d3be3e
independent_clean_source.source_python_sha256.analysis/44_full_system_mathematics/inverse_verify.py	7213543b1934d60abe08eff7b6adc208dd09c012d575e4374e125e095b372e54
independent_clean_source.source_python_sha256.analysis/44_full_system_mathematics/prepare_inverse_report.py	2608ef20e718c16c8328da97d844839d7b3c863ec48d63b9b9cfcc947fc186dc
independent_clean_source.source_python_sha256.analysis/44_full_system_mathematics/prepare_report.py	6249256f4734897412330556d27af71d4e2eaf21da8a965ebf1d09f7b909a2b7
independent_clean_source.source_python_sha256.analysis/44_full_system_mathematics/run_analysis.py	f26e66a93daea75655e97e2ca0339a3287c945d6fdb6ef5abbaedefbb6715ac
independent_clean_source.source_python_sha256.analysis/44_full_system_mathematics/sensitivity_analysis.py	03327b08223e7c0812aad6d90ed4bb06a2794bb01f850d682ac04e0d880b38eb

Field	Recorded value
independent_clean_source.source_python_sha256.analysis/44_full_system_mathematics/verify.py	1643e72a9c3892f49b7df9bce5c51dfea67cbc2d09776735897dc5cf8fc672a9
independent_clean_source.source_python_sha256.analysis/44_full_system_mathematics/verify_completion.py	e590ef11c92d83158155512d325a6438d178d9b317fe99b14090ac4361ec623d
independent_clean_source.source_python_sha256.analysis/44_full_system_mathematics/verify_exact.py	4f24701c32b7c0bb59f8616fa8f849987d5f3feafbd125265444dbf03d7c2617
independent_clean_source.source_python_sha256.src/ae4_identifiability.py	42a4d68c290b37e5ffd2b908af97c792b87336ed36db37d3137cd08b7dd0736b
independent_clean_source.source_python_sha256.src/ae4_mechanism_reconstruction/__init__.py	d587aa6f5453f262eab170603d655f844ead07b376f3862cbea2a92f2b1792c4
independent_clean_source.source_python_sha256.src/ae4_mechanism_reconstruction/chassis.py	c9c5fa4e98112d425c804a74c3c31638f06886b877c52e6e9b91ae3b0f0a4877
independent_clean_source.source_python_sha256.src/ae4_mechanism_reconstruction/comparison.py	686c90f7f13c6954625a0f75565eb33e0aabc911b20b7b5370feeb0187b934b4
independent_clean_source.source_python_sha256.src/ae4_mechanism_reconstruction/mechanisms.py	abc5e90aab2ee2f8f49a7bcb5c88b137ed7784049ee2d00ca997e2d76747a23
independent_clean_source.source_python_sha256.src/ae4_mechanism_reconstruction/run_analysis.py	6bdd73472709688f2cbad88c25f9ca00aeea44906796fe235ddbcbda8de69971c
independent_clean_source.source_python_sha256.src/ae4_mechanism_reconstruction/structure.py	b37cac0665a77c00e34a439bd65363aa68b9ea3b14e2a65c5496e5d2bba9e157
independent_clean_source.source_python_sha256.src/identifiability_discrimination/__init__.py	e1e5d65574a840def898aaf53b39aec8bf2fb6ff9900b22dcf4bf51eca6d39e9
independent_clean_source.source_python_sha256.src/identifiability_discrimination/reduced_model.py	29782581c8cdef5139a0379ffb76b38a86f014377859348df236b537917c368f
independent_clean_source.source_python_sha256.src/identifiability_discrimination/run_analysis.py	296a3b613865b2cab543572497f3dc22e49b9f0ef2a0aea4724affc50ab26430
independent_clean_source.source_python_sha256.src/modern_full_model/__init__.py	721ec58e52c171e5261d769f5136acb5566574710c909dd917a36d048240f8a5
independent_clean_source.source_python_sha256.src/modern_full_model/absolute_hydraulic_diagnostic.py	19affea6ff6c7d199cc028c71b541b5f24661a859d4130e77330538d1cef89f6
independent_clean_source.source_python_sha256.src/modern_full_model/acid_base.py	cf123937b6cf085b0cbad01fc464fca9eadbb56888802e427900473c1a737a31
independent_clean_source.source_python_sha256.src/modern_full_model/ae4_cacc_recruitment.py	1adb937058a0bc1fa27cc789b99f2491d7d99d7b7c6e3dec4356ee12158c55aa
independent_clean_source.source_python_sha256.src/modern_full_model/ae4_catalan2025.py	f3b488f6c0075d68178167b0145543af3f04e4db889002c11cde782fa868da11
independent_clean_source.source_python_sha256.src/modern_full_model/ae4_equal_cation_routing.py	fbfb6fd25f69d3e5f57ad995a465710670b7a3704f0cf797a0a6c018910fa3293
independent_clean_source.source_python_sha256.src/modern_full_model/ae4_routing_only.py	ef3a55d6a74fbdc1a07a6f89c19533376aec7bee98cfbafa5f73cf4e01f63a10
independent_clean_source.source_python_sha256.src/modern_full_model/calibration.py	fc839e1caf26c9d8143059b3945d77322e1e84584e495a17bb2c7d047a7ca9ca
independent_clean_source.source_python_sha256.src/modern_full_model/camp_pka.py	b03ef7c794017a1a312fc9bd47188a9bc430e63c8b9c6be4dbfb6c93147eaf03
independent_clean_source.source_python_sha256.src/modern_full_model/freeze_g5_joint.py	410e05b5d08c61dec4ca5a38c14027b94c0133327a0d5b2e00a093ccc03bc76a
independent_clean_source.source_python_sha256.src/modern_full_model/genotype_evaluation.py	03c1fc871741e1720b90a51a8b5a57de0786ec72803ac38fedd2bee633fd6cea
independent_clean_source.source_python_sha256.src/modern_full_model/independent.py	0fdbcbca373505fa4684769e4c297aae464cc814c3787092aec60241550f3385
independent_clean_source.source_python_sha256.src/modern_full_model/independent_genotype.py	a0deb1ccd57cbf702461446e92c00dc052fdb3c09ed24b75f69276ebbf6a3a61
independent_clean_source.source_python_sha256.src/modern_full_model/independent_native.py	849dd3e7d22c470884a1255f8d529e73d3006bf785eb217e1e7a043605f5676c
independent_clean_source.source_python_sha256.src/modern_full_model/membranes.py	c4200328179cd70156590dd10eb0859505ff03d2005652fd1ae2c9b692f85fea
independent_clean_source.source_python_sha256.src/modern_full_model/model.py	a391600986aed9204cddb3aede884f6c420a5d9ba7de232d255900ec9e84027c
independent_clean_source.source_python_sha256.src/modern_full_model/native_dynamic_contract.py	d11a1ea7c7c2f5c8ba9bf7939574975b724772cd837fc87ac75ef3ede91bd6cc
independent_clean_source.source_python_sha256.src/modern_full_model/native_source_panel.py	8693b91a69ca652784e4b21e14b8c3c6fb9030ebe87a33ade3147d191ea1a367
independent_clean_source.source_python_sha256.src/modern_full_model/nbc_minimal.py	48230ad9b6c0bbb017d8545d0c46ca6fc83eb0d6ade57f1ced96da7fd28bcfaf
independent_clean_source.source_python_sha256.src/modern_full_model/nhe1_cha2009.py	de76dccffadb75fcaecee90402161af1db48f5db88ea152164f504a974d0e5c1
independent_clean_source.source_python_sha256.src/modern_full_model/nkcc1_capacity_limited.py	5d0e5bdfbca6a874bfe055f162bf925ef4db70d415692271248f351c14bed0e8
independent_clean_source.source_python_sha256.src/modern_full_model/nkcc1_palk2010.py	2b4b58450589dbf2b4765c35dbeee9062815574903490654864640c2fff6d60a6

Field	Recorded value
independent_clean_source.source_python_sha256.src/modern_full_model/nkcc_stimulation.py	c272da41b93dea37986e293fdb0936f994dade13b4a65e093057575f58788475
independent_clean_source.source_python_sha256.src/modern_full_model/parameters.py	5a3037ec9849e8202134c664b2a87205f73e2e17edde781e4fef0bc57b1a39b
independent_clean_source.source_python_sha256.src/modern_full_model/pooled_ae4.py	92eb87df49bcde5a42d0f679c8d9dfd7b015a1d6b0ac03e8ee895b4d39334cf
independent_clean_source.source_python_sha256.src/modern_full_model/report_calcium_fast_screen.py	cda9bb274250b26ec1fa2af7e32c20c73b76ebe8ded75b61badd3649dd5ab165
independent_clean_source.source_python_sha256.src/modern_full_model/run_absolute_hydraulic_diagnostic.py	d99d5fd33ac4b592c815e44b38bf1d79f5db5d9c7c6e22c1cd49a72d5d99fa95
independent_clean_source.source_python_sha256.src/modern_full_model/run_ae4_carbon_nhe1_compensation.py	7057b97c088a0d78ce0c13a68382fa699eba4399579687aaf5d818e750de6c49
independent_clean_source.source_python_sha256.src/modern_full_model/run_ae4_routing_only.py	d742c8196cc87cb4786e0622e530073361f155aa48d4288793960fb3bd246ac9
independent_clean_source.source_python_sha256.src/modern_full_model/run_ae4_routing_re_equilibrated.py	59b6b615bf7b91cfd111c89860a93a9b33bb5e242c99a3d6bdf689703f9af80
independent_clean_source.source_python_sha256.src/modern_full_model/run_analysis.py	919a9bc843240550537b8a2019bec89dfa6dcc0e4e8a920d7f041bf26a608339
independent_clean_source.source_python_sha256.src/modern_full_model/run_calcium_fast_screen.py	62f2032dafebadb819d770ac37623178ec9f256f4f4cc3356b4af907e647238b
independent_clean_source.source_python_sha256.src/modern_full_model/run_g5_ae2_validation.py	08e753039974759904fee294fbac91689a98d94aa081c183f2401a07a5842769
independent_clean_source.source_python_sha256.src/modern_full_model/run_g5_cross_root_profile.py	c682dd163f2e997e86a344edee2b75a3dd185e8f49e515928afaf4300a3e0bf4
independent_clean_source.source_python_sha256.src/modern_full_model/run_g5_joint_repair_profile.py	c815b44021bab726a282461b9dbc228d5c89e0449e4512708dd82ef7b3f305cd
independent_clean_source.source_python_sha256.src/modern_full_model/run_g5_nearby_robustness.py	8ef1613ced73de593074eb06f32334e005f14dc10306ba24bd1f85179dfac187
independent_clean_source.source_python_sha256.src/modern_full_model/run_independent_native.py	ff647783872c07ac4ee940ae3720a613c6e9bd409ff7eaf54742f96141ebd957
independent_clean_source.source_python_sha256.src/modern_full_model/run_independent_native_ae2.py	5778a4358d9694e42575a3c87850147d98943ee0968c97acb8ed801414988c11
independent_clean_source.source_python_sha256.src/modern_full_model/run_independent_native_dynamic.py	8ed8f7262aaca6aaa6ddd0d322695d95d6220426c2ca90cdf7c5ddec5fae40b0
independent_clean_source.source_python_sha256.src/modern_full_model/run_independent_native_genotypes.py	0f171b69df639355d1cd7fd6f4174597033433130039bed92a71ba1a02b7243c
independent_clean_source.source_python_sha256.src/modern_full_model/run_native_dynamic_contract.py	f4380a27f859b6d76368630259d5b3988a3a693ffd8a58520c42457d46966c4f
independent_clean_source.source_python_sha256.src/modern_full_model/run_native_genotype_evaluation.py	db3d9f89ecbff3961c43188bbd13227aa672bd744496a126de960df4d65303ec
independent_clean_source.source_python_sha256.src/modern_full_model/run_native_source_panel.py	76847920661f7ec14e36b13f8c08dfdc7a2a101fd8e4731ff39751146cca88f
independent_clean_source.source_python_sha256.src/modern_full_model/run_nkcc_stimulation_profile.py	cbc08c882a1dcd0347cbc16a17cde03e285c00e6b84f4cbf3c24838c46a035a9
independent_clean_source.source_python_sha256.src/modern_full_model/run_vbeta_diagnostic.py	57ab6cbd860e3d22d3d5e90971b8241de94f3a4e02290d7b6e7a8cdf44659b5c
independent_clean_source.source_python_sha256.src/modern_full_model/run_wt_calibration.py	5224bd94638d915a6806933f233ff367e7650d0298c5fff630ece41a0e6c4f0e
independent_clean_source.source_python_sha256.src/modern_full_model/states.py	a037f9690960c3b52e738bc4051601adf76bf345b3a32b191770e2469255fd22
independent_clean_source.source_python_sha256.src/modern_full_model/task14_blind.py	d11b1b70c99589e62e8edfa54d60fa91165ee7cc4870bd0c64c95720fd296f94
independent_clean_source.source_python_sha256.src/modern_full_model/task14_compare.py	f8b758f4fd059e6150517aac15fb081b2fa9514552e49fd93f348f2480f7fb39
independent_clean_source.source_python_sha256.src/modern_full_model/task14_report.py	7db7eabfe5d1d029f77daf1bd301fedb2bb50879e1f919ba5bbebb5957e2277e
independent_clean_source.source_python_sha256.src/modern_full_model/task14_reveal.py	b387fa51991f6c6f1397e8e7273c9b47ecc93754b592ea849c032db2e591db48
independent_clean_source.source_python_sha256.src/modern_full_model/task14b_five_percent.py	d8c28629a89941a4fac6a5a7d2de3f4b4f67d54a96cbde61a72d0054146af256
independent_clean_source.source_python_sha256.src/modern_full_model/task16_postfreeze.py	f791f09c17165e5da89af8e66743f09307545e61143d90f89960ae61e4b65dad
independent_clean_source.source_python_sha256.src/modern_full_model/task16_wt_allocation.py	c611fc080836f86cb51da67ae324fbfcb95e7298b33a8da34f49285fc2760d77
independent_clean_source.source_python_sha256.src/modern_full_model/task16_wt_diagnostics.py	612f891b1ed39731bc47e805f7e89fe0205e793d2dba99c86116636f7c832d

Field	Recorded value
independent_clean_source.source_python_sha256.src/modern_full_model/task17_fixed_wt.py	45c3b4fac5e8ebca4f8f894bdba9218191d86904b9228c856e2a3cd6f9f4803
independent_clean_source.source_python_sha256.src/modern_full_model/task17_postfreeze.py	d127addd4e40d2f6b8efa8db2f20ccc714c504b19b197ff88a22ce0264d504ae0
independent_clean_source.source_python_sha256.src/modern_full_model/task18_fixed_wt.py	bddd5ae4b5d9f761d01191730e933437854d83206c9b757fd5a437d148ba78e0
independent_clean_source.source_python_sha256.src/modern_full_model/task18_postfreeze.py	50d723b43934585ba72820b26d99a42bb57126b22ee231a8eb9ddb5928df93c4
independent_clean_source.source_python_sha256.src/modern_full_model/task19_pump_coupling.py	51903e225e7471975dcc6ac013e687fcd7f556892890f4028ab3f50959edb869
independent_clean_source.source_python_sha256.src/modern_full_model/task19_reporting.py	ea3228e1671180c0380e7b3c6d19ac2f8e331662a01880c451c8f3ecb02fbd0
independent_clean_source.source_python_sha256.src/modern_full_model/task20_fixed_wt.py	159d7a822cfed6c6de1772a87e74e522bdebaf5d430c84ac7adf5fcd7d8fc7c3
independent_clean_source.source_python_sha256.src/modern_full_model/task20_postfreeze.py	4c2b6e2520fd87d46f0234bd248ceac30ab5a6382b26eea27d981d5f245b6989
independent_clean_source.source_python_sha256.src/modern_full_model/task30_nhe1_repair.py	7d48386bf081212c2e81aeb6f0d28e47c9d3c9edec4d2f4ab3b80f762f34b105
independent_clean_source.source_python_sha256.src/modern_full_model/task31_nhe1_repair.py	976b62f8aadf87dbb5badb156d40ea974b53eee0e21acd8fd7a08d64543e07d4
independent_clean_source.source_python_sha256.src/modern_full_model/task41_selected.py	56d55bbb196a5df2a86434f44a8be098eaa41345c0fbffa187dd72ccb44f7d67
independent_clean_source.source_python_sha256.src/modern_full_model/transporters.py	4177a444fa97e430c345fc1fb73f333b5b6623c3ea3cd84769cb9d6dca76061a
independent_clean_source.source_python_sha256.src/modern_full_model/validation.py	e8dd7d9bee588ecab6032a9fb35f41907a112b08d6e8448ca1d7ab60fee7b4b8
independent_clean_source.source_python_sha256.src/modern_full_model/vbeta_diagnostic.py	8cb8e303ad3d39e50ff0434d98494bc5822ad809a720b6ecfe476312ceb18f17
independent_clean_source.source_python_sha256.src/modern_full_model/water.py	d93494f93411123a9c9bdaea5c177c89714ee5ab0794717c586bfd87647671ba
independent_clean_source.source_python_sha256.src/run_identifiability_audit.py	9c2ca29a5f8b8b1f5fa05bc082d099ac047c01fa790baa3e727e85dc51b87001
independent_clean_source.source_python_sha256.src/state_resolved_ae4/__init__.py	72ec5bc9b8c7bad56a98e4458da95729b02c520e94fc144c2bd96a554c31cb73
independent_clean_source.source_python_sha256.src/state_resolved_ae4/adapters.py	314e5554314470efb07026cad42f030bedfd8a4ffcbe6b67be99a836fd38bc10
independent_clean_source.source_python_sha256.src/state_resolved_ae4/chassis.py	eca2b98676cb7b3b7c977bff5b8f32bd6437b567687f407d33bd2501640db7d
independent_clean_source.source_python_sha256.src/state_resolved_ae4/chemistry.py	cb8450a9e4422669d4ae87dc59c8b9cfd4b8376b6741aaa7941a08cbb0bd339
independent_clean_source.source_python_sha256.src/state_resolved_ae4/cycles.py	3f2db8788dde2bc24c36d09061732e69dded712583bfdd4ec97515429c108257
independent_clean_source.source_python_sha256.src/state_resolved_ae4/evidence.py	3d3be4dc0cf8d4d88778b780e8885a9ddc04ca7fb33780b553036a87a97d1e78
independent_clean_source.source_python_sha256.src/state_resolved_ae4/export_ctmc_reproduction.py	944a5f12e45f20e46a13ab754def94db7c66058d430f724b61f410fc4c0f1efc
independent_clean_source.source_python_sha256.src/state_resolved_ae4/fitting.py	f6be3be2cc21523e2a164edc216556f8854274464ece27ce2583546b8a99ca23
independent_clean_source.source_python_sha256.src/state_resolved_ae4/independent.py	370d266e3ee264225ec5890fba5d45d916d8dbbac56ebbbee9d9b8b2aec8231
independent_clean_source.source_python_sha256.src/state_resolved_ae4/localization.py	da8f44b85c37b5d0b88474f6600b4bef71827310c45b54fcadd8ee3b1ffef0a5
independent_clean_source.source_python_sha256.src/state_resolved_ae4/protocols.py	ee9b4ae0e83685ddd7594ccf7595673e9de71da33db6c32e6fb951a2f01616d6
independent_clean_source.source_python_sha256.src/state_resolved_ae4/run_amendment_01.py	d3800ead6bf28de70da7bb1933969c3f98ab51f7439cf783ad32a6ee073d431a
independent_clean_source.source_python_sha256.src/state_resolved_ae4/run_chassis_reconstruction.py	06bf55bfe8b8a7a037b865ef6b8fe0c7324c9fd8447ee8de8d64a09350ec533a
independent_clean_source.source_python_sha256.src/state_resolved_ae4/run_stage_a.py	b810af4432af5a392b701b8ee50659c6651f086722df2af70160a7c831db7bde
independent_clean_source.source_python_sha256.src/state_resolved_ae4/solvers.py	bc66a087229c9a062173e3b2cd1a0f965231a3b15dcd05f5b544dc8dcb31585b
independent_clean_source.source_python_sha256.src/state_resolved_ae4/validation.py	00deb5128df5de0d29e1ee230e09721a1f7a95b3fda646546f36bd32ace35c2c
independent_clean_source.source_sha	f783785a863440df459e9c5530beba4c7eb59110
numeric_values_compared	331157
numeric_values_with_tolerated_difference	0
numerical_cache_reuse	No
protected_paths_unchanged	Yes
regenerated_inverse_trial_cache_files	532
regeneration_scope.authoritative_files_compared	47

Field	Recorded value
regeneration_scope.explanation	The historical recovery record documents an earlier check-point and is explicitly identified as such; it is compared unchanged, while every current numerical product is re-generated.
regeneration_scope.numerical_output_files_regenerated	46
source_files_unchanged	Yes
status	passed
supplementary_worktree_audit.all_task44_numerical_sources_unchanged	Yes
supplementary_worktree_audit.status	passed
supplementary_worktree_audit.tracked_files_outside_task44	1751
tolerance_policy.conservations_tolerance_ratio_rollback.atol	1e-05
tolerance_policy.conservations_tolerance_ratio_rollback.rtol	1e-08
tolerance_policy.default.atol	1e-14
tolerance_policy.default.rtol	1e-08
tolerance_policy.electrical_current_ampere.atol	1e-19
tolerance_policy.electrical_current_ampere.rtol	1e-08
tolerance_policy.rationale	The same runtime should reproduce deterministic solves nearly exactly. Tiny differences of conservation cancellation residuals are allowed far below their physiological gates. No failed gate, boolean, class, root count or experimental observable is excluded.
tolerance_policy.small_equilibrium_or_charge_residual.atol	1e-10
tolerance_policy.small_equilibrium_or_charge_residual.rtol	1e-08
trial_cache_scope	Inverse trial caches are regenerated from zero and are not authoritative report files. Every top level inverse summary/CSV and every output JSON/CSV is compared. No cache value is read by a solver.
verification_revision_provenance.clean_recomputed_check_sha	f783785a863440df459e9c5530beba4c7eb59110
verification_revision_provenance.published_check_is_ancestor_of_clean_checkpoint	Yes
verification_revision_provenance.published_precommit_check_sha	5589fa36e2c7d1e5bf21869aad82909a123e759b

Source record: evidence/task44/output/clean_recomputation_verification.json.

clean sensitivity gate verification

Table 42: Scalar receipt fields.

Field	Recorded value
architecture_gate_revision.all_384_case_records_unchanged	Yes
architecture_gate_revision.current_script_sha256	7219d49b5e4819adfa6eade20f83ce44bd8ece9637cdae997ef3c5a06dae62c6
architecture_gate_revision.numerical_values_unchanged	Yes
architecture_gate_revision.prior_script_sha256	0d53ed3d2340638ad87f35ee43756d180708de5b5181e97915b1136b361185ac
base_numerical_checkpoint	f783785a863440df459e9c5530beba4c7eb59110
byte_identical	Yes
comparison_policy	Exact equality for every key, scalar and list element; no numeric tolerance or metadata exclusion.
expected_json_sha256	dbd522586ccdbc1be521a2dc7d7827a06e813cccc5b29e4598c71705bc792316
groups.compensation_expression.attempted	256
groups.compensation_expression.expected	256
groups.compensation_expression.failed	0
groups.compensation_expression.passed	256

Field	Recorded value
groups.parameter_WT_null.attempted	128
groups.parameter_WT_null.expected	128
groups.parameter_WT_null.failed	0
groups.parameter_WT_null.passed	128
maximum_conservation_tolerance_ratio	4.085620731e-05
maximum_independent_rhs	6.879770757e-12
numeric_values_compared_exactly	7567
protected_worktree_paths_unchanged	Yes
recomputed_json_sha256	dbd522586ccdbc1be521a2dc7d7827a06e813cccc5b29e4598c71705bc792316
scalar_values_compared_exactly	10265
scope_note	Calls complete Task 40 diagnose: all inherited Task 37 physiological/conservation/NBC checks plus equal_routing_evaluator_active and equal_routing_source_signature. No gate is bypassed or relaxed.
source_acquisition.base_commit	f783785a863440df459e9c5530beba4c7eb59110
source_acquisition.clean_output_absent_before_execution	Yes
source_acquisition.elapsed_s	9.274271503
source_acquisition.expected_output_sha256	dbd522586ccdbc1be521a2dc7d7827a06e813cccc5b29e4598c71705bc792316
source_acquisition.returncode	0
source_acquisition.revision_note	Supplement expanded to include both Task 40 equal-routing architecture gates.
source_acquisition.script_sha256	7219d49b5e4819adfa6eade20f83ce44bd8ece9637cdae997ef3c5a06dae62c6
status	passed
supplementary_script	analysis/44_full_system_mathematics/verify_sensitivity_gates.py
supplementary_script_sha256	7219d49b5e4819adfa6eade20f83ce44bd8ece9637cdae997ef3c5a06dae62c6

Source record: evidence/task44/output/clean_sensitivity_gate_verification.json.

report verification

Table 43: Scalar receipt fields.

Field	Recorded value
clean_replay_numeric_values	331157
figures	5
independently_reviewed_required_topics	17
numerical_source_commit	f783785a863440df459e9c5530beba4c7eb59110
pdf_pages	26
references_resolved	Yes
report_artifact_sha256.AE4_full_system_mathematics.pdf	3a86965653ff2cc4834ee99772f48edc4dbbba8b6b6b0960ff3110c91b63c60d
report_artifact_sha256.build.log	6eea687535a191b910b72de9771990173dec001d45655158af479a7578c8c9b3
report_artifact_sha256.build.sh	fe580d1e98c0c351c33c9b3a89a4fcd55486df8232d8e84a2dee09e64c829f73
report_artifact_sha256.equilibrium_results.tex	18665662d250a809ebfbc7de1df8a2194702f29847645e462f48a3a138d6d576
report_artifact_sha256.exact_results.tex	f71ac326b8723641b55a74049c5ed5861cdfdd0f8f5c76d5d66da5ea684333c3
report_artifact_sha256.integrated_context.tex	9c8c9dc93922d3e82917731d0cbd31c71d8a023ab5e2ec3cb36c82130d19efb8
report_artifact_sha256.inverse_results.tex	55995704db8c817ad47284ec9d8c09922947b0a79568144bcd4a409a1f2f1d07
report_artifact_sha256.literature_comparison.tex	8d730d3dae86111402938b40319f9e63d47e57d8d9411986da1fede03cae9065
report_artifact_sha256.output/additional_groups_table.tex	24bc5373a76678d888b510e22c28f18f152f24deaebd9b20b3d62284883d1a0e
report_artifact_sha256.output/class_table.tex	f1ad436b949807789e26bb4d529314182ef8b456f866717761ab48e07e75ae1d

Field	Recorded value
report_artifact_sha256.output/equilibrium_flow.pdf	6671a8c6287066f399e650a5b71d8dda9ec44242779ad7df75c157ef4d60a5da
report_artifact_sha256.output/equilibrium_flow.png	72aaccacf57f93e350fc25c0906b3fd43780c34a2aa33abd3047028224dcf0c13
report_artifact_sha256.output/inverse_ph.pdf	f1583732dd4787d8bcf646e9c22a55daffd3715e07b1699e4fb98c4fa932416a
report_artifact_sha256.output/inverse_ph.png	8273b8f9897cc09f004e77957f6acc29cee4e6700c252d83bd515c05ff8b4614
report_artifact_sha256.output/inverse_threshold.pdf	9c704437cd5062d67a70a4a2bd6ff8458115e6929c2875d64bc2f8276448ab62
report_artifact_sha256.output/inverse_threshold.png	a42d72a6420446b3790867b195c9371350ae3f1e95f4305e016bdececefefb50
report_artifact_sha256.output/linear_response.pdf	908dd7e220b7af6c9171a8857b676fe00232ae5f9a1bc54df62c1818de64d0bd
report_artifact_sha256.output/linear_response.png	6f3cfaee2945d68748f215f853e9455ef193f2726e8873d1f90965c67edc83f6
report_artifact_sha256.output/nbc_table.tex	ae9ea8821f91af4e7126286b7a4ccf6dba857b40ecb89f1c5e9a43aledafadcd
report_artifact_sha256.output/projection_table.tex	fa5809b8065255de3c5f330307f8a06932cfd8d9f266bc0b1e122de45ba38e06
report_artifact_sha256.output/scale_table.tex	2960f90a792c7b93ede8deda9b3c9213908d3a4e75defe478a869fe7595850c8
report_artifact_sha256.output/sensitivity_table.tex	65d6572a3e6aebac49ef20f0b728c931868d5572d34cf1fe04801c98f92cd238
report_artifact_sha256.output/slow_modes.pdf	171a813e74560a67dea841ecb56625a5e91179b976e7d3cd057606b26c5311ae
report_artifact_sha256.output/slow_modes.png	ce93878bf86d31f065662c59283dc1dc62b4a4c85f0f796bec2d1eac46ff9f64
report_artifact_sha256.output/spectrum_table.tex	a65cfc329eea128dec23701891d00e13f78dee8e4063824e9b42f67053ad03a8
report_artifact_sha256.report.tex	384795a43b0fc7becc02313ec2f68cbeaee4c974c22960427c874ce407d8273f
report_artifact_sha256.sensitivity_results.tex	be01d6d49a36bb6bf6c99f98f35ae0fc18f56f935216ed77b6dfce84de30f0a9
report_artifact_sha256.sources.bib	05d7cc09976860539b440649d8125c357ec749368b069043f82f3e0c05385634
scope	Artifact consistency and unchanged numerical payload checks. Visual and scientific review is recorded separately; no new scientific model solves.
status	passed
supplementary_clean_numeric_values	7567
supplementary_gate_roots	384
verified_numerical_payload_sha256.inverse_detail/crossing_long_time.csv	05f0c18e96d13dd1cf073dc8f2bd1e42d912465864197de00ed7b385b2f8e776
verified_numerical_payload_sha256.inverse_detail/crossings.json	91095cdf07abf3faf5e9264f4f7e3869d07b66ff2bd6584bdf27109318c951a26
verified_numerical_payload_sha256.inverse_detail/independent_verification.json	386898553ac8a9817686427cf7bbf421eaf4ab0fc6226a8822c0c26f50e0963f
verified_numerical_payload_sha256.inverse_detail/long_time_summary.json	8615e26807c3468e66868dfd23d07ff59a36f9713184da7fd8f243c8d335e502
verified_numerical_payload_sha256.inverse_detail/ordered_family_scan.json	b6620d04217b911b64178b6db87e899fabcac1ba07ed061ecc855ae78982ebe1
verified_numerical_payload_sha256.inverse_detail/protocol.json	aaf620bdb42aa90c76309379d48edf85a7139a50dfe855cb0d100bcf0b272db3
verified_numerical_payload_sha256.inverse_detail/selected_long_time.csv	5b265bfa7498d84b17fb64f1623e477c7c902cbd290620c2b82e20686e488a46
verified_numerical_payload_sha256.inverse_detail/selected_reproduction.json	2b34f7942e59699b502452f0d4c9a237a8961c0339c51f6336018974fec12d27
verified_numerical_payload_sha256.inverse_detail/sensitivity_AE4_carrier.json	7f789e95fd6d9d122cdc051e2616f8190399866ed92dc5fe71caa257c623d5d1
verified_numerical_payload_sha256.inverse_detail/sensitivity_NHE_carrier.json	d998b11871d319e02d98c522455d3e1c69086e23bc93ea837062bf7f21a965c5
verified_numerical_payload_sha256.inverse_detail/sensitivity_NBC_capacity.json	08855428f9b346456f83ef8767841a9d8ceec6ec6b52ffda c22fefc9fdef07b6
verified_numerical_payload_sha256.inverse_detail/sensitivity_NHE_carrier.json	22dc5c868b6d6acc10ec2f9ca1c3947423c17228981adc2c bc9a37664fa12666
verified_numerical_payload_sha256.inverse_detail/sensitivity_NKCC_scale.json	c7d63cb915d49f5205459057b7ab003121b084d4ad58a57c 5f92c42ad1cc693a
verified_numerical_payload_sha256.inverse_detail/sensitivity_buffer_pool.json	26b2b757672eb86f882cf0fd0aca3bca6cc427b0a36e2c20 18d9cae86404a78b
verified_numerical_payload_sha256.inverse_detail/sensitivity_pump_capacity.json	ba4048ab72be49257e8f64c233b22e190a794cc14eb2fa29 79c2ea46fa0540f1

Field	Recorded value
verified_numerical_payload_sha256.inverse_detail/sensitivity_water_apical.json	f25a6765d7d74af6c074697ac532e69ba122408e1e9f53ba933bc81f0ac3188e
verified_numerical_payload_sha256.inverse_detail/threshold_sensitivities.csv	b7551d60ecb463367d5b19ae3e3b9aa519ea727f267d3ba25165a9a0d4837a65
verified_numerical_payload_sha256.inverse_detail/threshold_sensitivities.json	73149e626bc608c998b7dc23ca261b8b9520f0361a5f50b16a364ae19b7876ac
verified_numerical_payload_sha256.output/active_parameters.json	1a4a4708de6d1752add85b99135b421f96c54c141a018390c64de50368ec774e
verified_numerical_payload_sha256.output/algebra_verification.json	f652361c64e015167165c1da1656733871b9c4840df7ed5cb442f257f61a4dbb
verified_numerical_payload_sha256.output/compensation_comparison.json	ec2a57700e742b3c824816a0df48cf4e9e4eb3db44b0b84815bbcff1d8c36b2c
verified_numerical_payload_sha256.output/completion_verification.json	92338be4399e4d1837f2033efd9d371f646a5908e18d01ca63b7ef7de82c4a5f
verified_numerical_payload_sha256.output/coordinate_attribution.csv	c0f3605d0c4081364e5543aed5f072299321fda4bb872fd0c05b23374a7092e6
verified_numerical_payload_sha256.output/dimensionless_groups.json	c639fde60da5fef593f19e8154a80178ba7bfdec0696571d607599141f558a3d
verified_numerical_payload_sha256.output/equilibrium.json	d5814e8a55b00fd5075e81a9d8021235c7dc1ffa5a52b88b3d02b9ced2fd3d2f
verified_numerical_payload_sha256.output/equilibrium_review.json	029f64cd334febf16304e5bc9e358cd1660304a86faf81a8771756354cbd569f
verified_numerical_payload_sha256.output/equilibrium_summary.csv	0bc4d74b45720e2fb706f97cc8b2cd297fc334141f68746feb0f29581c985cb3
verified_numerical_payload_sha256.output/exact_verification.json	7e6a9815695ff279204ad188df09fac37a5b00b0da315933cb99044389a2508a
verified_numerical_payload_sha256.output/execution_summary.json	ddd9fbf93443a273c14bf38e97345345906942e37ff46694d8cba41ae1a11773
verified_numerical_payload_sha256.output/expression_sensitivities.csv	7bd17f65f8899dbeeb2b47fa000de2df8c4c2709ecc99ca7b288e2d283df4fe0
verified_numerical_payload_sha256.output/inverse_legacy_threshold.json	f67d6dc3a66b7f307597179c61dee123588481e40ba6d47e829e833ff28f8c96
verified_numerical_payload_sha256.output/inverse_long_time.csv	6b7cc6f04833de136809b83f47905d78a9f3f2d330913da02fcc5035be41ae9
verified_numerical_payload_sha256.output/inverse_long_time_summary.json	41a1b8590f1d73ebc74e6316d1db28914e7000caa521a263024848254294102f
verified_numerical_payload_sha256.output/inverse_samples.json	b8b78ba698f62466a929f232862579259d32509c10ca580c9137d0eae7de786e
verified_numerical_payload_sha256.output/inverse_threshold.json	418366d8c49ed44ef1002e971007491c2a9fe2d11c6380660109438d2cfebb2c
verified_numerical_payload_sha256.output/local_step_response.csv	bc5c64d461549d298908d8fb3b78b30d2430e78fffe348ff2f6f88d43ae6d935
verified_numerical_payload_sha256.output/nbc_continuation.csv	10bb57e113cdcefaf18dc7047412f5b240e73ed4f99d90032dc800594414ae36
verified_numerical_payload_sha256.output/parameter_sensitivity.csv	e72a4d01296a40be04a8067d058135343bca57c1764b7f048b1ce443516e1ba1
verified_numerical_payload_sha256.output/recovery_verification.json	b0cfb323bde3a37fd6d9ec0b97db72254c9165482bc20734376988d44e10a2f7
verified_numerical_payload_sha256.output/rescaling_verification.json	9a10f81de7a2920518c99a5b1bf8c2ea462be283039dd0cf3550049f3bfa24d6
verified_numerical_payload_sha256.output/sensitivity_extended.csv	992742a4a365c8964780a4a341ae9fcac78b94a1aa185b448d90fc5f2f62f4fe
verified_numerical_payload_sha256.output/sensitivity_extended.json	712fe98b8c75a63028c3bb697f1a3a6e901a3653ede82e6f670c4703dde145a
verified_numerical_payload_sha256.output/sensitivity_verification.json	940451d234d1beaf1b95aa2e02ef3b459878d43b5a530955206e630ac113713d
verified_numerical_payload_sha256.output/source_audit.json	52aafdfc2da3665c29bf4834209036c1e9a4c41f6522b87bfd452e57619ee829
verified_numerical_payload_sha256.output/source_projections.csv	1f6f0a3cdf96b6ee5db4f435bcc9d7bd24dabfac3298e6e5e2b96b0e1a74d4f
verified_numerical_payload_sha256.output/test_summary.json	512bc2c2d7e5cce5e8353a09bd221ca3979dbfeee354254a83f4a496f1009e49
verified_numerical_payload_sha256.output/wt_trajectory.csv	64aa41573f9d9841cd1bed007cf7b823df9d405411500dd996cc7cf79c5244ae
visual_review_scope	Root inspected pages 1–8 and all five standalone figures; independent reviewers inspected pages 9–16 and 17–26, and all changed pages were reinspected.

Source record: evidence/task44/output/report_verification.json.

independent verification

Table 44: Scalar receipt fields.

Field	Recorded value
all_inherited_gates_checked	Yes
diagnostic_scope	Long extension checked separately; no retrospective rejection or alteration of Task 41
sensitivity_rows	16
status	PASS
threshold_tolerance_checks.adaptive	1.36557432e-13
threshold_tolerance_checks.sampled	1.715638742e-11
verification_method	Raw 13-state Radau integration plus independent adaptive dense-flow quadrature; original sampled trapezoid also reproduced

Source record: `evidence/task44/inverse_detail/independent_verification.json`.

42 All audited numerical controls

Table 45: Production tolerances and inversion settings. These are not bounds on biological model error.

Control	Recorded value
classification	numerical controls and unit conventions, not physiological uncertainty
conservation_tolerances.ae4_charge_fmols	1e-10
conservation_tolerances.apical_current_A	1e-20
conservation_tolerances.basolateral_current_A	1e-20
conservation_tolerances.buffer_site_accounting_fmols	1e-10
conservation_tolerances.carbon_accounting_fmols	1e-10
conservation_tolerances.cell_bulk_charge_rate_fmols	1e-10
conservation_tolerances.cell_speciation_alkalinity_mM	1e-09
conservation_tolerances.homeostasis_charge_fmols	1e-10
conservation_tolerances.lumen_bulk_charge_rate_minus_outflow_fmols	1e-10
conservation_tolerances.lumen_speciation_alkalinity_mM	1e-09
conservation_tolerances.water_volume_accounting_pLs	1e-12
note	Tolerances are audited from retained source or production specifications. They do not bound model error or parameter uncertainty.
ph_inversion.lower	3
ph_inversion.rtol	1e-13
ph_inversion.source	<code>src/modern_full_model/acid_base.py:speciate</code>
ph_inversion.upper	11
ph_inversion.xtol	1e-12
production_solver.amount_atol_fmols	1e-10
production_solver.label	production_radau
production_solver.max_step_s	2
production_solver.method	Radau
production_solver.regulatory_atol_fraction	1e-10
production_solver.rtol	1e-07
production_solver.volume_atol_pL	1e-12
regulatory_trial_tolerance	0.0001
stimulus_discontinuity.initial_observable_time_s	0
stimulus_discontinuity.integration_start_s	1e-06
stimulus_discontinuity.shared_initial_state_preserved	Yes
stimulus_discontinuity.source	<code>analysis/40_ae4_equal_cation_routing/run_case.py:152</code>
unit_conventions.fmols_per_mol	1e+15
unit_conventions.mM_times_pL_equals_fmols	Yes
unit_conventions.ms_per_s	1000
voltage_inversion.lower_V	-0.5
voltage_inversion.rtol	1e-13

Control	Recorded value
voltage_inversion.source	src/modern_full_model/nbc_minimal.py:evaluate _membrane_closure_with_nbc
voltage_inversion.upper_V	0.5
voltage_inversion.xtol_V	1e-14

Source record: evidence/task43/output/numerical_controls.json.

43 Complete electronic evidence catalogue

Every file below is included in the report folder as an unchanged copy from the pinned scientific commit. The full SHA256, source path, source commit and byte count are recorded in **SOURCE_MANIFEST.json**. No cache, missing historical seed or new experimental observation is inferred from this catalogue.

Table 46: Complete task43 snapshot.

File within snapshot	Bytes
AE4_parameter_provenance.pdf	287482
CHECKPOINT_02.md	1396
FINAL_STATUS.md	5831
PROTOCOL.md	1264
audit.py	16615
build.sh	856
import_crosswalk.py	6122
output/audit_counts.tex	145
output/audit_summary.json	2089
output/build_provenance.json	873
output/inventory_table.tex	10449
output/numerical_controls.json	1702
output/parameter_inventory.csv	357834
output/parameter_inventory.json	460253
output/sensitivity_table.tex	1312
output/source_manifest.json	11220
output/task44_sensitivity_crosswalk.csv	17601
output/task44_sensitivity_crosswalk.json	81226
output/verification.json	1131
prepare_report.py	2301
provenance_metadata.py	18562
report.tex	24609
sources.bib	22761
verify.py	7763

Table 47: Complete task44 snapshot.

File within snapshot	Bytes
.gitignore	111
AE4_full_system_mathematics.pdf	555772
CHECKPOINT_01.json	1409
FINAL_STATUS.md	10624
PROTOCOL.md	3490
README.md	3293
RECOVERY.md	2325
build.log	27284
build.sh	1336
core.py	10775
equilibrium_results.tex	6131
equilibrium_review.py	22532
exact_results.tex	24136
integrated_context.tex	11776
inverse_analysis.py	19157
inverse_detail/crossing_long_time.csv	2066705

File within snapshot	Bytes
inverse_detail/crossings.json	48226
inverse_detail/independent_verification.json	2519
inverse_detail/long_time_summary.json	26562
inverse_detail/ordered_family_scan.json	93366
inverse_detail/protocol.json	16188
inverse_detail/selected_long_time.csv	2063008
inverse_detail/selected_reproduction.json	17141
inverse_detail/sensitivity_AE4_carrier.json	2871
inverse_detail/sensitivity_CaCC_conductance.json	2844
inverse_detail/sensitivity_NBC_capacity.json	2878
inverse_detail/sensitivity_NHE_carrier.json	2869
inverse_detail/sensitivity_NKCC_scale.json	2872
inverse_detail/sensitivity_buffer_pool.json	2862
inverse_detail/sensitivity_pump_capacity.json	2862
inverse_detail/sensitivity_water_apical.json	2896
inverse_detail/threshold_sensitivities.csv	5426
inverse_detail/threshold_sensitivities.json	22933
inverse_results.tex	7803
inverse_verify.py	4027
literature_comparison.tex	8358
output/active_parameters.json	3845
output/additional_groups_table.tex	959
output/algebra_verification.json	99
output/class_table.tex	450
output/clean_recomputation_verification.json	41356
output/clean_sensitivity_gate_verification.json	2514
output/compensation_comparison.json	396
output/completion_verification.json	940
output/coordinate_attribution.csv	799
output/dimensionless_groups.json	1926
output/equilibria.json	178467
output/equilibrium_flow.pdf	14708
output/equilibrium_flow.png	47674
output/equilibrium_review.json	265862
output/equilibrium_summary.csv	17493
output/exact_verification.json	4008
output/execution_summary.json	347
output/expression_sensitivities.csv	1140
output/final_repository_audit.json	10465
output/inverse_legacy_threshold.json	5687
output/inverse_long_time.csv	383361
output/inverse_long_time_summary.json	428
output/inverse_ph.pdf	15563
output/inverse_ph.png	47583
output/inverse_samples.json	214828
output/inverse_threshold.json	23417
output/inverse_threshold.pdf	16996
output/inverse_threshold.png	48225
output/linear_response.pdf	14270
output/linear_response.png	56496
output/local_step_response.csv	577
output/nbc_continuation.csv	3008
output/nbc_table.tex	344
output/parameter_sensitivity.csv	690
output/projection_table.tex	471
output/recovery_verification.json	215
output/report_independent_review.json	11158
output/report_verification.json	9222
output/rescaling_verification.json	275
output/scale_table.tex	435
output/sensitivity_extended.csv	16719
output/sensitivity_extended.json	316717
output/sensitivity_full_gate_verification.json	522030
output/sensitivity_table.tex	467
output/sensitivity_verification.json	682
output/slow_modes.pdf	13543
output/slow_modes.png	44784
output/source_audit.json	27304
output/source_projections.csv	447
output/spectrum_table.tex	461
output/test_summary.json	69
output/wt_trajectory.csv	326554

File within snapshot	Bytes
prepare_inverse_report.py	9383
prepare_report.py	5461
prepare_summary.py	1795
recompute_pinned_checkpoint.py	13229
report.tex	17192
run_analysis.py	16898
sensitivity_analysis.py	25679
sensitivity_results.tex	4707
sources.bib	3247
verify.py	5126
verify_completion.py	3772
verify_exact.py	13677
verify_report.py	5412
verify_sensitivity_gates.py	6690

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